

Quantitative Aptitude



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Module - 1: Algebra

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1.1 Algebra

1.1.1 Introduction to Indices, properties and application in real life

1.1.2 Introduction to Surds, properties and application in real life

1.1.3 Introduction to Logarithms, properties and application of common logarithms

1.1.4 Introduction to equations - Linear and Quadratic, roots of quadratic equation

1.1.5 Methods to solve quadratic equations

1.1.6 Introduction to simultaneous equations and methods to solve them with two or three unknowns

1.1.7 Introduction to Inequalities with Graphs

Notes

Unit - 1: Algebra

Objectives:

At the end of this unit, you will be able to understand:

- Indices, Properties and Application in Real Life
- Surds, Properties and Application in Real Life
- Logarithms, Properties and Application of Common Logarithms
- Equations- Linear and Quadratic, Roots of Quadratic Equations
- Methods to Solve Quadratic Equations
- Simultaneous Equations and Methods to solve with 2 or 3 unknowns
- Inequalities with Graphs

Introduction

In this unit, we will discuss about indices, surds, logarithms and their properties and applications in real life; Linear and Quadratic equations; Simultaneous equations; and Inequalities with graphs.

Introduction to Indices, Properties and Application in Real Life

Index (indices) in Maths is the power or exponent which is raised to a number or a variable. For example, in number 24, 4 is the index of 2. The plural form of index is indices. In algebra, we come across constants and variables. The constant is a value which cannot be changed. Whereas a variable quantity can be assigned any number or we can say its value can be changed. In algebra, we deal with indices in terms of numbers. Let us learn the laws/rules of the indices along with formulas and solved examples.

Indices

A number or a variable may have an index. Index of a variable (or a constant) is a value that is raised to the power of the variable. The indices are also known as powers or exponents. It shows the number of times a given number has to be multiplied. It is represented in the form:

$$a^m = a \times a \times a \times \dots \times a \text{ (m times)}$$

Here, a is the base and m is the index.

The index says that a particular number (or base) is to be multiplied by itself, the number of times equal to the index raised to it. It is a compressed method of writing big numbers and calculations.

Example: $2^3 = 2 \times 2 \times 2 = 8$

In the example, 2 is the base and 3 is the index.

If n is a positive integer, then a^n means the continued product of n factors, each equal to a . Here a is called the base and n is termed as the index or exponent or power of a^n .

Notes

Rules of Indices For $a \neq 0, b \neq 0$	
Rule	Example
$a^x \times a^y = a^{x+y}$	$a^3 \times a^2 = a^{3+2} = a^5$
$a^x \div a^y = a^{x-y}$	$a^6 \div a^2 = a^{6-2} = a^4$
$(a^x)^y = a^{xy}$	$(a^2)^3 = a^{2 \times 3} = a^6$
$a^0 = 1$	$a^0 = 1$
$a^{-x} = \frac{1}{a^x}$	$a^{-5} = \frac{1}{a^5}$
$a^{\frac{x}{y}} = \sqrt[y]{a^x} = (\sqrt[y]{a})^x$	$a^{\frac{3}{2}} = \sqrt[2]{a^3} = (\sqrt[2]{a})^3$

Thus, 4^5 , 4, 4, 4 where 4 is the base and 5 is the index.

Laws of indices (for positive integral exponents)

If m, n are positive integers and a, b are any two non-zero real numbers, then

i. $a^m \times a^n \times a^p = a^{m+n+p}$

ii. $a^m \div a^n = \begin{cases} a^{m-n}, & \text{if } m > n \\ \frac{1}{a^{n-m}}, & \text{if } m < n \end{cases}$

iii. $(a^m)^n = a^{m \cdot n}$

iv. $(a \times b)^m = a^m \times b^m$

v. $\left(\frac{a}{b}\right)^m = a^m \div b^m$

vi. $(a)^{m/n} = (\sqrt[n]{a^m}) = (\sqrt[n]{a})^m$

Note:

i. $a^0 = 1$; 0^0 is undetermined.

ii. $a^{-m} = \frac{1}{a^m}$

Ex. 1.1: Simplify: $(3 \times 10^8)^3$.

Solution:

$$(3 \times 10^8)^3 = 3^3 \times (10^8)^3$$

$$= 27 \times 10^{24}$$

$$= 2.7 \times 10^1 \times 10^{24}$$

Notes

$$= 2.7 \times 10^{25}$$

Ex. Simplify: $\frac{3^{10} \times 10^8}{3^6 \times 10^5}$

Laws of Indices

There are some fundamental rules or laws of indices which are necessary to understand before we start dealing with indices. These laws are used while performing algebraic operations on indices and while solving the algebraic expressions, including it.

Rule 1: If a constant or variable has index as '0', then the result will be equal to one, regardless of any base value.

$$a^0 = 1$$

Example: $5^0 = 1$, $12^0 = 1$, $y^0 = 1$

Rule 2: If the index is a negative value, then it can be shown as the reciprocal of the positive index raised to the same variable.

$$a^{-p} = 1/a^p$$

Example: $5^{-1} = 1/2$, $8^{-3} = 1/8^3$

Rule 3: To multiply two variables with the same base, we need to add its powers and raise them to that base.

$$a^p \cdot a^q = a^{p+q}$$

Example: $5^2 \cdot 5^3 = 5^{2+3} = 5^5$

Rule 4: To divide two variables with the same base, we need to subtract the power of denominator from the power of numerator and raise it to that base.

$$a^p/a^q = a^{p-q}$$

Example: $10^4/10^2 = 10^{4-2} = 10^2$

Rule 5: When a variable with some index is again raised with different index, then both the indices are multiplied together raised to the power of the same base.

$$(a^p)^q = a^{pq}$$

Example: $(8^2)^3 = 8^{2 \cdot 3} = 8^6$

Rule 6: When two variables with different bases, but same indices are multiplied together, we have to multiply its base and raise the same index to multiplied variables.

$$a^p \cdot b^p = (ab)^p$$

Example: $3^2 \cdot 5^2 = (3 \times 5)^2 = 15^2$

Rule 7: When two variables with different bases, but same indices are divided, we are required to divide the bases and raise the same index to it.

$$a^p/b^p = (a/b)^p$$

Example: $3^{2/5} = (3/5)^2$

Rule 8: An index in the form of a fraction can be represented as the radical form.

$$a^{p/q} = q\sqrt[q]{a^p}$$

Example: $6^{1/2} = \sqrt{6}$

Notes

Introduction to Surds, Properties and Application in Real Life

In Mathematics, surds are the values in square root that cannot be further simplified into whole numbers or integers. Surds are irrational numbers. The examples of surds are $\sqrt{2}$, $\sqrt{3}$, $\sqrt{5}$, etc., as these values cannot be further simplified. If we further simplify them, we get decimal values, such as:

$$\sqrt{2} = 1.4142135\dots$$

$$\sqrt{3} = 1.7320508\dots$$

$$\sqrt{5} = 2.2360679\dots$$

Surds Definition

Surds are the square roots ($\sqrt{}$) of numbers that cannot be simplified into a whole or rational number. It cannot be accurately represented in a fraction. In other words, a surd is a root of the whole number that has an irrational value. Consider an example, $\sqrt{2} \approx 1.414213$. It is more accurate if we leave it as a surd $\sqrt{2}$.

In the set of irrational numbers some are algebraic (like $\sqrt{2}$, $\sqrt[3]{9}$ etc.) and these numbers are termed as surds. Of course, non-algebraic irrational numbers, such as π , e etc. are not surds. Thus, all surds are irrational numbers, but all irrational numbers are not surds.

Let m and p respectively denote any positive rational number and any positive integer. Then any p^{th} root of m (i.e., $\sqrt[p]{m}$) is known as a surd, if it is irrational.

A surd is called a pure surd if no rational number (except 1) can be extracted out of its radical (e.g. $\sqrt{5}$, $\sqrt[3]{6}$ etc.). Any surd which is not pure is known as a mixed surd. Two mixed surds are called similar if their irrational factors are same.

The order of a surd is indicated by the index of the radical present in the surd. Thus $\sqrt{5}$, $\sqrt[3]{6}$, $\sqrt[4]{7}$

are respectively surds of the second, third and fourth order. A given surd is called a quadratic, cubic, quadratic according as the order of the surd is respectively 2, 3, 4.

Types of Surds

The different types of surds are as follows:

Simple Surds – A surd that has only one term is called a simple surd. Example: $\sqrt{2}$, $\sqrt{5}$, ...

Pure Surds – Surds which are completely irrational. Example: $\sqrt{3}$

Similar Surds – The surds having the same common surds factor

Notes

Mixed Surds – Surds that are not completely irrational and can be expressed as a product of a rational number and an irrational number

Compound Surds – An expression which is the addition or subtraction of two or more surds

Binomial Surds – A surd that is made of two other surds

Understanding the Forms of Surds

1. Mixed Surds: A surd of the form $k\sqrt[n]{a}$ where k is a rational number, $k\sqrt[n]{0}$ and $k\sqrt[n]{1}$ is called a mixed surd.

$2\sqrt[3]{7}$, $8\sqrt[3]{7}$ etc are mixed surds. For

Note: A surd can be expressed as a pure surd. For e.g. $5\sqrt{2} = \sqrt{25 \times 2} = \sqrt{50}$

2. Pure Surds: A surd of the form $k\sqrt[n]{a}$ where k is a rational number, $k = \pm 1$ is called a pure surd. For e.g. $\sqrt{3}$, $\sqrt{5}$ and $\sqrt[3]{2}$ etc are pure surd.

FORMS OF SURDS

1. Mixed Surds- A surd of the form $k\sqrt[n]{a}$ where k is a rational number, $k \neq 0$ and $k \neq \pm 1$ is called a mixed surd.

For ex. $2\sqrt[3]{7}$, $\frac{1}{4}\sqrt{5}$ etc are mixed surds.

NOTE: A mixed surd can be expressed as a pure surd.
For ex. $5\sqrt{2} = \sqrt{25 \times 2} = \sqrt{50}$

2. Pure Surds- A surd of the form $k\sqrt[n]{a}$ where k is a rational number, $k = \pm 1$ is called a Pure surd.

For ex. $\sqrt{3}$, $\sqrt{5}$, $\sqrt[3]{2}$ etc are Pure surds.

Rules of Surds:

Given below are some rules that one needs to follow:

Any rational number cannot be a surd.

All irrational numbers are surds.

Root of a positive real quantity can be called a surd, if its value remains undetermined.

$$\sqrt{a} \times \sqrt{a} = a \Rightarrow \sqrt{5} \times \sqrt{5} = 5$$

The sum along with the difference of two quadratic surds will be called complementary or conjugate surds to each other.

THE RULES OF SURDS

1) $\sqrt{a} \times \sqrt{b} = \sqrt{ab}$

2) $\sqrt{a} / \sqrt{b} = \sqrt{a/b}$

3) $\sqrt{a} + \sqrt{b}$ - can't simplify

4) $(a+\sqrt{b})^2 = (a+\sqrt{b})(a+\sqrt{b})$
 $= a^2 + 2a\sqrt{b} + b$

5) $(a+\sqrt{b})(a-\sqrt{b})$
 $= a^2 + a\sqrt{b} - a\sqrt{b} - b = a^2 - b$

6) $a / \sqrt{b} = (a\sqrt{b})/b$

1.1.3 Introduction to Logarithms, Properties and Applications of Common Logarithms**Meaning of logarithm**

If $a^x = n$, a and n are positive real number such that $a \neq 1$, then x is said to be the logarithm of the number n to the base ' a ' symbolically it can be expressed as follows:

$$\log_a n = x$$

For example

1. $4^3 = 64 \Rightarrow \log_4 64 = 3$

2. $3^3 = 27 \Rightarrow \log_3 27 = 3$

Evaluate $\log_4 8$

Write an equation in logarithmic form

$$\log_4 8 = x$$

Convert the equation in exponential form

$$4^x = 8$$

Write each side with a base of 2

$$(2^2)^x = 2^3$$

$$(2)^{2x} = 2^3$$

$$2x = 3$$

$$x = \frac{3}{2}$$

$$\text{So } \log_4 8 = \frac{3}{2}$$

Example 1 Find the value of $\log_2 8 + \log_2 4$.

Notes

Solution:

$$\begin{aligned}\log_2 8 + \log_2 4 &= \log_2 2^3 + \log_2 2^2 \\ &= 3\log_2 2 + 2\log_2 2 \\ &= 3 + 2 \quad (\log_2 2 = 1)\end{aligned}$$

Example 2 Find the value of $\log_3 162 - \log_3 2$.

Solution:

$$\begin{aligned}\log_3 162 - \log_3 2 &= \log_3 \left(\frac{162}{2} \right) \\ &= \log_3 81 \\ &= \log_3 3^4 \\ &= 4\log_3 3 \quad (\because \log_x x = 1) \\ &= 4\end{aligned}$$

Example 3 If $\log_6 (2x - 4) + \log_6 4 = \log_6 40$, then find the value of x .

Solution:

$$\begin{aligned}\log_6 (2x - 4) + \log_6 4 &= \log_6 40 \\ \log_6 4(2x - 4) &= \log_6 40 \\ 8x - 16 &= 40 \\ 8x &= 56 \\ x &= 7\end{aligned}$$

Two types of logarithm are used:

1. **Natural logarithm:** Logarithm of numbers to base e are known as natural logarithm.

Common Logarithm: Logarithms of numbers to the base 10 are known as common logarithm.

Applications of Common Logarithms:

The base of a common logarithm is always 10. The common log of a number N is written as follows:

$\log N$ or $\log_{10} N$: Decadic logarithms and decimal logarithms are other names for common logarithms.

If **$\log N = x$** , then this logarithmic form can be represented in exponential form, i.e., $10^x = N$.

Common logarithms are widely used in science and engineering.

The Richter scale, which is used to measure earthquakes, and the decibel scale, which is used to measure sound, are both usually expressed in logarithmic form. It is so common that if you find no base written, you can assume it is $\log x$ or common log.

Laws of Logarithm

$$\log_a(mn) = \log_a m + \log_a n$$

$$\log_a\left(\frac{m}{n}\right) = \log_a m - \log_a n$$

$$\log_a(m^n) = n(\log_a m)$$

$$\log_a m = \frac{1}{\log_m a}$$

$$\log_a m = \frac{\log_b m}{\log_b a}$$

Notes**1.1.4. Introduction to Equations- Linear and Quadratic, Roots of Quadratic Equation**

An equation having the form $ax + b = 0$ where $a \neq 0$, a and b are arbitrary constants, is called a linear equation in x . This equation is also called first degree equation, the highest power of the unknown factor x being 1. Again, two linear equations of the form $a_1x + b_1y + c_1 = 0$, $a_2x + b_2y + c_2 = 0$ are known as simultaneous equations in x and y .

Linear equations are widely used in different places in Physics, Engineering and Mathematics. It is partly due to the fact that it can help to approximate non-linear equations. Linear equations are equations for a straight line and are of the first order

Equation:

An equation is a statement where two algebraic expressions are equal.

For Example,

$$3x + 4 = 16$$

$$x^2 - 7x + 12 = 0$$

$$x - y = -1$$

$$2x - y = 7$$

In each of the above equations the parts separated by the sign $=$ are termed as sides of the equation.

Quadratic equation

If $f(x)$ is a quadratic polynomial. Then $f(x) = 0$ is known as quadratic equation. The general form of a quadratic equation is $ax^2 + bx + c = 0$, $a, b, c \in R$ and $a \neq 0$.

Following image is an example of quadratic equation:

Notes

$$4x^2 - 22 = x^2 + 5$$

Add 22 to each side of the equation.

$$4x^2 = x^2 + 27$$

Subtract x^2 from each side of the equation.

$$3x^2 = 27$$

Divide each side of the equation by 3.

$$x^2 = 9$$

Take the square root of each side of the equation.

$$x = 3, -3$$

Roots of an equation

The value of a variable in an equation which satisfied the given equation are known as roots of an equation i.e., if $f(x) = 0$ is a polynomial equation and $f(a) = 0$, then a is a root of $f(x) = 0$.

Roots of a quadratic equation

Roots of the quadratic equation $ax^2 + bx + c = 0$ are $\frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$, where $b^2 - 4ac$ is known as discriminant and it is denoted by D .

If the value of discriminant is negative, then the solution of the quadratic equation is not possible.

If the value of discriminant is positive, then the solution of the quadratic equation is possible.

Nature of roots

If $D > 0$, then the roots are real and distinct.

If $D = 0$, then the roots are real and equal.

If $D < 0$, then the roots are complex with non-zero imaginary part.

The roots are rational, if a, b, c are rational and D is perfect square.

The roots are irrational, if a, b, c are rational and D is not a perfect square.

Relation between coefficients and roots of an equation

If α and β are the roots of the quadratic equation $ax^2 + bx + c = 0$, then

Sum of the roots $\alpha + \beta = -\frac{b}{a}$ and

Products of the roots $\alpha\beta = \frac{c}{a}$

Notes

1.1.5. Methods to Solve Quadratic Equation-

If $f(x)$ is a quadratic polynomial. Then $f(x) = 0$ is known as quadratic equation. The general form of a quadratic equation is $ax^2 + bx + c = 0$, $a, b, c \in R$ and $a \neq 0$.

For Ex. $3x^2 + 2x + 4$, $5x^2 + 8x + 9$ and $2x^2 + 8x + 13$.

Roots of quadratic equation

Roots of the quadratic equation $ax^2 + bx + c = 0$ are $\frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ where $b^2 - 4ac$ is known as discriminant and it is denoted by D .

If the value of discriminant is negative, then the solution of the quadratic equation is not possible.

If the value of discriminant is positive, then the solution of the quadratic equation is possible.

How to Solve quadratic equations?

We have two methods by which a quadratic equation can be solved.

1. Solving quadratic equations by factor using:

To solve a quadratic equation, we have to follow following steps

Step 1: Express the given equation in the form $ax^2 + bx + c = 0$.

Step 2: Factorise $ax^2 + bx + c = 0$.

Step 3: Put each factor equals to 0.

Step 4; Solve each resulting equation.

2. Solving quadratic equations by using formula:

Let the given quadratic equation be $ax^2 + bx + c = 0$, $a \neq 0$.

So, the roots of the given equation are

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Ex.1.15: Solve the equation $x^2 + 6x + 8$.

Solution:

$$\begin{aligned} x^2 + 6x + 8 &= x^2 + 2x + 4x + 8 \\ &= (x^2 + 2x) + (4x + 8) \\ &= x(x + 2) + 4(x + 2) \\ &= (x + 2)(x + 4) \\ &\Rightarrow x = -2, -4 \end{aligned}$$

Ex.1.16: Solve the equation $x^2 + 4x - 21$.

Notes

Solution:

$$\begin{aligned}
 x^2 + 4x - 21 &= x^2 + 7x - 3x - 21 \\
 &= (x^2 + 7x) - (3x + 21) \\
 &= x(x + 7) - 3(x + 7) \\
 &= (x + 7)(x - 3) \\
 \Rightarrow x &= -7, 3
 \end{aligned}$$

Ex. 1.17: Solve the equation $3x^2 - 10x + 3 = 0$.

Solution: Here, $a = 3$, $b = -10$, $c = 3$

So, the roots are

$$\begin{aligned}
 x &= \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \\
 &= \frac{10 \pm \sqrt{100 - 36}}{6} \\
 &= \frac{10 \pm \sqrt{64}}{6} \\
 &= \frac{10 \pm 8}{6} \\
 &= \frac{10 + 8}{6} \text{ or } \frac{10 - 8}{6} \\
 &= 3 \text{ or } \frac{1}{3}
 \end{aligned}$$

Ex. 1.18: Solve the equation $2x^2 + 2x - 3 = 0$.

Solution: Here, $a = 2$, $b = 2$, $c = -3$

So, the roots are

$$\begin{aligned}
 x &= \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \\
 &= \frac{-2 \pm \sqrt{4 + 24}}{4} \\
 &= \frac{-2 \pm \sqrt{28}}{4} \\
 &= \frac{-2 \pm 2\sqrt{7}}{4} \\
 &= \frac{-1 + \sqrt{7}}{2} \text{ or } \frac{-1 - \sqrt{7}}{2}
 \end{aligned}$$

1.1.6. Introduction to Simultaneous Equations and Methods to Solve them with Two or Three Unknowns-

Simultaneous (linear equation) equation

An equation having the form $ax + b = 0$ where $a \neq 0$, a and b are arbitrary constants, is called a linear equation in x . This equation is also called first degree equation, the highest power of the unknown factor x being 1. Again, two linear equations of the form $a_1x + b_1y + c_1 = 0$, $a_2x + b_2y + c_2 = 0$ are known as simultaneous equations in x and y .

Linear equations are widely used in different places in Physics, Engineering and Mathematics. It is partly due to the fact that it can help to approximate non-linear equations. Linear equations are equations for a straight line and are of the first order.

The graph of Simultaneous linear equation in two variables is given below:

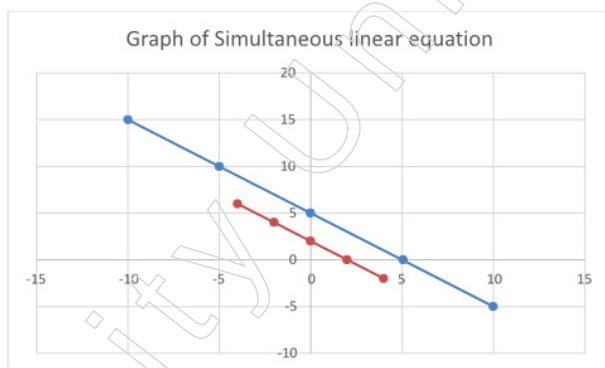
Suppose the linear equation in two variables are $x + y = 5$ and $x + y = 2$.

So, the roots of the linear equation $x + y = 5$ is

X	-10	-5	0	5	10
$y = 5 - x$	15	10	5	0	-5

So, the roots of the linear equation $x + y = 2$ is

X	-4	-2	0	2	4
$y = 2 - x$	6	4	2	0	-2



Here blue line shows the graph of $x + y = 5$ and red line shows the graph of $x + y = 2$.

Given below is a basic example of how this equation is graphed:

Notes

How to Graph Equations in Point-Slope Form?

1. Find the y-intercept, by setting $x = 0$ and plot the point.
2. From the y-intercept, use the slope to find the second point and plot it.
3. Join the two points with a straight line.

Example:

Graph the equation $y = \frac{6}{5}x - 2$

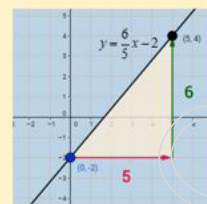
Let $x = 0$

$$y = \frac{6}{5}(0) - 2$$

$$y = -2$$

The y-intercept is $(0, -2)$

Use the slope $\frac{6}{5}$ to plot the second point



Methods of solving simultaneous equations

There are three methods to solve any simultaneous equations. These methods are as follows:

1. Method of elimination by substitution.
2. Method elimination by equating coefficients.
3. Method of cross multiplication.

1. Method of elimination by substitution.

Step 1:

We obtain the value of one variable in terms of the other from any of the given two equations.

Step 2:

Substitute the value of variable, obtained in step 1, in the other equation and solve it.

Step 3:

Substitute the value of variable, obtained in step 2, in the result of step 1 and get the value of the remaining unknown variable.

2. Method of elimination by equating coefficient.

Step 1:

Multiply one or both of the equations by a suitable number or numbers so that either the coefficients of x or the coefficients of y in both the equations become numerically equal.

Step 2:

Add both the equations, as obtained in step 1, or subtract one equation from the other, so that the terms with equal numerical coefficients cancel mutually.

Step 3:

Solve the resulting equation to find the value of one of the unknowns.

Step 4:

Substitute this value in any of the two given equations and find the value of the other unknowns.

3. Method of cross-multiplication.

Step 1.

Express the given simultaneous equations as:

$$a_1x + b_1y + c_1 = 0$$

and, $a_2x + b_2y + c_2 = 0$

Step 2.

Write the equation in the following form in order to obtain the solution of the given equation.

$$\frac{x}{\begin{pmatrix} b_1 & c_1 \\ b_2 & c_2 \end{pmatrix}} = \frac{y}{\begin{pmatrix} c_1 & a_1 \\ c_2 & a_2 \end{pmatrix}} = \frac{1}{\begin{pmatrix} a_1 & b_2 \\ a_2 & b_1 \end{pmatrix}}$$

$$\frac{x}{b_1c_2 - b_2c_1} = \frac{y}{c_1a_2 - c_2a_1} = \frac{x}{a_1b_2 - a_2b_1}$$

1.1.7. Introduction to Inequalities with Graphs

Inequalities

Mathematical sentences of the type $5 \neq 7, 7 > 5, -1 < 2, 3y < 15, 5t > 20$ are called inequalities or inequations. These sentences say that one thing is not equal to another.

Universal set

Consider the sentences

$$3x = 12$$

$$2x < 3$$

If the variable x in the equation (a) is replaced by the number 4, it yields a true statement. We say that the equation is satisfied. Similarly, the inequation (b) is satisfied if the variable x is replaced by the numbers

$$\dots, -3, -2, -1, 0, 1$$

Sometimes it is necessary to specify a set from which the replacement for the variable should be chosen. For example, if we write the open sentences. "_____ is a beautiful city", it would be difficult to give any precise answer, unless the set of cities is specified.

Notes

Notes

The set of a elements from which the replacement for the variable is taken is called the replacement set.

Solution Set or truth Set

Consider the inequation $1 + x > 5$. We can obtain the following solutions for the replacement sets shown.

Replacement set	Solutions
$A = \{0, 1, 2, 5, 7, 8\}$	5, 7 and 8
$B = \{0, 1, 2, 3, 4, 6\}$	6
$C = \{0, 1, 2, 3, 4\}$	ϕ

The above Ex. illustrates the fact that the solution of an inequation depend upon the replacement set used. It also follows from the above that an equation may have one, many or no solution, depending upon the replacement set. The solution or solutions of a given inequation from a set, which we call the solution set or the truth set. It is obviously a subset of the replacement set.

Properties:

1. Adding to or subtracting from both sides of any inequality any non-zero number produces an equivalent inequality.

(i) $x - 3 > 5$ is equivalent to

$$x - 3 + 3 > 5 + 3$$

$$x > 8$$

(ii) $x + 1 > 3$ is equivalent to

$$x + 1 - 1 > 3 - 1$$

$$x > 2$$

2. Multiplying or dividing both sides of an inequalities by the same positive number produces an equivalent inequality.

(i) $5x > 15$ is equivalent to

$$\frac{5x}{5} > \frac{15}{5}$$

$$x > 3$$

(ii) $\frac{x}{3} > 2$ is equivalent to

$$\frac{x}{3} \times 3 > 2 \times 3$$

$$x > 6$$

3. Multiplying both sides of an inequality by the same negative number produces an inequality with its direction reverse.

(i) $x > 8$ is equivalent to $-x < -8$

Notes

(ii) $-x > -8$ is equivalent to $x < 8$

4. (i) Squaring both sides of an inequality produces an equivalent inequality if both sides are positive.

In other words,

If $x > y, x > 0, y > 0$ then $x^2 > y^2$:

For Ex.: $5 > 3$ produces

$$5^2 > 3^2$$

$$25 > 9$$

(ii) Squaring both sides of an inequality produces an inequality with its direction reversed if both sides are negative.

In other words,

If $x > y, x < 0, y < 0$ then $x^2 < y^2$ or

If $x < y, x < 0, y < 0$ then $x^2 > y^2$

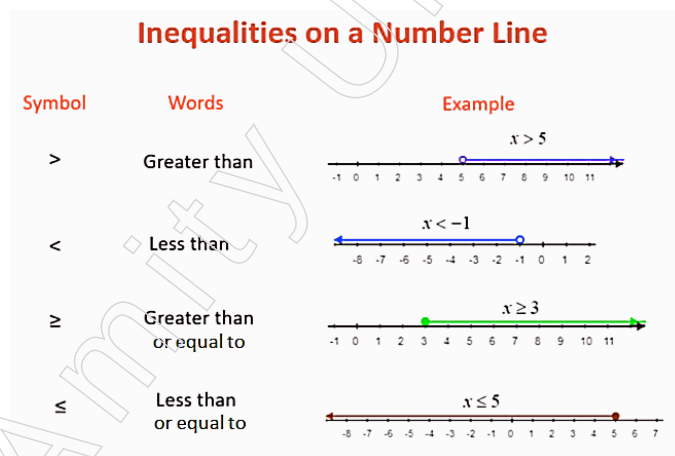
For Ex.: $-3 > -5$ produces

$$(-3)^2 < (-5)^2$$

$$9 < 25$$

Graphing the inequality

Place the variable on the left and the inequality symbol will point to the direction to be shaded. The arrows at the ends of the number line will match the inequality sign.



a. When the $\leq$ or $\geq$ signs are used, the point is included $\rightarrow$ use the solid circle $\bullet$.

b. When the $<$ or $>$ signs are used, the point is not included $\rightarrow$ use the open circle $\circ$.

c. The line goes to the right (greater than symbol) if $>$ is used

Notes

- d. the lines goes to the left (less than the symbol) if $<$ is used
Here are some Example of inequalities which shows how such inequalities are represented.

Solve and graph.

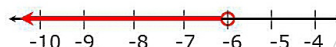
1. $4x - 6 > 10$

$x > 4$



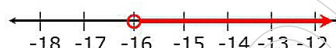
2. $7x + 9 < 3x - 15$

$x < -6$



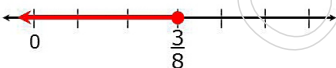
3. $w - 3w < 32$

$w > -16$



4. $\frac{2}{3}w + \frac{1}{4} \leq \frac{1}{2}$

$w \leq \frac{3}{8}$



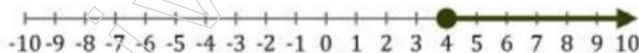
1. $x < 1$



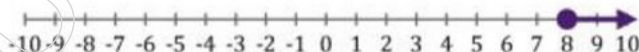
2. $x \geq -6$



3. $x \geq 4$



4. $x \geq 8$



5. $x \leq 1$



Check your Understanding

- _____ of a variable (or a constant) is a value that is raised to the power of the variable.
- _____ are the square roots ($\sqrt{\quad}$) of numbers that cannot be simplified into a whole or rational number.
- Thus, all surds are irrational numbers, but all irrational numbers are not surds. True / False
- An _____ is a statement where two algebraic expressions are equal.
- Similar surds are surds that are not completely irrational and can be expressed as a product of a rational number and an irrational number. True / False

Notes

6. Logarithms of numbers to the base 10 are known as _____ logarithm.
7. An equation having the form $ax + b = 0$ where $a \neq 0$, a and b are arbitrary constants, is called a _____ equation in x .
8. Linear equations are equations for a _____ line and are of the first order.
9. The set of elements from which the replacement for the variable is taken is called the _____ set.
10. The value of a variable in an equation which satisfied the given equation are known as _____ of an equation

Multiple Choice Questions

1. Solve $4^{\log_9 3} + 9^{\log_2 4} = 10^{\log_x 83}$
 - a. 3
 - b. 11
 - c. 4
 - d. 9
2. Find the value of $\left(\frac{x^a}{x^b}\right)^{a+b} \times \left(\frac{x^b}{x^c}\right)^{b+c} \times \left(\frac{x^c}{x^a}\right)^{c+a}$
 - a. 1
 - b. 6
 - c. 9
 - d. 0
3. If $x = \frac{\sqrt{3}}{2}$, then find the value of $\frac{\sqrt{1+x} + \sqrt{1-x}}{\sqrt{1+x} - \sqrt{1-x}}$
 - a. $\sqrt{3}$
 - b. $\sqrt{5}$
 - c. $\sqrt{7}$
 - d. 1
4. Find the value of $\log_{0.01} 10000$
 - a. 2
 - b. 6
 - c. -2
 - d. -8
5. Find the value of $\log_3 \log_2 \log_2 256$
 - a. 7
 - b. 5
 - c. 3
 - d. 1

Notes

Summary

- The index says that a particular number (or base) is to be multiplied by itself, the number of times equal to the index raised to it. It is a compressed method of writing big numbers and calculations.
- Surds are the square roots ($\sqrt{\quad}$) of numbers that cannot be simplified into a whole or rational number. It cannot be accurately represented in a fraction. In other words, a surd is a root of the whole number that has an irrational value.
- Logarithm of numbers to base e are known as nature logarithm.
- Logarithms of numbers to the base 10 are known as common logarithm.
- An equation is a statement where two algebraic expressions are equal.
- The value of a variable in an equation which satisfied the given equation are known as roots of an equation i.e., if $f(x)=0$ is a polynomial equation and $f(a)=0$, then a is a root of $f(x)=0$.
- An equation having the form $ax+b=0$ where $a \neq 0$, a and b are arbitrary constants, is called a linear equation in x . This equation is also called first degree equation, the highest power of the unknown factor x being 1.
- There are three methods to solve any simultaneous equations. These methods are as follows:
 - Method of elimination by substitution.
 - Method elimination by equating coefficients.
 - Method of cross multiplication.
- Mathematical sentences of the type $5 \neq 7, 7 > 5, -1 < 2, 3y < 15, 5t > 20$ are called inequalities or in inequations. These sentences say that one thing is not equal to another.
- The solution or solutions of a given inequation from a set, which we call the solution set or the truth set. It is obviously a subset of the replacement set.

Activity

1. Simplify: $3^{10} \times 10^8 / 3^6 \times 10^5$
2. Solve the equation: $4x^2 + 5x - 10 = 0$ What are the roots?
3. Solve the equation: $6x^2 - 3x + 9 = 0$ What are the roots?
4. Describe the 3 methods of solving simultaneous equations. Explain in detail and provide examples that are not provided in the book.
5. $2x^2 + 15 < 25$ Solve and graph the answer

Questions and Exercises

1. What is the law of indices for positive integral exponents?
2. List the eight Laws of Indices.
3. Give two examples of irrational numbers that are not considered surds.
4. List the rules of surds.

5. What are the steps to solving a quadratic equation?
6. How do you graph an equation in point-slope form?

Glossary

- **Binomial Surds:** A surd that is made of two other surds
- **Common Logarithm:** Logarithms of numbers to the base 10 are known as common logarithm.
- **Compound Surds:** An expression which is the addition or subtraction of two or more surds
- **Index:** is a value that is raised to the power of the variable(or a constant).
- **Natural logarithm:** Logarithm of numbers to base e are known as nature logarithm.
- **Simple Surds** – A surd that has only one term is called a simple surd.
- **Surds:** are the square roots ($\sqrt{}$) of numbers that cannot be simplified into a whole or rational number.
- **Pure Surds** – Surds which are completely irrational.

Further Reading

1. McCune, Sandra Luna. McGraw-Hill Education Algebra I Review and Workbook. McGraw Hill; 1st edition (January 8, 2019)
2. Selby, Peter H.; Slavin, Steve. Practical Algebra: A self-Teaching Guide, Second Edition. John Wiley & Sons; 2nd edition (February 14, 1991)

Check your Understanding: Answer

1. Index
2. Surds
3. True
4. Equation
5. False
6. Common
7. Linear
8. Straight
9. Replacement
10. Roots

Multiple Choice

- | | | |
|------|------|------|
| 1. B | 2. A | 3. A |
| 4. C | 5. D | |

Module - 2: Data Arrangement

Structure:

2.1 Arithmetic Progression

2.1.1 Definition of Arithmetic Progression and the n th term of AP.

2.1.2 Sum of n terms of AP

2.1.3 Representation of terms in AP

2.1.4 Arithmetic mean between a and b

2.1.5 Arithmetic Progression and its applications in business

2.2 Geometric Progression

2.2.1 Definition of Geometric Progression, n th term of GP

2.2.2 Sum of n terms of GP

2.2.3 Representation of terms in GP

2.2.4 Geometric mean between a and b

2.2.5 Geometric Progression and its applications in business

2.3 Permutation and Combination

2.3.1 Introduction to Permutations and Combination and its applications in business

Unit - 2.1: Arithmetic Progression

Notes

Objectives:

At the end of this unit, you will be able to understand

- Definition of Arithmetic Progression and the n th term
- Sum of n terms of AP
- Representation of terms in AP
- Arithmetic mean between a and b
- Arithmetic Progression and its applications in Business

Introduction

In mathematics, the word sequence is used in the same way as it is in ordinary English. When we say that a collection of objects is listed in a sequence, we usually mean that the collection is ordered in such a way that it first identifies the first member, then the second member, then the third member and so on. In this unit, we will learn about arithmetic progression, n th term of an A.P., the sum of first n terms of A.P. and arithmetic mean also.

2.1.1 Definition of Arithmetic Progression and the n th Term of AP

Arithmetic Progression

An arithmetic progression (A.P.) is a sequence whose terms increase or decrease by a fixed number is called the common difference of the A.P.

In mathematical form, if $a_1, a_2, a_3, a_4, \dots, a_n$ are in A.P., then

$$a_2 - a_1 = a_3 - a_2 = \dots = a_n - a_{n-1} = d$$

If a is the first term and d is a common difference, then A.P. can be written as

$$a, a + d, a + 2d, \dots, [a + (n-1)d]$$

Ex. (i) 2, 4, 6, 8,

(ii) 4, 8, 12, 16,

Some characteristics of an Arithmetic series

If the sum of n terms of any series is a quadratic expression, then that series will be arithmetic series.

If the same number is added or subtracted from each term of an arithmetic progression, then new series formed will also be in an arithmetic progression.

If the same number is multiplied or divided by each term of an arithmetic progression, then new series formed will also be in the form of arithmetic series.

If corresponding terms of two series are added or subtracted, then new series formed will also be in an arithmetic series.

Notes

Series formed by multiplying or dividing corresponding terms of two arithmetic progression, then series formed will be not be in an arithmetic series.

2.1.2 Sum of n terms of AP

The n^{th} term of an A.P.

Suppose a be the first term, d be a common difference and a_n be the last term of an A.P., then the n^{th} term is given by

$$a_n = a + (n-1)d, \text{ where } d = a_n - a_{n-1}$$

The r^{th} term from the end of finite A.P. is the $(n-r+1)^{\text{th}}$ term from the beginning.

The sum of n terms of an A.P.

Suppose there are n terms of a sequence, whose first term is a , the common difference is d and the last term is a_n , then the sum of n terms is given by

$$S_n = \frac{n}{2} [2a + (n-1)d]$$

Example

Find the 10th term of the following sequence:

Sequence: 1, 3, 5, 7, ...

Answer:

Given: 1, 3, 5, 7, ...

$$a_1 = 1, a_2 = 3, a_3 = 5, a_4 = 7, a_{10} = ?$$

Common difference,

$$\begin{aligned} d_1 &= a_2 - a_1 \\ &= 3 - 1 \end{aligned}$$

$$= 2$$

$$\begin{aligned} d_2 &= a_3 - a_2 \\ &= 5 - 3 \\ &= 2 \end{aligned}$$

$$d = d_1 = d_2 = 2$$

So, the given sequence is A.P.

$$A_n = a + (n-1)d$$

$$\begin{aligned} A_{10} &= 1 + 9d \\ &= 19 \end{aligned}$$

Example

Find the sum of the first 10 terms of the sequence

Sequence: 10, 15, 20, 25, ...

$a_1 = 10, a_2 = 15, a_3 = 20, a_4 = 25, S_{10} = ?$

Common difference,

$$D_1 = a_2 - a_1$$

$$= 15 - 10$$

$$= 5$$

$$D_2 = a_3 - a_2$$

$$= 20 - 15$$

$$= 5$$

So, the given sequence is A.P.

$$S_n = \frac{n}{2} [2a + (n-1)d]$$

$$S_{10} = 5(20 + 45)$$

$$= 325$$

2.1.3 Representation of Terms in AP

A progression is a type of sequence for which a formula for the n th term can be found. The Arithmetic Progression is the most commonly used sequence in mathematics, with simple formulas.

Arithmetic Progression Notation

In AP, we will encounter three main terms, which are denoted as follows:

- Common difference (d)
- n th Term (a_n)
- Sum of the first n terms (S_n)

Arithmetic Progression is represented by all three terms. In the following section, we will learn more about these three properties.

Common Variation in Arithmetic Progression

The terms used in this progression for a given series are the first term, the common difference between the two terms, and the n th term. Let's suppose, $a_1, a_2, a_3, \dots, a_n$ is an AP, then; the **common difference 'd'** can be obtained as:

$$d = a_2 - a_1 = a_3 - a_2 = \dots = a_n - a_{n-1}$$

Where 'd' is a common distinction. It can be either positive or negative, or it can be zero.

Notes

First Term of Arithmetic Progression

The AP can also be expressed in terms of common difference, as shown below:

$$a, a + d, a + 2d, a + 3d, a + 4d, \dots, a + (n - 1) d$$

where 'a' is the progression's first term.

General Form of an Arithmetic Progression

Consider an Arithmetic Progression to be: $a_1, a_2, a_3, \dots, a_n$

Position of Terms	Representation of Terms	Values of Term
1	a_1	$a = a + (1-1) d$
2	a_2	$a + d = a + (2-1) d$
3	a_3	$a + 2d = a + (3-1) d$
4	a_4	$a + 3d = a + (4-1) d$
.	.	.
.	.	.
N	a_n	$a + (n-1)d$

2.1.4 Arithmetic mean between a and b

If we insert one arithmetic mean between two numbers a and b, then

$$\text{Arithmetic mean} = \frac{a + b}{2}$$

Arithmetic mean of a group of observations is the quotient obtained by dividing the sum of all the observations by their number. Thus, arithmetic mean denoted by is

$$x = \frac{\text{Sum of all the observations}}{\text{Number of Observations}}$$

$$\begin{aligned} A.M. &= \frac{x_1 + x_2 + \dots + x_n}{n} \\ &= \frac{\sum x}{n} \end{aligned}$$

Discrete Frequency Distribution: if the value x_1 occurs f_1 times, the value x_2 occurs f_2 times, and so on, then

$$\begin{aligned} A.M. &= \frac{f_1 x_1 + f_2 x_2 + \dots + f_n x_n}{f_1 + f_2 + \dots + f_n} \\ &= \frac{\sum_{i=1}^n f_i x_i}{N} \end{aligned}$$

Where $N = f_1 + f_2 + \dots + f_n = \text{Total frequency}$.

Methods of calculating Arithmetic mean

There are three methods of calculating arithmetic mean

- Direct method
- Short-Cut method
- Step-deviation method

These methods are applicable to any type of series.

a. Direct method:

$$A.M. = \frac{\sum x}{n}$$

- Short cut method: In this method first we assumed any number say A (often called assumed mean). Then

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

- By Stem-Deviation Method

$$\begin{aligned}\bar{x} &= A + \frac{\sum f(x-A)}{\sum f} \\ &= A + \frac{\sum f(x-A)}{N}\end{aligned}$$

Where.

h=Class size

$$u_i = \frac{x_i - A}{h}$$

A = Assumed mean

$\sum_{i=1}^n f_i$ = Sum of the frequencies given, can be denoted by N.

Properties of arithmetic mean

- The algebraic sum of the deviations of all the variate values from their mean is zero.
- If every value of the variable is increased by same constant a, then arithmetic mean is also increased by a.
- Arithmetic mean is not independent of the change of origin and scale.
- The sum of the squares of the deviations of all the values taken about their mean is minimum.

Example

Evaluate $\sum_{n=1}^{10} 3(-2)^{n-1}$.

Notes

Answer: $\sum_{n=1}^{10} 3(-2)^{n-1} = 3 - 6 + 12 - 24 + \dots$

First term, $a = 3$

Common ratio, $r = -2$

$$\begin{aligned} S_n &= \frac{a(1-r^n)}{1-r} \\ S_{10} &= \frac{3(1-(-2)^{10})}{1-(-2)} \\ &= \frac{3(1-1024)}{3} \\ &= -1023 \end{aligned}$$

Example

Insert one arithmetic mean between 3 and 5.

Answer:

Given $a = 3$ and $b = 5$

$$\begin{aligned} \text{Arithmetic mean} &= \frac{a+b}{2} \\ &= \frac{3+5}{2} \\ &= 4 \end{aligned}$$

The required arithmetic mean is 4.

2.1.5 Arithmetic Progression and its Applications in Business

If a similar number is added or subtracted from each term of an Arithmetic Progression, the following terms in the succession are likewise in Arithmetic Progression with similar basic contrast. If each term in an Arithmetic Progression is divided or multiplied by the equivalent non-zero number, the following sequence is also in an Arithmetic Progression.

If $2y = x + z$, three numbers x , y , and z are in an Arithmetic Progression.

If the n th term of a sequence is a linear expression, the sequence is an Arithmetic Progression.

If we select terms from an Arithmetic Progression that are in the ordinary stretch, these terms will also be in Arithmetic Progression.

Example: Mr. Singh wants to start a savings plan. He decides that the bare minimum he should save in any given year is Rs. 100,000. He saves Rs. 100,000 at the end of the first year and saves an additional Rs. 5000 each year after that. How much money will he have saved by the end of his 30th year?

Solution: $a = \text{Rs. } 100,000$

$d = \text{Rs. } 5000$

$n = 30$

$$S_n = \frac{n}{2} [2a + (n-1)d]$$

$$S_{30} = \frac{30}{2} [2(100000) + (30-1)5000]$$

$$= \frac{30}{2} [200000 + 29(5000)]$$

$$= 15 [200000 + 29(5000)]$$

$$= 15 [200000 + 145000]$$

$$= 15 (345000) = \underline{\underline{5175000 \text{ frs}}}$$

Some real-life applications of an arithmetic progression include:

- Arithmetic Progression is used in the calculation of straight-line depreciation.
- Arithmetic Progression is used to forecast any sequence while someone is waiting for a cab. If the traffic is moving at a steady pace, they can predict when the next cab will arrive.
- Arithmetic Progression is used in Pyramid-like patterns where things are constantly changing, among other things, and in a variety of other applications.
- Online account checking is a fundamental application of everyday arithmetic. With the risk of identity theft and online banking, a general understanding of fundamental mathematics is essential.
- You can see a real-life application of arithmetic progression when you take a taxi. Once you board a taxi, you will be charged an initial fee followed by a per mile or kilometre fee. This diagram shows an arithmetic sequence in which you will be charged a fixed (constant) rate plus the starting rate for each kilometre travelled.

Check your Understanding

1. An arithmetic progression (A.P.) is a _____ whose terms increase or decrease by a fixed number is called the common difference of the A.P.
2. Arithmetic _____ of a group of observations is the quotient obtained by dividing the sum of all the observations by their number.

Notes

3. If the same number is added or subtracted from each term of an arithmetic progression, then new series formed will also be in an arithmetic progression. True / False
4. Arithmetic mean is independent of the change of origin and scale. True / False
5. Arithmetic Progression is used in Pyramid-like patterns where things are constantly changing, among other things, and in a variety of other applications. True / False

Multiple Choice

1. Find the 10th term of the following sequence: 2,4,6,8,.....
 - a. 12
 - b. 10
 - c. 40
 - d. 20
2. Find the sum of the first 10 terms of the following sequence: 1,3,5,7,.....
 - a. 250
 - b. 88
 - c. 100
 - d. 125
3. Find the number of all-natural numbers between 20 and 80, which are divisible by 3.
 - a. 15
 - b. 20
 - c. 6
 - d. 18
4. The sum of three terms in A.P. is 33 and their product is 1155, Find the terms.
 - a. 15,11,7
 - b. 1,4,3
 - c. 6,8,4
 - d. 4,2,9

Summary

- An arithmetic progression (A.P.) is a sequence whose terms increase or decrease by a fixed number is called the common difference of the A.P.
- Some characteristics of an Arithmetic series
- If the sum of n terms of any series is a quadratic expression, then that series will be arithmetic series.

- If the same number is added or subtracted from each term of an arithmetic progression, then new series formed will also be in an arithmetic progression.
- If the same number is multiplied or divided by each term of an arithmetic progression, then new series formed will also be in the form of arithmetic series.
- If corresponding terms of two series are added or subtracted, then new series formed will also be in an arithmetic series.
- Series formed by multiplying or dividing corresponding terms of two arithmetic progression, then series formed will be not be in an arithmetic series.
- The terms used in this progression for a given series are the first term, the common difference between the two terms, and the n th term.
- Arithmetic mean of a group of observations is the quotient obtained by dividing the sum of all the observations by their number.

Activity

1. Rose wants to start saving money. She decides that the minimum amount she needs to save in any year is Rs. 90,000. She deposits Rs. 90,000 at the end of the first year and saves an additional Rs. 8000 each year following that. How much money will she have saved after 25 years?
2. How does one calculate the arithmetic mean using the direct method; Short-cut method; and Step-deviation method. Explain the steps and give original example problems to illustrate how each method is used and in which scenarios.

Questions and Exercises

1. List five characteristics of an arithmetic series.
2. What are the three terms most commonly used in arithmetic progression notation?
3. What are the steps to calculating the arithmetic mean?
4. List 3 examples of real-life applications of arithmetic progression.

Glossary

- **Arithmetic progression (A.P.):** is a sequence whose terms increase or decrease by a fixed number
- **Progression:** is a type of sequence for which a formula for the n th term can be found.
- Further Reading
- Carnielli, Antonio Lucio. Mathematics on Several Subjects: Theory and tests: Combinatorics, Logarithms, Arithmetic Progressions and Geometric Progressions and other topics. Independently published (November 1, 2016)
- Future Point Coaching Center. Important questions of Quadratic Equations and Arithmetic Progression for competitive exams and CBSE Xth class board exam 2020: Useful for achievers. Kindle Edition January 25, 2020

Notes

Check your Understanding: Answers

1. Sequence
2. Mean
3. True
4. False
5. True

Multiple Choice

1. D
2. C
3. B
4. A

Unit -2.2: Geometric Progression

Notes

Objectives:

At the end of this unit, you will be able to understand:

- Definition of Geometric Progression and the nth term of GP
- Sum of terms in GP
- Representation of terms in GP
- Geometric mean between a and b
- Geometric Progression and its Applications in Business

Introduction

In this unit, we will learn about geometric progression, nth term of a G.P., the sum of first n terms of G.P. and geometric mean also.

2.2.1 Definition of Geometric Progression, nth Term of GP

A geometric progression (G.P.) is a sequence of numbers, whose first term is non-zero and each of the terms is obtained by multiplying its just preceding term is obtained by a constant quantity. This constant quantity is called the common ratio of the G.P.

In mathematical term, if $a_1, a_2, a_3, \dots, a_n$ are in G.P., then

$$\frac{a_2}{a_1} = \frac{a_3}{a_2} = \dots = \frac{a_n}{a_{n-1}} = r$$

Here, r is known as the common ratio of G.P.

If a is the first and r is the common ratio, then G.P. can be written as $a, ar, ar^2, \dots, ar^{n-1}$, where $a \neq 0$.

Example: Find the Geometric Progression ratio.

3, 6, 12, 24,

$a_1 = 3, a_2 = 6, a_3 = 12, a_4 = 24$

$$\frac{a_2}{a_1} = \frac{a_3}{a_2} = \dots = \frac{a_n}{a_{n-1}} = r$$

$$(6/3) = (12/6) = 2$$

2, 6, 18, 54,

$a_1 = 2, a_2 = 6, a_3 = 18, a_4 = 54$

$$\frac{a_2}{a_1} = \frac{a_3}{a_2} = \dots = \frac{a_n}{a_{n-1}} = r$$

$$(6/2) = (18/6) = 3$$

Notes

2.2.2 Sum of n Terms of GP

It is also called the general term of a G.P. Let a be the first term r be the common ratio and a_n be the last term of a G.P., then the n^{th} term is given by

$$a_n = ar^{n-1}, \text{ where } r = \frac{a_n}{a_{n-1}}$$

Suppose there are n terms of a sequence, whose first term is a , the common ratio is r , and the last term is a_n , then the sum of n terms is given by

$$S_n = \frac{a(r^n - 1)}{r - 1}, \text{ when } r > 1$$

$$S_n = \frac{a(r^n - 1)}{r - 1}, \text{ when } r < 1$$

$$S_n = \text{not define, when } r = 1$$

If terms are infinite, then the sum of G.P. is:

$$S_{\infty} = \frac{a}{1 - r}, \text{ when } |r| < 1$$

Example Find the sum of 5 terms of the G.P.: $3 + 6 + 12 + 24 + \dots$

Answer:

Given $a = 3$

$$\begin{aligned} \text{Common ratio } r &= \frac{6}{3} \\ &= 2 \end{aligned}$$

$\therefore r > 1$. So, the sum of 5 terms

$$S_n = \frac{a(r^n - 1)}{r - 1}$$

$$S_5 = \frac{3(2^5 - 1)}{2 - 1}$$

$$S_5 = 3 \times 31$$

$$S_5 = 93$$

Hence, the required sum is 93.

Example 1.21 Find the sum of 10 terms of the geometric progression

$$1 + \sqrt{3} + 3 + 3\sqrt{3} + \dots$$

Answer:

Given $a = 1$

$$\begin{aligned}\text{Common ratio } r &= \frac{\sqrt{3}}{1} \\ &= \sqrt{3}\end{aligned}$$

$\therefore r > 1$. So, the sum of 5 terms

$$\begin{aligned}S_n &= \frac{a(r^n - 1)}{r - 1} \\ S_{10} &= \frac{1((\sqrt{3})^{10} - 1)}{\sqrt{3} - 1} \\ &= \frac{3^5 - 1}{\sqrt{3} - 1} \times \frac{\sqrt{3} + 1}{\sqrt{3} + 1} \\ &= \frac{243 - 1}{3 - 1} \times (\sqrt{3} + 1) \\ &= 121(\sqrt{3} + 1)\end{aligned}$$

Hence, the required sum is $121(\sqrt{3} + 1)$.

2.2.3 Representations of terms in GP

A geometric progression is a sequence where each term has the same fixed ratio, known as the common ratio. G.P. is an abbreviation for geometric progression. The geometric sequence is generally written as $a, ar, ar^2, \dots$, where a is the first term and r is the common ratio of the sequence. The common ratio can have both negative and positive values. In a geometric progression, each successive term is obtained by multiplying the common ratio by the preceding term.

Let a be the first term and r be the common ratio for a Geometric Sequence.

Then the second term, $a_2 = a \times r = ar$

Third term, $a_3 = a_2 \times r = ar \times r = ar^2$

Similarly, n th term, $a_n = ar^{n-1}$

The common ratio of a geometric progression $r = \frac{a^2}{a^1}$

The formula for the n th term of a geometric progression with the first term a and a common ratio of r .

$$a^n = ar^{n-1}$$

The formula for the sum of geometric progression formula,

$$\text{If } |r| < 1, S_n = \frac{a(1-r^n)}{1-r}$$

$$\text{If } |r| > 1, S_n = \frac{a(r^n-1)}{r-1}$$

Notes

The sum of infinite geometric formula $S_{\infty} = \frac{a}{1-r}$ where $r < 1$.

2.2.4 Geometric Mean Between a and b

The geometric mean is defined as the n^{th} root of the product of n numbers, i.e., for a set of numbers $a_1, a_2, a_3, \dots, a_n$, the geometric mean is defined as

$$\text{Geometric mean} = \sqrt[n]{a_1 \cdot a_2 \cdot a_3 \cdot \dots \cdot a_n}$$

And also if we insert one geometric mean between two numbers a and b , then

$$\text{Geometric mean} = \sqrt{ab}$$

2.2.5 Geometric Progression and its Applications in Business

- Geometric progression is a common method for calculating interest earned.
- Geometric progression is a popular method for calculating the balance in our savings account.

Characteristics of Geometric Progression

- If each term of geometric series is multiplied or divided by the same quantity, then new series formed will also be in geometric progression.
- Series formed by multiplying or dividing by corresponding terms of two geometric series will also be in geometric progression.
- Series formed by reciprocal of the term of geometric series is also in geometric progression.

Solved Examples:

Example What is the geometric mean of 10, 51.2 and 8?

Answer:

$$\begin{aligned} \text{Product of all given numbers} &= 10 \times 51.2 \times 8 \\ &= 4096 \\ \text{G.M.} &= \sqrt[3]{4096} \\ &= \sqrt[3]{(16)^3} \\ &= 16 \end{aligned}$$

Example What is the geometric mean of 1, 3, 9, 27 and 81?

Answer:

Product of all given numbers = $1 \times 3 \times 9 \times 27 \times 81$

$$= 59049$$

$$\text{G.M.} = \sqrt[5]{59049}$$

$$= \sqrt[5]{(9)^5}$$

$$= 9$$

Exmple:

3.9 If the first two terms of a G.P. are 125 and 25. Find its 5th term.

Answer:

Given $a = 125$ and $r = 25$

$$\therefore \text{Common ratio} = \frac{ar}{a}$$

$$\frac{ar}{a} = \frac{25}{125}$$

$$r = \frac{1}{5}$$

So, the fifth term of G.P.

$$ar^4 = 125 \times \left(\frac{1}{5}\right)^4$$

$$= \frac{1}{5}$$

Hence, the fifth term is $\frac{1}{5}$

Check your Understanding

1. A _____ progression (G.P.) is a sequence of numbers, whose first term is non-zero and each of the terms is obtained by multiplying its just preceding term is obtained by a constant quantity.
2. A geometric progression is a sequence where each term has the same _____ ratio, known as the common ratio.
3. The geometric sequence is generally written as $a, ar, ar^2, \dots$, where a is the first term and r is the _____ ratio of the sequence. True / False
4. In a geometric progression, each successive term is obtained by multiplying the common ratio by the preceding term. True / False
5. The geometric _____ is defined as the n^{th} root of the product of n numbers
6. Geometric progression is a common method for calculating _____ earned.

Notes

Notes

7. Series formed by multiplying or dividing by corresponding terms of two geometric series will also be in geometric progression. True / False
8. In a geometric progression, each successive term is obtained by multiplying the common ratio by the following term. True / False

Multiple Choice

1. If the first two terms of a G.P. are 125 and 25. Find its 5th term.
 - a. $1/5$
 - b. 5
 - c. 1
 - d. 2
2. Find the sum of 5 terms of the G.P.: $3 + 6 + 12 + 24 + \dots$
 - a. 65
 - b. 78
 - c. 44
 - d. 93
3. Find the sum of 10 terms of the geometric progression $1 + \sqrt{3} + 3 + 3\sqrt{3} + \dots$
 - a. $154(\sqrt{3}+3)$
 - b. $100(\sqrt{3}+5)$
 - c. $121(\sqrt{3}+1)$
 - d. $145(\sqrt{3}+3)$
4. The sum of few terms of any ratio series is 728, if common ratio is 3 and last term is 486, then find the first term of the series.
 - a. 5
 - b. 2
 - c. 9
 - d. 22
5. Calculate the following geometric series: $5 + \frac{5}{3} + \frac{5}{9} + \frac{5}{27} + \dots$
 - a. 7.5
 - b. 9
 - c. 4
 - d. 3.5

Summary

- A geometric progression (G.P.) is a sequence of numbers, whose first term is non-zero and each of the terms is obtained by multiplying its just preceding term is obtained by a constant quantity. This constant quantity is called the common ratio of the G.P.

- The geometric sequence is generally written as $a, ar, ar^2, \dots$, where a is the first term and r is the common ratio of the sequence. The common ratio can have both negative and positive values. In a geometric progression, each successive term is obtained by multiplying the common ratio by the preceding term.
- The geometric mean is defined as the n^{th} root of the product of n numbers
- Characteristics of Geometric Progression
- If each term of geometric series is multiplied or divided by the same quantity, then new series formed will also be in geometric progression.
- Series formed by multiplying or dividing by corresponding terms of two geometric series will also be in geometric progression.
- Series formed by reciprocal of the term of geometric series is also in geometric progression.

Activity

1. The 6th term and 8th term of a G.P. are 32 and 128 respectively. Find the common ratio of the G.P.
2. If three geometric means are put between 2 and 32, find the value of the third geometric mean.

Questions and Exercises

1. What is the formula for the n^{th} term of a geometric progression?
2. What is the formula for the sum of geometric progression?
3. List the characteristics of geometric progression.

Glossary

- **Common Ratio:** This constant quantity is called the common ratio of the G.P.
- **Geometric mean:** is defined as the n^{th} root of the product of n numbers
- **Geometric progression (G.P.)** is a sequence of numbers, whose first term is non-zero and each of the terms is obtained by multiplying its just preceding term is obtained by a constant quantity.

Further Readings

1. Carnielli, Antonio Lucio. Mathematics on Several Subjects: Theory and tests: Combinatorics, Logarithms, Arithmetic Progressions and Geometric Progressions and other topics. Independently published (November 1, 2016)
2. Future Point Coaching Center. Important questions of Quadratic Equations and Arithmetic Progression for competitive exams and CBSE Xth class board exam 2020: Useful for achievers. Kindle Edition January 25, 2020

Check your Understanding: Answers

1. Geometric
2. Fixed

Notes

3. Common

4. True

5. Mean

6. Interest

7. True

8. False

Multiple Choice

1. A

2. D

3. C

4. B

5. A

Unit 2.3 Permutation and Combination

Notes

Objectives:

At the end of this unit, you will be able to understand:

- Introduction to Permutations and Combination and its applications in Business

Introduction

In today's business world scenarios, there are many such problems which we can solve with the help of Permutation and combination. This unit will cover in-depth about the problems presented in businesses and how to solve them.

2.3.1 Introduction to Permutations and Combinations and its Applications in Business

Each of different arrangements, which can be made by taking some or all of a number of things is called a permutation. e.g., Arrangements of objects taking 2 at a time from given 3 objects (a, b, c) are ab, bc, ca, cb, ac, ba, then total number of arrangements is 6, each of which is known as permutation.

Meaning of ${}^n P_r$

Number of permutations of n distinct objects taking r at a time is denoted by ${}^n P_r$.

$${}^n P_r = \frac{n!}{(n-r)!}, \forall 0 \leq r \leq n$$

$$= n(n-1)(n-2).....(n-r+1), \forall n \in N \text{ and } r \in W$$

Properties of ${}^n P_r$

- The number of permutations of n distinct object taken all at a time is ${}^n P_n = n!$.
- ${}^n P_0 = 1$, ${}^n P_1 = n$, and ${}^n P_{n-1} = n!$,
- ${}^n P_r = n \cdot {}^{n-1} P_{r-1} = r \cdot {}^{n-1} P_{r-1} + {}^{n-1} P_r$
- ${}^{n-1} P_r = (n-r) {}^{n-1} P_{r-1}$

Ex.: In how many of the distinct permutations of the letters in the 'MISSISSIPPI' do the four I's not come together?

Solution: The word MISSISSIPPI has 11 letters in which

The number of permutations of the word 'MISSISSIPPI' in which 4 I's and 4's are like

$$= \frac{11!}{4!4!2!}$$

If all the I's are together, then it will be considered as one letter and remaining 7 letters and 1 I's letter will be considered as 8 letters. So, the number of permutations is

$$= \frac{8!}{4!2!}$$

Hence, total number of arrangements

Notes

$$\begin{aligned}
 &= \frac{11!}{4!4!2!} - \frac{8!}{4!2!} \\
 &= \frac{11 \times 10 \times 9 \times 8 \times 7 \times 6 \times 5 \times 4!}{4!4!2!} - \frac{8 \times 7 \times 6 \times 5 \times 4!}{4!2!} \\
 &= 34650 - 840 \\
 &= 33810
 \end{aligned}$$

Application of Permutation and Combination

In today's business world scenarios, there are many such problems which we can solve with the help of Permutation and combination. Few of them are discussed below:

Ex.: If it is required to seat 5 men and 4 women in a row so that the women occupy the even places. How many such arrangements are possible?

Solution: We are given that there are 5 men and 4 women.

There are 9 positions.

The even positions are: 2nd, 4th, 6th, and 8th places.

These four places can be occupied by 4 women in $P(4, 4)$ ways = $4! = 4 \times 3 \times 2 \times 1 = 24$

The remaining 5 positions can be occupied by 5 men in $P(5, 5)$ ways = $5! = 5 \times 4 \times 3 \times 2 \times 1 = 120$ ways

Therefore, by the fundamental counting principle,

Total number of ways of seating arrangements = $24 \times 120 = 2880$

Ex.: How many different ways can five male and five females form a circle for a business meeting such that the male and female set alternate?

Solution: After fixing up one boy on the table, the remaining can be arranged in $4!$ ways but male and female are to sit alternate.

There will be 5 places, between two males, these 5 places can be filled by 5 females in $5!$ ways.

The required number of ways = $4! \times 5! = 2880$

Ex.: A board meeting of a company is organised in a room for 24 persons along the two sides of a table with 12 chairs on each side, 6 persons want to sit on a particular side and persons want to sit on the other side. In how many ways can they be seated?

Solution: 6 persons can be arranged on the 12 chairs on the particular sides in ${}^{12}P_6$ ways and 3 persons can be arranged on the 12 chairs on the other side in ${}^{12}P_3$.

Remaining persons = $24 - 6 - 3 = 15$

Now the remaining 15 persons can be arranged on the remaining 15 chairs in ${}^{15}P_{15}$, i.e., $15!$ Ways.

Therefore, the required number of ways = ${}^{12}P_6 \times {}^{12}P_3 \times 15!$

Notes

Ex.: Mr. Rajan has 10 employee and he wants to invite 6 of them to a party. How many times 3 particular employee never attend the party?

Solution: Excluding the 3 particular friends. Mr. Rajan can invite 6 friends from remaining 7 friends. This can be done in ${}^7C_6 = 7$ ways.

Each of the different groups or selections which can be made by some or all of a number given, things without reference to the order of the things in each group is called a combination. e.g., The groups made by taking 2 objects at a time from three objects (a,b,c) are ab, bc, ca. then, the number of groups is 3 each of which is known as combination.

Meaning of nC_r

The number of combinations of n different things taken r at a time is denoted by nC_r or $\binom{n}{r}$

$$\begin{aligned} {}^nC_r &= \frac{n!}{r!(n-r)!} = \frac{{}^nP_r}{r!}, \forall 0 \leq r \leq n \\ &= \frac{n(n-1)(n-2)\dots(n-r+1)}{r(r-1)(r-2)\dots 2.1}, \forall n \in N \text{ and } r \in W \end{aligned}$$

Properties of nC_r

nC_r is a natural number.

$${}^nC_0 = {}^nC_n = 1, {}^nC_1 = n$$

$${}^nC_r = {}^nC_{n-r}$$

$${}^nC_r + {}^nC_{r-1} = {}^{n+1}C_r$$

$${}^nC_0 + {}^nC_1 + {}^nC_2 + \dots + {}^nC_n = 2^n$$

Ex.1.32: Find n, if ${}^{n-1}P_3 : {}^nP_4 = 1:9$

Solution:

$$\begin{aligned} {}^{n-1}P_3 : {}^nP_4 &= 1:9 \\ \frac{(n-1)!}{(n-1-3)!} : \frac{n!}{(n-4)!} &= 1:9 \\ \frac{(n-1)!}{(n-4)!} : \frac{n(n-1)!}{(n-4)!} &= 1:9 \\ \frac{(n-1)!}{(n-4)!} \times \frac{(n-4)!}{n(n-1)!} &= 1:9 \\ \frac{1}{n} &= \frac{1}{9} \\ n &= 9 \end{aligned}$$

Ex.: In how many of the distinct permutations of the letters in the 'MISSISSIPPI' do the four I's not come together?

Solution: The word MISSISSIPPI has 11 letters.

Notes

The number of permutations of the word 'MISSISSIPPI' in which 4 I's and 4's are like

$$= \frac{11!}{4!4!2!}$$

If all the I's are together, then it will be considered as one letter and remaining 7 letters and 1 I's letter will be considered as 8 letters. So, the number of permutations is

$$= \frac{8!}{4!2!}$$

Hence, total number of arrangements

$$\begin{aligned} &= \frac{11!}{4!4!2!} - \frac{8!}{4!2!} \\ &= \frac{11 \times 10 \times 9 \times 8 \times 7 \times 6 \times 5 \times 4!}{4!4!2!} - \frac{8 \times 7 \times 6 \times 5 \times 4!}{4!2!} \\ &= 34650 - 840 \\ &= 33810 \end{aligned}$$

Ex.1.34: In a polygon the number of diagonals is 54. The number of sides of the polygon is ____?

Solution: Let number of sides of polygon is n. Since number of sides of the polygon is equal to the number of vertices of polygon.

$\therefore$ Number of diagonals of polygon = 54

$$\begin{aligned} {}^nC_2 - n &= 54 \\ \frac{n(n-1)}{2} - n &= 54 \\ n^2 - 3n &= 108 \\ n^2 - 3n - 108 &= 0 \\ (n+9)(n-12) &= 0 \\ n &= 12 \end{aligned}$$

Because n cannot be negative.

Ex.1.35: Find the number of ways of selecting 9 balls from 6 red balls, 5 white balls and 5 blue balls, if each selection consist of 3 balls of each colour.

Solution: Total number of ways selecting 9 balls (3 balls of each colour) is

$$\begin{aligned} &= {}^6C_3 \times {}^5C_3 \times {}^5C_3 \\ &= {}^6C_3 \times {}^5C_2 \times {}^5C_2 \\ &= \frac{6 \times 5 \times 4}{6} \times \frac{5 \times 4}{2} \times \frac{5 \times 4}{2} \\ &= 20 \times 10 \times 10 \\ &= 2000 \end{aligned}$$

Ex.1.36: Four boys picked up 30 mangoes. In how many can they divide them, if all mangoes are identical?

Solution: 30 mangoes can be distributed among 4 boys such that each boy can receive any number of mangoes.

Hence, total number of ways

$$\begin{aligned}
 &= {}^{30+4-1}_{4-1} C_{4-1} \\
 &= {}^{33}_3 C_3 \\
 &= \frac{33.32.31}{1.2.3} \\
 &= 5456
 \end{aligned}$$

Ex.1.37: How many different ways can five boys and five girls form a circle such that the boys and girls set alternate?

Solution: After fixing up one boy on the table, the remaining can be arranged in 4! ways but boys and girls are to sit alternate.

There will be 5 places, between two boys, these 5 places can be filled by 5 girls in 5! ways.

The required number of ways

$$\begin{aligned}
 &= 4! \times 5! \\
 &= 2880
 \end{aligned}$$

Application of Permutation and Combination

In today's business world scenarios, there are many such problems which we can solve with the help of Permutation and combination. Few of them are discussed below:

Ex. 1.38: If it is required to seat 5 men and 4 women in a row so that the women occupy the even places. How many such arrangements are possible?

Solution: We are given that there are 5 men and 4 women.

There are 9 positions.

The even positions are: 2nd, 4th, 6th, and 8th places.

These four places can be occupied by 4 women in $P(4, 4)$ ways $= 4! = 4 \times 3 \times 2 \times 1 = 24$

The remaining 5 positions can be occupied by 5 men in $P(5, 5)$ ways $= 5! = 5 \times 4 \times 3 \times 2 \times 1 = 120$ ways

Therefore, by the fundamental counting principle,

Total number of ways of seating arrangements $= 24 \times 120 = 2880$

Ex. 1.39: How many different ways can five male and five females form a circle for a business meeting such that the male and female set alternate?

Solution: After fixing up one boy on the table, the remaining can be arranged in 4! ways but male and female are to sit alternate.

There will be 5 places, between two males, these 5 places can be filled by 5 females in 5! ways.

The required number of ways

Notes

$$= 4! \times 5!$$

$$= 2880$$

Ex. 1.40: A board meeting of a company is organised in a room for 24 persons along the two sides of a table with 12 chairs on each side, 6 persons want to sit on a particular side and persons want to sit on the other side. In how many ways can they be seated?

Solution: 6 persons can be arranged on the 12 chairs on the particular sides in ${}^{12}P_6$ ways and 3 persons can be arranged on the 12 chairs on the other side in ${}^{12}P_3$.

Remaining persons = $24 - 6 - 3 = 15$ and remaining chairs = $24 - 6 - 3 = 15$.

Now the remaining 15 persons can be arranged on the remaining 15 chairs in ${}^{15}P_{15}$, i.e., 15! Ways.

Therefore, the required number of ways = ${}^{12}P_6 \times {}^{12}P_3 \times 15!$

Ex. 1.41: Mr. Rajan has 10 employee and he wants to invite 6 of them to a party. How many times 3 particular employee never attend the party?

Solution: Excluding the 3 particular friends. Mr. Rajan can invite 6 friends from remaining 7 friends. This can be done in ${}^7C_6 = 7$ ways.

Check your Understanding

- Each of different arrangements, which can be made by taking some or all of a number of things is called a _____.
- _____ of objects taking 2 at a time from given 3 objects (a, b, c) are ab, bc, ca, cb, ac, ba.
- Each of the different groups or selections which can be made by some or all of a number given, things without reference to the order of the things in each group is called a _____.
- In today's business world scenarios, there are many such problems which we can solve with the help of Permutation and combination. True / False

Multiple Choice

- How many different ways can five boys and five girls form a circle such that the boys and girls set alternate?
 - 3456
 - 2880
 - 4321
 - 2112
- Four boys picked up 30 mangoes. In how many can they divide them, if all mangoes are identical?
 - 5456
 - 6666

Notes

- c. 4347
d. 2112
3. Find the number of ways of selecting 9 balls from 6 red balls, 5 white balls and 5 blue balls, if each selection consist of 3 balls of each colour
- a. 5000
b. 2000
c. 1500
d. 4000
4. Find n, if ${}^{n-1}P_3 : {}^nP_4 = 1:9$
- a. 9
b. 5
c. 7
d. 11

Summary

- Each of different arrangements, which can be made by taking some or all of a number of things is called a permutation. e.g., Arrangements of objects taking 2 at a time from given 3 objects (a, b, c) are ab, bc, ca, cb, ac, ba, then total number of arrangements is 6, each of which is known as permutation.
- Each of the different groups or selections which can be made by some or all of a number given, things without reference to the order of the things in each group is called a combination. e.g., The groups made by taking 2 objects at a time from three objects (a,b,c) are ab, bc, ca. then, the number of groups is 3 each of which is known as combination.
- Each of the different groups or selections which can be made by some or all of a number given, things without reference to the order of the things in each group is called a combination.

Activity

1. How many ways can you arrange the letters of the word 'Leading' where all the vowels always come together?
2. How many three constant and two vowel words can you form if you have seven constants and four vowels?
3. How many three digit numbers which are divisible by five can be formed out of the numbers 2, 3, 5, 6, 7, 9?

Questions and Exercises

1. What is the equation that is used to calculate the number of permutations?
2. What are two examples on how permutations solve business related problems?

Notes

Glossary

- **Combination:** each of the different groups or selections which can be made by some or all of a number given, things without reference to the order of the things in each group.
- **Permutation:** Each of different arrangements, which can be made by taking some or all of a number of things.

Further Readings

1. Carnielli, Antonio Lucio. Mathematics on Several Subjects: Theory and tests: Combinatorics, Logarithms, Arithmetic Progressions and Geometric Progressions and other topics. Independently published (November 1, 2016)
2. Future Point Coaching Center. Important questions of Quadratic Equations and Arithmetic Progression for competitive exams and CBSE Xth class board exam 2020: Useful for achievers. Kindle Edition January 25, 2020

Check your Understanding: Answers

1. Permutation
2. Arrangement
3. Combination
4. True

Multiple Choice

1. B
2. A
3. B
4. A

Module - 3: Differential Calculus

Notes

Structure:

- 3.1 Sets
 - 3.1.1 Definition of Sets and Subsets.
 - 3.1.2 Representation of a Set - Roster form, Descriptive form and Set Builder form.
 - 3.1.3 Types of Sets - Empty, Singleton, Finite, Infinite Sets, Equal, Power and Universal Sets..
- 3.2 Relations and Functions
 - 3.2.1 Relations and functions
 - 3.2.2 Function - Definition and Types
- 3.3 Limit and Continuity
 - 3.3.1 Limit of a function - Definition and Methods
 - 3.3.2 Methods to solve limit
 - 3.3.3 Continuous and discontinuous functions - Introduction and Application

Notes

Unit - 3.1: Sets

Objectives:

At the end of this unit, you will be able to understand:

- Definition of Set and Subset
- Representation of Set - Roster Form, Descriptive Form and Set Builder Form
- Types of Sets – Empty, Singleton, Finite, Infinite Sets, Equal, Power, Universal

3.1.1 Definition of Sets and Subsets

The set theory was developed by a German Mathematician Georg Cantor (1845-1918). Nowadays, set theory is used in almost all branches of mathematics. We also use sets to define Relation and Functions. The knowledge of sets is required in the study of geometry, sequence, probability, etc. In this unit, we will discuss some basic definitions related to sets.

Definition of a Set

“A well-defined collection of objects is called a set”.

The term “well-defined” implies that in a given set, it would be possible to decide if certain objects belong to the set. The term “distinct” implies that a given object should not be repeated in a collection or group. The object in the set is called its member or element. A set is represented by $\{ \}$.

Generally, sets are denoted by capital letters X, Y, Z.... and its element are denoted by small letters x, y, z.....

Let X is a non-empty set. If x is an element of X, then we write $x \in X$ and it can be read as ‘x is an element of X’ or ‘x belongs to X’. If x is not an element of X, then we write $x \notin X$ and read as ‘x is not an element of X’ or ‘x does not belong to X’.

Example: Suppose we have a set X that is defined in this way X = Set of all days in a week. In this set Sunday, Monday, Tuesday, Wednesday, Thursday, Friday, Saturday are members of the set.

Definition of a Subset

In the previous unit, we learned what sets are, how they are represented and they are classified. In this unit, we will learn what are Subset, Superset and Proper subset.

Let X and Y be two non-empty sets. If each element of set X is an element of set Y, then set X is known as subset of set Y.

If set X is a subset of set Y, then set Y is called the superset of X.

Also, if X is a subset of Y, then it is denoted as $X \subseteq Y$ and read as ‘X is a subset of Y’.

If $x \in X \Rightarrow x \in Y$ then $X \subseteq Y$ If $x \in X \Rightarrow x \notin Y$ Then X will not be a subset of Y.

Example 2.2.1 If $X = \{a, b\}$ and $Y = \{a, b, c, d\}$

Here each element of X is an element of Y . Thus $X \subseteq Y$ i.e. X is a subset of Y and Y is a superset of X .

Proper Subset If each element of X is in set Y but set Y has at least one element which is not in X , then set X is known as proper subset of set Y . If X is a proper subset of Y , then it is written as $X \subset Y$ and read as X is a proper subset of Y .

Example:

If $N = \{1, 2, 3, 4, \dots\}$

And $W = \{0, 1, 2, 3, 4, \dots\}$

Then $N \subset W$

3.1.2 Representation of a Set- Roster Form, Descriptive Form and Set Builder Form

Sets can be represented by following two methods:

1. Tabular Method or Roster Method
2. Set Builder Method

Tabular Method or Roster Method – In this method, elements are listed and put with in a brace $\{\}$ and separated by commas.

Example Suppose we have a set X that is defined in this way $X = \{x : x \text{ is an even number and } x < 15\}$ $X = \{2, 4, 6, 8, 10, 12, 14\}$

Set Builder Method: - In this method, instead of listing all elements of a set, we list the property or properties satisfied by the elements of a set and write it as $X = \{x : P(x)\}$ It is read as “ X is the set of all elements x such that x has the property $P(x)$.” The symbol ‘:’ stands for such that.

Example: Suppose we have a set X that is defined in this way $X = \text{Set of all even number less than 15}$. $X = \{x : x = 2n, n \in N \text{ and } 1 \leq n \leq 7\}$ Or $X = \{x : x \text{ is an even number less than 15}\}$ This method is also known as Rule method.

3.1.3 Types of Sets- Empty, Singleton, Finite, Infinite Sets, Equal, Power and Universal Sets

Types of Set:

- (i) **Empty (Void/Null) Set:** A set which has no element, is called an empty set. It is denoted by ϕ or $\{\}$. Example, Let $X = \text{Set of all even prime numbers greater than 3}$.

Example, Let $Y = \text{Set of all prime numbers less than 2}$.

- (ii) **Singleton Set:** A set which has only one element or member, is known as singleton set.

Example, Let $X = \{x : x \text{ is an even prime number}\}$ and $Y = \{a\}$

Notes

(iii) Finite Set: A set which has finite number of element or member, is known as finite set.

Example, Let $X = \{x: x \text{ is an even number less than } 9\}$ and $Y = \{1, 3, 5, 7, 11, 13, 15\}$

(iv) Infinite Set: A set which have infinite number of element or member, is known as infinite set. Example, Let $X = \{x: x \text{ is a natural number}\}$ and $Y = \{2, 4, 6, 8, 10, 12, 14, \dots\}$

(v) Equivalent Sets: If two finite sets X and Y have same number of elements, then the sets are known as equivalent set.

Example, Let $X = \{2, 4, 6, 8\}$ and $Y = \{1, 3, 5, 7\}$

(vi) Equal Sets: If X and Y are two non-empty sets and each element of X is an element of set Y and each element of set Y is an element of set X , then sets X , and Y are called equal sets.

Example, Let $X = \{x: x = 2n, n \in \mathbb{N} \text{ and } 1 \leq n \leq 5\}$ and $Y = \{2, 4, 6, 8, 10\}$

(vii) Universal Set: If there are some sets under consideration, then there happens to be a set which is a superset of each one of the given sets. Such a set is known as the universal set and it is denoted by U .

Example, Suppose we have three set $X = \{a, b\}$, $Y = \{c, d, e\}$, and $Z = \{f, g, h, i, j\}$.
 $\therefore U = \{a, b, c, d, e, f, g, h, i, j\}$ is a universal set for all given sets.

(viii) Power Set: If X is a non-empty set, then the collection of all possible subsets of set X can be referred to as power set. It is denoted by $P(X)$. The total number of elements in a power set of X , containing n elements, is 2^n .

Example, Let $X = \{a, b, c\} \therefore P(X) = \{\phi, \{a\}, \{b\}, \{c\}, \{a, b\}, \{b, c\}, \{c, a\}, \{a, b, c\}\}$

Check your Understanding

1. The set theory was developed by a German Mathematician Georg Cantor (1845-1918).
2. We also use sets to define Relation and Functions. True / False
3. The term "_____ " implies that a given object should not be repeated in a collection or group.
4. The term "_____ - _____ " implies that in a given set, it would be possible to decide if certain objects belong to the set.
5. In the _____ method(roster method), elements are listed and put with in a brace $\{\}$ and separated by commas.
6. A set which has no element, is called an _____ set.
7. If two finite sets X and Y have same number of elements, then the sets are known as universal set. True / False
8. If X is a non-empty set, then the collection of all possible subsets of set X can be referred to as _____ set.

Notes

Summary:

- A set is well-defined collection of distinct objects.
- Set can be represented in two ways (i) Tabular form or Roster form (ii) Set builder method.
- In the tabular form, the elements of a set are actually written down, separated by commas and enclosed within braces.
- In the set builder method, a set is described by a characterizing property of its element.
- A set that does not contain a single element or member is called a null or empty set.
- A set with only one element or member is known as singleton set.
- A set, which has a finite number of element or member, is known as finite set; otherwise it is called non-finite set.
- Two sets X and Y are said to be equal, if every element of set X is in set Y and every element of set Y is in set X.
- The collection of all subset of a set X is called Power set of X.
- Two sets X and Y are said to be equivalent, if the number of element in both sets are equal.
- All the sets under consideration are likely to be subsets of a set is called the universal set

Activity

1. Find ten real life examples of where set are used. What method was used to represent them? What kind of set are they?

Questions and Exercises

1. What is a subset?
2. List the types of sets.
3. How does one represent a set?
4. What are the two methods of representing a set?

Glossary

- **Set:** A well-defined collection of objects
- **Set Builder Method:** In this method, instead of listing all elements of a set, we list the property or properties satisfied by the elements of a set
- **Tabular Method or Roster Method:** In this method, elements are listed and put with in a brace {} and separated by commas.

Further Reading

1. Boelkins, Matthew. Active Calculus 2018: Single Variable. CreateSpace Independent Publishing Platform; 2018th edition (August 13, 2018)

Notes

2. Edwards, Bruce H.; Larson, Ron. Calculus: Early Transcendental Functions Cengage Learning; 7th edition (January 1, 2018)
3. Stewart, James. Calculus 8th Edition. Cengage Learning; 8th edition (May 19, 2015)

Check your Understanding: Answers

1. Set
2. True
3. Distinct
4. Well-defined
5. Tabular
6. Empty
7. False
8. Power

Unit - 3.2: Relations and Functions

Notes

Objective:

At the end of this unit, you will be able to understand:

- Relations and Functions

Introduction

In mathematics, a function is a relation between a set of inputs and a set of possible outputs. Functions have a property that they have an output for every input. In this unit, we will learn what functions are and how they are classified, and we will also discuss the inverse function.

3.2.1 Relations and Functions

Definition of Function

A function f from a set X to another set Y is said to be a function (or mapping) from X to Y if, with every element of X , the relation f relates a unique element of Y . The element of set Y is called f -image of the element of set X . Also, the element of set X is called pre-image of the element of set Y under f .

Classification of Functions

Constant Function: A function which does not change as its parameters vary, i.e., the function whose rate of change is zero.

Or

Let c be a constant, then the function $f(x) = c, \forall x \in R$ is known as a constant function.

The domain of $f(x) = R$ and range of $f(x) = [c]$

Example 2.2.1 The graph of a constant function $f(x) = 2$.

Table for function at a different value of x .

x	1	2	3	4	5
$y = f(x)$	2	2	2	2	2

Now we can draw a graph according to the obtain data from the above table

Notes

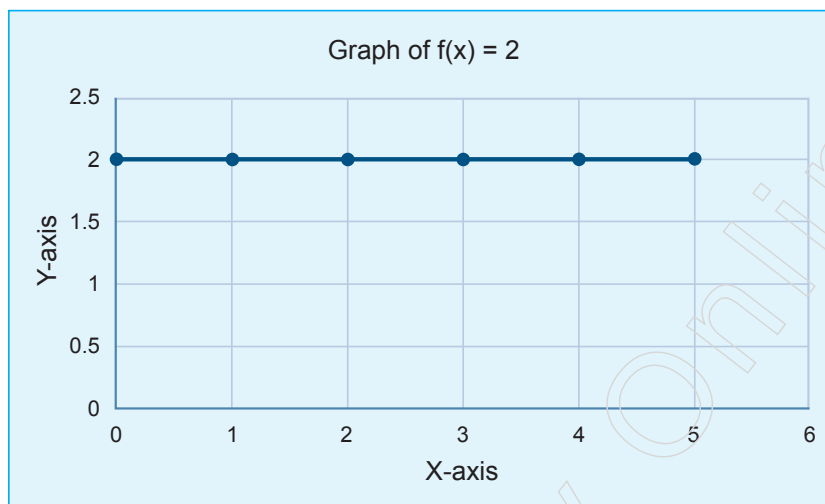


Fig.3.1.3 The graph of constant function $f(x) = 2$

Polynomial Function: The function $y = f(x) = a_0x^n + a_1x^{n-1} + \dots + a_n$, where $a_0, a_1, a_2, \dots, a_n$ are real coefficients, and n is a non-negative integer, is known as a polynomial function.

If $a_0 \neq 0$, then the degree of a polynomial function is n .

The domain of $f(x) = R$ and range varies from function to function.

Example 3.2.2 The graph of the function $f(x) = x^2$

Table for functions at different values of x .

x	-5	-4	-3	-2	-1	0	1	2	3	4	5
$f(x) = x^2$	25	16	9	4	1	0	1	4	9	16	25

Now we can draw a graph according to the obtain data from the above table.

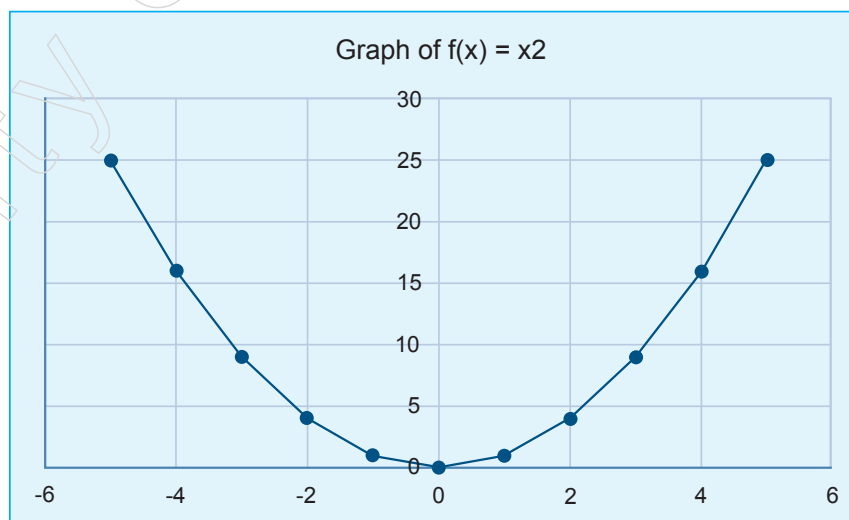


Fig. 3.1.4 The graph of function $f(x) = x^2$

Rational Function: If $P(x)$ and $Q(x)$ are polynomial functions, $Q(x) \neq 0$,

Notes

then the function $f(x) = \frac{P(x)}{Q(x)}$ is known as a rational function.

The domain of $f(x) = R - \{x : Q(x) = 0\}$ and range varies from function to function.

Irrational Function: The function containing one or more terms having non-integral rational powers of x are called irrational function.

Example 3.2.3.1 $y = f(x) = \sqrt{x}$

Domain varies from function to function.

Identity Function: Function $f(x) = x, \forall x \in R$ is known as identity function. It is a straight line passing origin and having slope unity.

The domain of $f(x) = R$ and Range of $f(x) = R$

Example 3.2.3 The graph of a function $f(x) = x$

Table for function at the different value of x .

x	-5	-4	-3	-2	-1	0	1	2	3	4	5
$f(x) = x$	-5	-4	-3	-2	-1	0	1	2	3	4	5

Now we can draw a graph according to the obtain data from the above table.

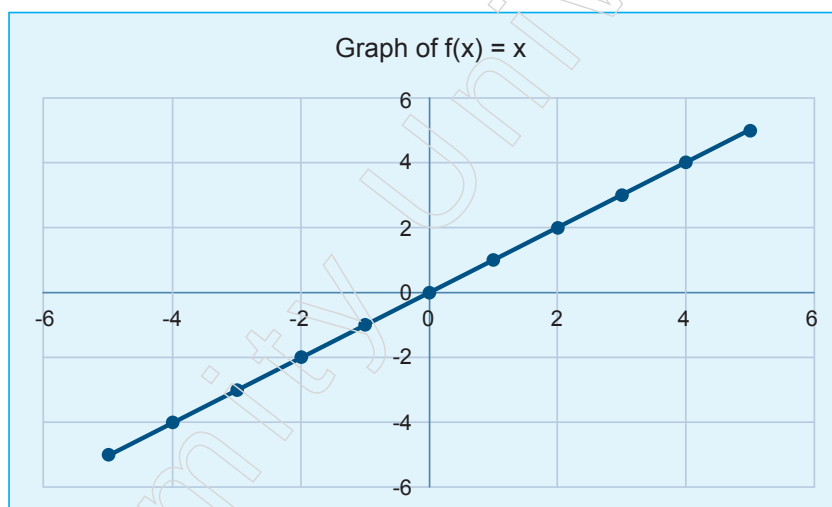


Fig. 3.1.5 The graph of function $f(x) = x$

Square Root Function: The function that associates every positive real number x to $+\sqrt{x}$ is called the square root function, i.e., $f(x) = +\sqrt{x}$

Range of $f(x) = [0, \infty)$.

Exponential Function: A function of the form $f(x) = a^x$, a is a positive real number, is an exponential function. The value of the function depends upon the value of

Notes

a for $0 < a < 1$, the function is decreasing and for $a > 1$, the function is increasing.

Domain of $f(x) = R$

And Range of $f(x) = [0, \infty)$

Example 3.2.4 The graph of the function $f(x) = e^x$

Table for function at the different value of x.

x	$f(x) = e^x$
-5	0.0067379
-4	0.0183156
-3	0.0497871
-2	0.1353353
-1	0.3678794
0	1
1	2.7182818
2	7.3890561
3	20.085537
4	54.59815
5	148.41316

Now we can draw a graph according to the obtain data from the above table.

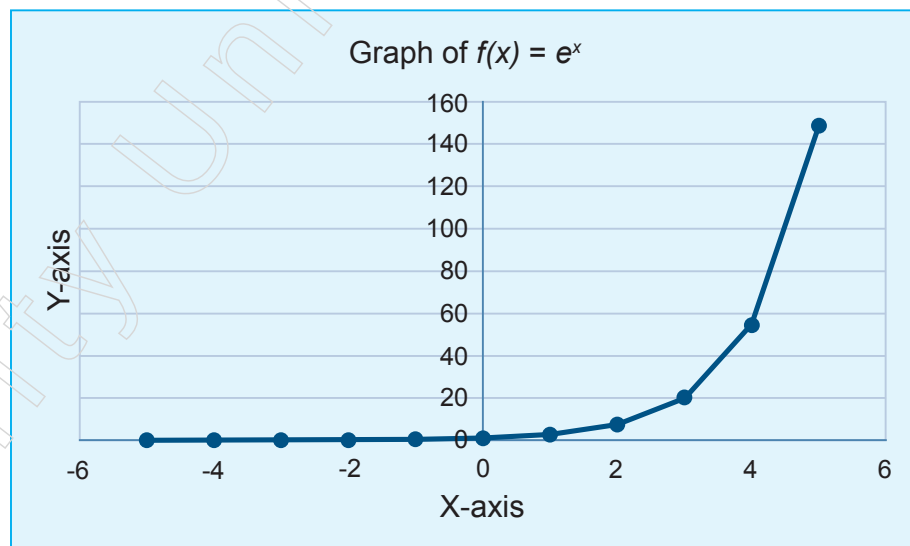


Fig. 3.1.6 The graph of function $f(x) = e^x$

Logarithmic Function: Function $f(x) = \log_a x$, ($x, a > 0$) and $a \neq 1$ is known as the logarithmic function.

Domain of $f(x) = (0, \infty)$

And Range of $f(x) = R$

Example 3.2.5 The graph of the function $f(x) = \log_{10} x$

Table for function at the different value of x .

x	0.1	1	10	25	50	75	100
$f(x) = \log_{10} x$	-1	0	1	2	3	4	5

Now we can draw a graph according to the obtain data from the above table

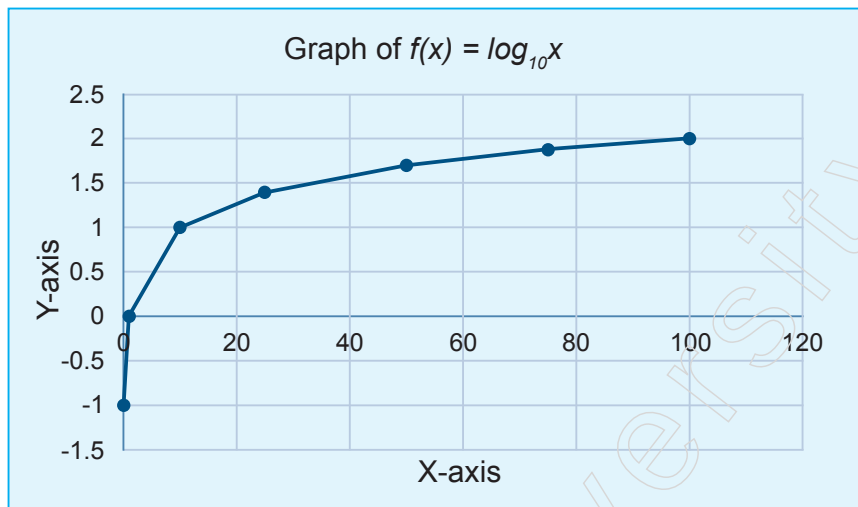


Fig. 3.1.7 The graph of the function $f(x) = \log_{10} x$

Modulus Function: Function $y = f(x) = |x|$ is known as modulus function.

$$y = f(x) = |x| = \begin{cases} -x, & x < 0 \\ x, & x \geq 0 \end{cases}$$

Example 3.2.6 The graph of the function $f(x) = |x|$

Table for function at the different value of x .

x	-5	-4	-3	-2	-1	0	1	2	3	4	5
$f(x) = x $	5	4	3	2	1	0	1	2	3	4	5

Now we can draw a graph according to the obtain data from the above table

Notes

Notes

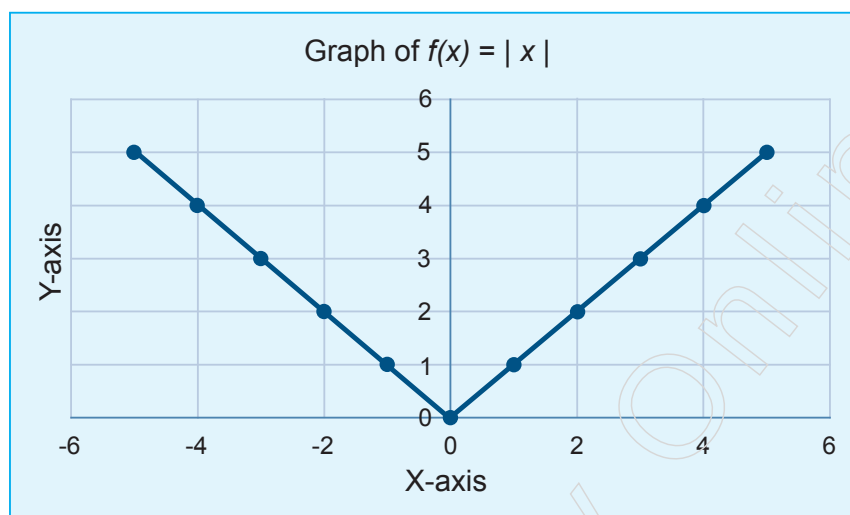


Fig. 3.1.8 The graph of Function $f(x) = |x|$

Inverse Function: Let f be defined as a function from X to Y such that for every element of Y there exists an image. Let y be an arbitrary element of Y . Then, f being onto, there exists an element $x \in X$ such that $f(x) = y$. Also, f being one-one, this x must be unique. Thus, for each $y \in Y$, there exists a unique element $x \in X$ such that $f(x) = y$. So, we may define a function,

$$f^{-1} : Y \rightarrow X$$

$$\therefore f^{-1}(y) = x \Leftrightarrow f(x) = y$$

The above function f^{-1} is called the inverse of f .

Example 3.2.7 Suppose we have a function $y = f(x) = x + 3$ and

Table for function and inverse of the given function at the different value of x and y , respectively.

We have given $y = f(x) = x + 3$

So, to find the inverse of this function, we need to find the value of x in terms of y .

So, $f^{-1}(x) = y - 3$

x	0	1	2	3	4	5
$f(x)$	3	4	5	6	7	8
Y	0	1	2	3	4	5
$f^{-1}(x)$	-3	-2	-1	0	1	2

Now we can draw a graph according to the obtain data from the above table.

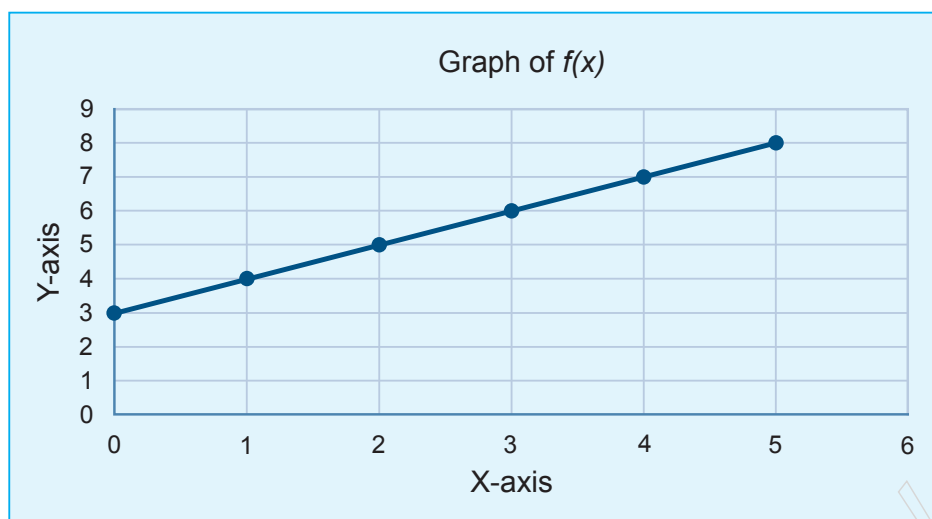


Fig. 3.1.9 The graph of function $f(x) = x + 3$

Now we can draw a graph of the inverse of the above function according to the obtain data from the above table.

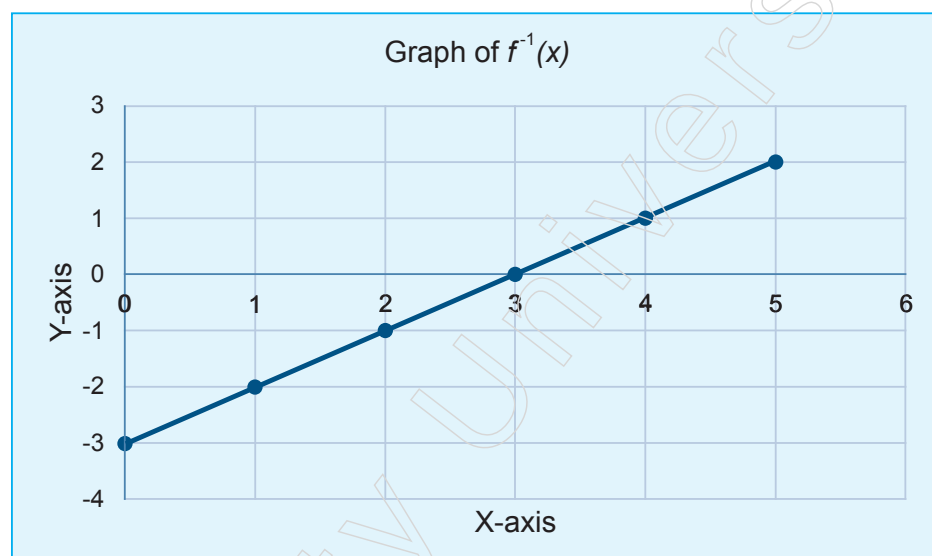


Fig. 3.1.10 Graph of $f^{-1}(x)$

It is clear from the above figure that the points of the line of the two graphs are the image of each other.

Check your Understanding

1. In mathematics, a _____ is a relation between a set of inputs and a set of possible outputs.
2. A _____ function is a function which does not change as its parameters vary.
A function of the form $f(x) = a^2$, where (a) is a positive real number, is an _____ function.
3. Functions have the property that they have an output for every input. True / False

Questions and Exercises

1. What is a polynomial function?
2. Describe a rational function.
3. How is a square root function depicted?
4. What does a logarithmic function look like graphed out?
5. Describe a modulus function.

Glossary

- Constant Function: a function which does not change as its parameters vary.
- Exponential Function: a function of the form $f(x)=a^x$, where (a) is a positive real number.
- Irrational Function: function containing one or more terms having non-integral rational powers of x.

Further Reading

1. Boelkins, Matthew. Active Calculus 2018: Single Variable. CreateSpace Independent Publishing Platform; 2018th edition (August 13, 2018)
2. Edwards, Bruce H.; Larson, Ron. Calculus: Early Transcendental Functions Cengage Learning; 7th edition (January 1, 2018)
3. Stewart, James. Calculus 8th Edition. Cengage Learning; 8th edition (May 19, 2015)

Check your Understanding: Answers

1. Function
2. Constant
3. Exponential
4. True

Multiple Choice

1. A
2. D
3. D
4. B

Notes

Unit - 3.3: Limit and Continuity

Objectives:

At the end of this unit, you will be able to understand:

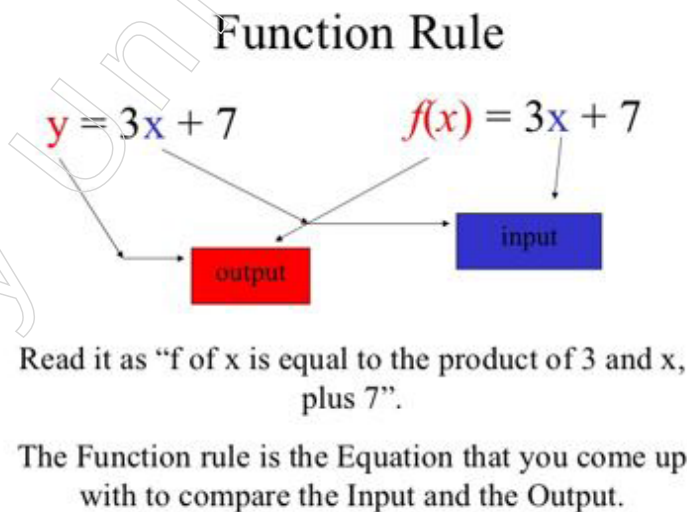
- Function – Definition and Types
- Limit of a function – Definition and Methods
- Methods to Solve Limit
- Continuous and Discontinuous Functions – Introduction and Application

3.3.1 Function – Definition and Types

In mathematics, a function is a relation between a set of inputs and a set of possible outputs. Functions have a property, which is they have an output for every input. In this unit, we will learn what functions are and how they are classified, and we will also discuss the inverse function.

Definition of Function

A function f from a set X to another set Y is said to be a function (or mapping) from X to Y if, with every element of X , the relation f relates a unique element of Y . The element of set Y is called f -image of the element of set X . Also, the element of set X is called pre-image of the element of set Y under f . The image given below is an Example of function in mathematics:



Constant Function: A function which does not change, as its parameters vary, i.e., the function whose rate of change is zero.

Or

Let c be a constant, then the function $f(x) = c, \forall x \in R$ is known as a constant function.

The domain of $f(x) = R$ and range of $f(x) = [c]$

Notes

Ex. 3.1 The graph of a constant function $f(x) = 2$.

Table for function at a different value of x.

Table for function at a different value of x.

X	1	2	3	4	5
y = f(x)	2	2	2	2	2

Now we can draw a graph according to the obtain data from the above table

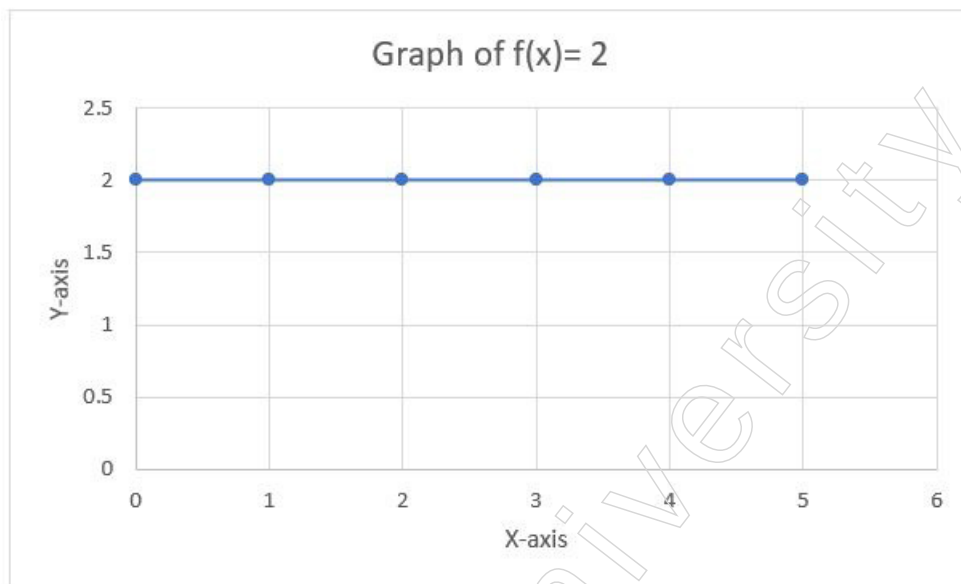


Fig.3.1.1 The graph of constant function $f(x) = 2$

Polynomial Function: The function $y = f(x) = a_0x^n + a_1x^{n-1} + \dots + a_n$, where $a_0, a_1, a_2, \dots, a_n$ are real coefficients, and n is a non-negative integer, is known as a polynomial function.

If $a_0 \neq 0$, then the degree of a polynomial function is n .

The domain of $f(x) = R$ and range varies from function to function.

Ex. 3.2 The graph of the function $f(x) = x^2$

Table for functions at different values of x.

x	-5	-4	-3	-2	-1	0	1	2	3	4	5
f(x) = x^2	25	16	9	4	1	0	1	4	9	16	25

Now we can draw a graph according to the obtain data from the above table.

Notes

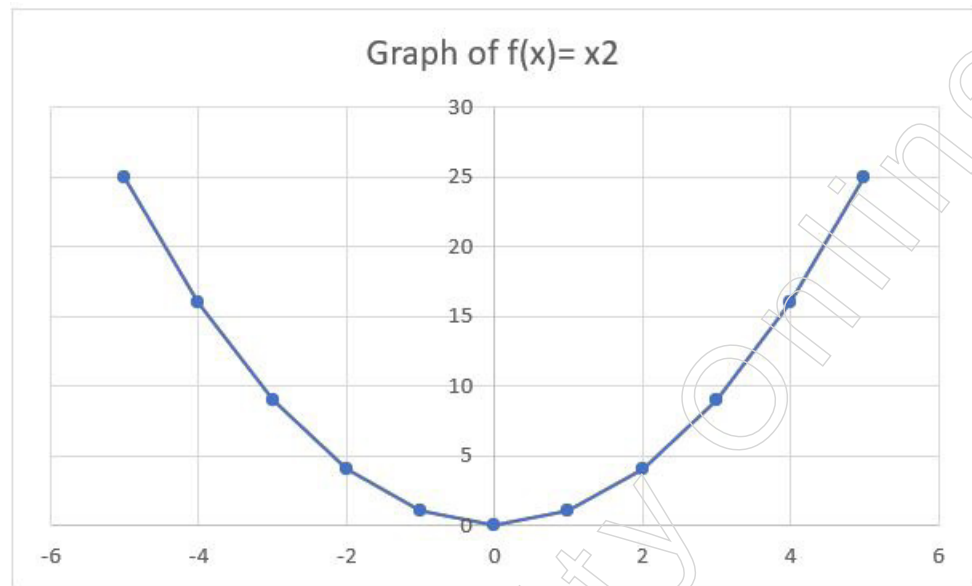


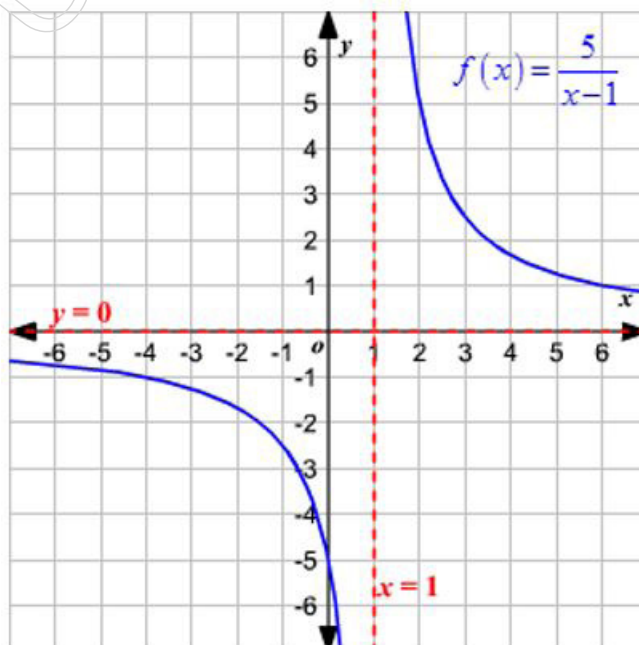
Fig. 3.1.2 The graph of function $f(x) = x^2$

Rational Function: If $P(x)$ and $Q(x)$ are polynomial functions, $Q(x) \neq 0$, then the function $f(x) = \frac{P(x)}{Q(x)}$ is known as a rational function.

The domain of $f(x) = R - \{x : Q(x) = 0\}$ and range

varies from function to function. In other words, domain is the set of all values for which we get the function defined. Range of the function will be a set of all the values taken by f .

A graph for the function $f(x) = \frac{5}{x-1}$ is shown below.



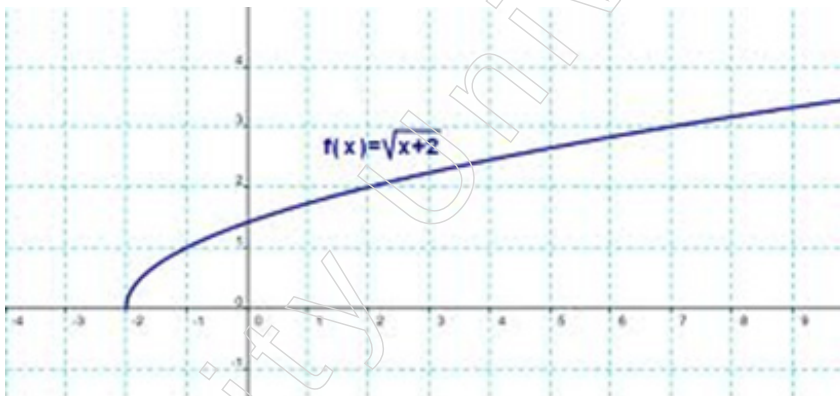
Irrational Function: The function containing one or more terms having non-integral rational powers of x are called irrational function.

Ex. 3.3 $y = f(x) = \sqrt{x}$



Domain varies from function to function.

For Example, if we are to find the graph of function $f(x) = \sqrt{x+2}$, the following will be the solution:



Identity Function: Function $f(x) = x, \forall x \in R$ is known as identity function. It is a straight-line passing origin and having slope unity.

The domain of $f(x) = R$ and Range of $f(x) = R$

Ex. 3.4 The graph of a function $f(x) = x$

Table for function at the different value of x .

x	-5	-4	-3	-2	-1	0	1	2	3	4	5
$f(x) = x$	-5	-4	-3	-2	-1	0	1	2	3	4	5

Now we can draw a graph according to the obtain data from the above table.

Notes

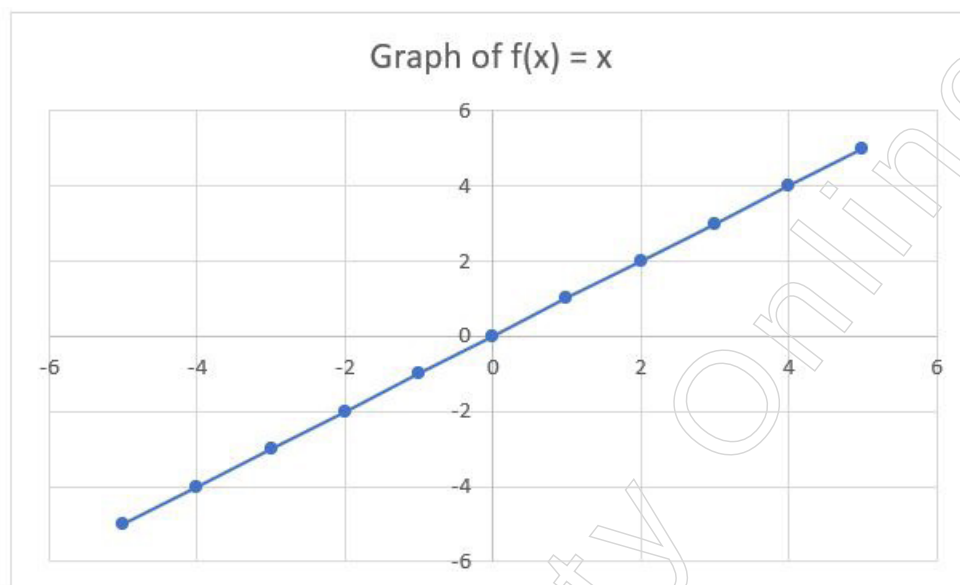


Fig. 3.1.5 The graph of function $f(x) = x$

Square root Function: The function that associates every positive real number x to $+\sqrt{x}$ is called the square root function, i.e., $f(x) = +\sqrt{x}$

Range of $f(x) = [0, \infty)$, which means that the range of a function can be 0 but not ∞ .

Exponential Function: A function of the form $f(x) = a^x$, where a is a positive real number and is an exponential function. The value of the function depends upon the value of a for $0 < a < 1$, the function is decreasing and for $a > 1$, the function is increasing.

Domain of $f(x) = R$

And Range of $f(x) = [0, \infty)$

Ex. 3.5 The graph of the function $f(x) = e^x$

Table for function at the different value of x .

x	$f(x) = e^x$
-5	0.0067379
-4	0.0183156
-3	0.0497871
-2	0.1353353
-1	0.3678794
0	1
1	2.7182818
2	7.3890561
3	20.085537
4	54.59815
5	148.41316

Now we can draw a graph according to the obtain data from the above table.

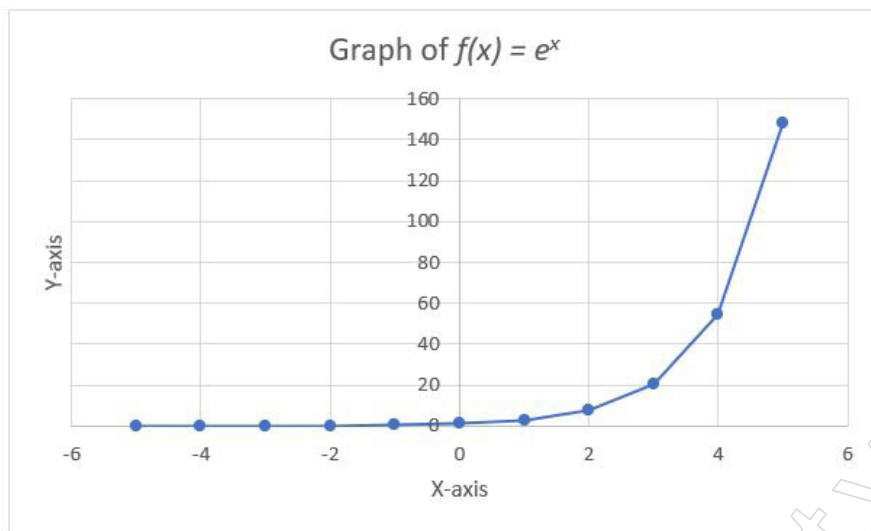


Fig. 3.1.6 The graph of function $f(x) = e^x$

Logarithmic Function: Function $f(x) = \log_a x$, ($x, a > 0$) and $a \neq 1$ is known as the logarithmic function.

Domain of $f(x) = (0, \infty)$

And Range of $f(x) = R$

Ex. 3.6 The graph of the function $f(x) = \log_{10} x$

Table for function at the different value of x .

x	0.1	1	10	25	50	75	100
$f(x) = \log_{10} x$	-1	0	1	2	3	4	5

Now we can draw a graph according to the obtain data from the above table

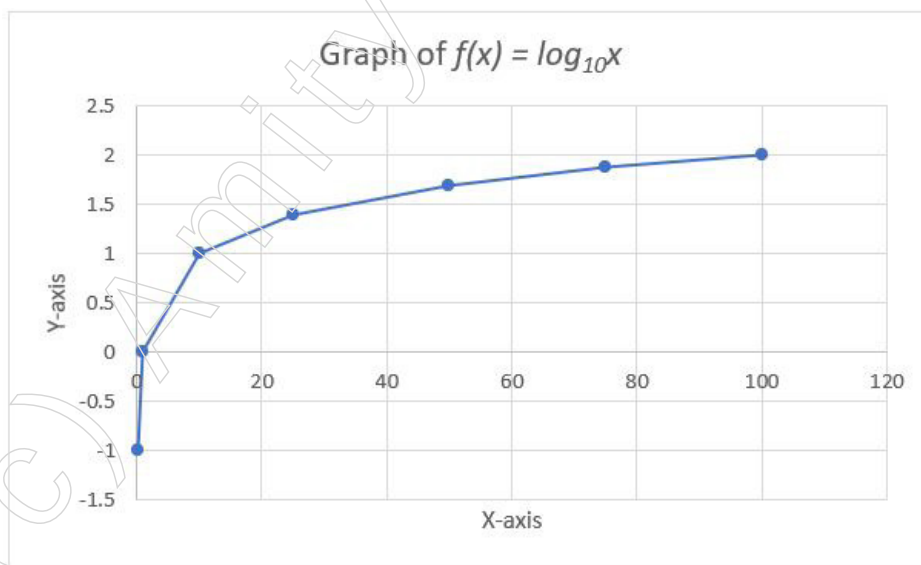


Fig. 3.1.7 The graph of the function

Notes

Notes

Modulus Function: Function $y = f(x) = |x|$ is known as modulus function.

Ex. 3.7 The graph of the function $f(x) = |x|$

Table for function at the different value of x .

x	-5	-4	-3	-2	-1	0	1	2	3	4	5
$f(x) = x $	5	4	3	2	1	0	1	2	3	4	5

Now we can draw a graph according to the obtained data from the above table

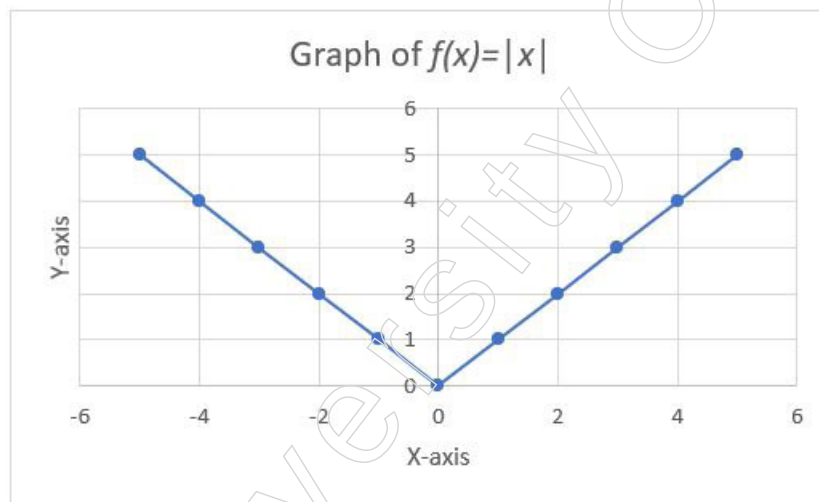


Fig. 3.1.8 The graph of function $f(x) = |x|$

Inverse Function: Let f be defined as a function from X to Y such that for every element of Y there exists an image. Let y be an arbitrary element of Y . Then, f being onto, there exists an element $x \in X$ such that $f(x) = y$. Also, f being one-one, this x must be unique. Thus, for each $y \in Y$, there exists a unique element $x \in X$ such that $f(x) = y$. So, we may define a function,

$$f^{-1}: Y \rightarrow X$$

$$\therefore f^{-1}(y) = x \Leftrightarrow f(x) = y$$

The above function f^{-1} is called the inverse of f .

Ex. 3.8 Suppose we have a function $y = f(x) = x + 3$ and

Table for function and inverse of the given function at the different value of x and y respectively.

We have given $y = f(x) = x + 3$

So, to find the inverse of this function, we need to find the value of x in terms of y .

So, $f^{-1}(x) = y - 3$

X	0	1	2	3	4	5
$f(x)$	3	4	5	6	7	8
Y	0	1	2	3	4	5
$f^{-1}(x)$	-3	-2	-1	0	1	2

Now we can draw a graph according to the obtain data from the above table.

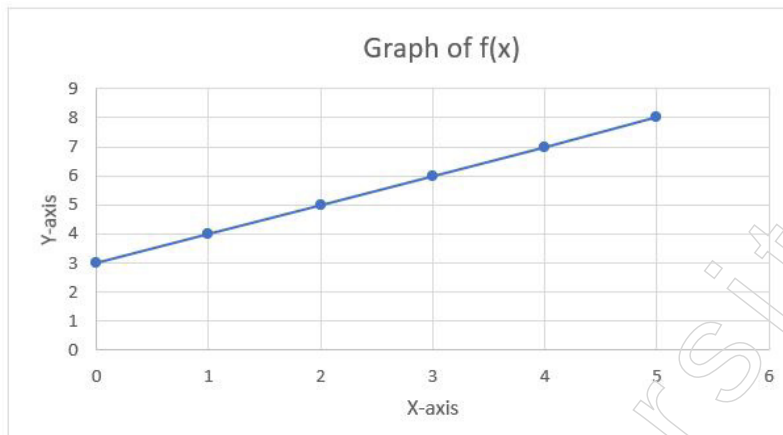


Fig. 3.1.9 The graph of function $f(x) = x + 3$

We can now draw a graph of the inverse, of the above function according to the obtained data from the above table.

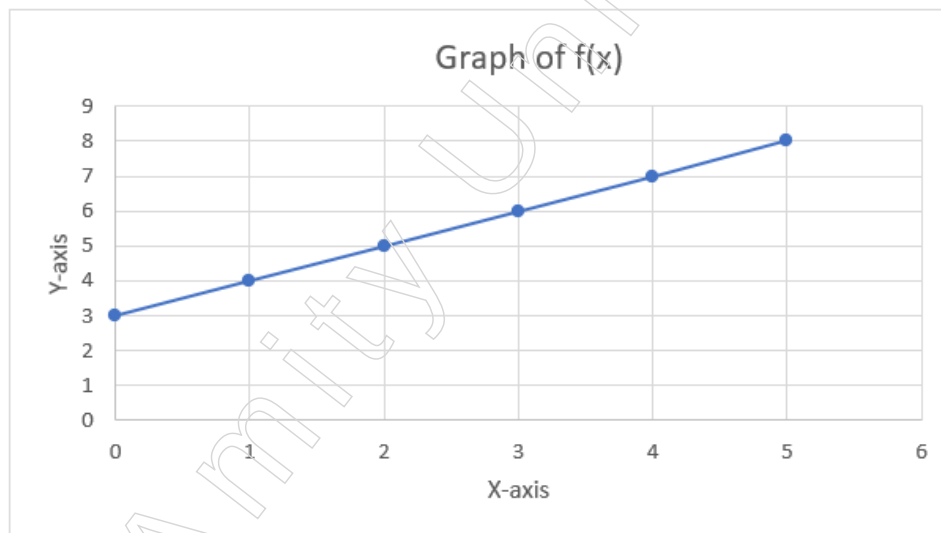


Fig. 3.1.10 Graph of $f^{-1}(x)$

It is clear from the above figure that the points of the line of the two graphs are the image of each other.

Notes

Notes

3.3.2 Limit of a Function – Definition and Methods

Concepts of limits and continuity

If $f(x)$ is a function of x such that, if x approaches to a constant value 'a', then the value of $f(x)$ also approaches to another constant k , then constant k is known as limit of $f(x)$ at $x=a$. Limit is defined as

$$\lim_{x \rightarrow a} f(x) = k$$

Or

The theoretical explanation and diagram does not match correctly. Please discuss individual concepts with different diagram i.e. LHL, RHL and existence of limit, using the exactly same notations as in theory, to specify where they exist in diagram.

A real number b is called the limit of the function f , if for every $\varepsilon > 0$, however small, there exists $\delta > 0$ such that $|f(x) - b| < \varepsilon$, whenever $0 < |x - a| < \delta$

$$\lim_{x \rightarrow a} f(x) = b$$

Left Hand Limit

A function $f(x)$ is said to approach 'b' as x approaches 'a' from the left, if corresponding to an arbitrary positive number ε , there exists a positive number δ such that $|f(x) - b| < \varepsilon$, whenever $a - \delta < x < a$.

It is written as $\lim_{x \rightarrow a^-} f(x) = b$ or $f(a-0) = b$

The working rule for finding the left hand limit is put $a-h$ for x in $f(x)$, where h is positive and very small and make h approach zero.

i.e.,

$$f(a-0) = \lim_{h \rightarrow 0} f(a-h)$$

Right Hand Limit

A function $f(x)$ is said to approach 'b' as x approaches 'a' from the right, if corresponding to an arbitrary positive number ε , there exists a positive number δ such that $|f(x) - b| < \varepsilon$, whenever $a < x < a + \delta$.

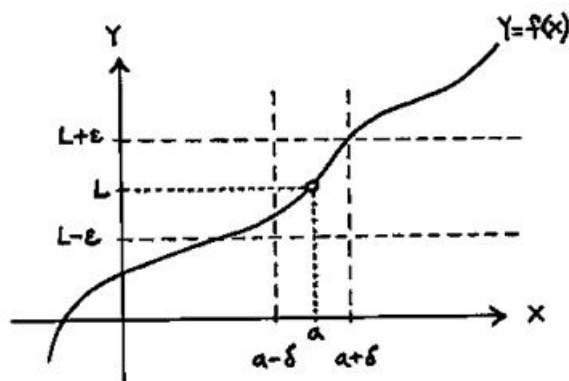
It is written as $\lim_{x \rightarrow a^+} f(x) = b$ or $f(a+0) = b$.

The working rule for finding the right-hand limit is put $a+h$ for x in $f(x)$, where h is positive and very small and make h approach zero.

i.e.,

$$f(a+0) = \lim_{h \rightarrow 0} f(a+h)$$

The right and left limits are shown in the diagram below. The region to the right of a shows the right-hand limit and the region to the left of a shows the left-hand limit.



Existence of Limit

If both right hand limit and left hand limit exist and are equal, then their common value, evidently will be the limit of f as $x \rightarrow a$ i.e.,

$$\text{If } \lim_{x \rightarrow a^+} f(x) = \lim_{x \rightarrow a^-} f(x) = b, \text{ then } \lim_{x \rightarrow a} f(x) = b,$$

If however, either both of these limits do not exist or both these limits exist but are not equal in value, then $\lim_{x \rightarrow a} f(x)$ does not exist.

Algebra of Limits

Let f and g be two functions such that both $\lim_{x \rightarrow a} f(x)$ and $\lim_{x \rightarrow a} g(x)$ exist. Then,

Limit of sum of two functions is sum of limits of the functions, i.e.,

$$\lim_{x \rightarrow a} [f(x) + g(x)] = \lim_{x \rightarrow a} f(x) + \lim_{x \rightarrow a} g(x)$$

Limit of difference of two functions is difference of the limits of the functions, i.e.,

$$\lim_{x \rightarrow a} [f(x) - g(x)] = \lim_{x \rightarrow a} f(x) - \lim_{x \rightarrow a} g(x)$$

Limit of product of two functions is product of the limits of the functions, i.e.,

$$\lim_{x \rightarrow a} [f(x) \cdot g(x)] = \lim_{x \rightarrow a} f(x) \cdot \lim_{x \rightarrow a} g(x)$$

Limit of a quotient of two functions is quotient of the limits of the functions (whenever denominator is non-zero), i.e.,

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{\lim_{x \rightarrow a} f(x)}{\lim_{x \rightarrow a} g(x)}$$

Ex. 3.9: Find the value of $\lim_{x \rightarrow 3} (x^4 + x^3 - 3x)$.

Solution:

$$\begin{aligned} \lim_{x \rightarrow 3} (x^4 + x^3 - 3x) &= \lim_{x \rightarrow 3} x^4 + \lim_{x \rightarrow 3} x^3 - 3 \lim_{x \rightarrow 3} x \\ &= 3^4 + 3^3 - 9 \\ &= 81 + 27 - 9 \\ &= 99 \end{aligned}$$

Notes

Ex. 3.10: Find the value of $\lim_{x \rightarrow 4} \frac{4x+3}{x-2}$.

Solution: Taking $x = 4$

$$\begin{aligned}\lim_{x \rightarrow 4} \frac{4x+3}{x-2} &= \frac{4 \times 4 + 3}{4 - 2} \\ &= \frac{19}{2}\end{aligned}$$

Ex. 3.11: Find the value of $\lim_{x \rightarrow 4} \frac{|x-4|}{x-4}$.

Solution: Let $f(x) = \lim_{x \rightarrow 4} \frac{|x-4|}{x-4}$

At $x = 4$,

$$\begin{aligned}RHL &= \lim_{x \rightarrow 4^+} f(x) \\ &= \lim_{h \rightarrow 0} f(4+h) \\ &= \lim_{h \rightarrow 0} \frac{|4+h-4|}{4+h-4} \\ &= \lim_{h \rightarrow 0} \frac{(4+h-4)}{4+h-4} \\ &= 1\end{aligned}$$

At $x = 4$,

$$\begin{aligned}LHL &= \lim_{x \rightarrow 4^-} f(x) \\ &= \lim_{h \rightarrow 0} f(4-h) \\ &= \lim_{h \rightarrow 0} \frac{|4-h-4|}{4-h-4} \\ &= \lim_{h \rightarrow 0} \frac{-(4-h-4)}{4-h-4} \\ &= -1\end{aligned}$$

$\therefore RHL \neq LHL$

$\therefore$ At $x = 4$, limit does not exist.

Ex. 3.12: If $\lim_{x \rightarrow 1} \frac{x^4-1}{x-1} = \lim_{x \rightarrow k} \frac{x^3-k^3}{x^2-k^2}$, then find the value of k .

Solution: Given, $\lim_{x \rightarrow 1} \frac{x^4-1}{x-1} = \lim_{x \rightarrow k} \frac{x^3-k^3}{x^2-k^2}$

$$\Rightarrow 4(1)^{4-1} = \frac{3}{2}k^{3-2} \quad \left(\because \lim_{x \rightarrow a} \frac{x^n - a^n}{x - a} \right)$$

$$\Rightarrow 4 = \frac{3}{2}k$$

$$\Rightarrow k = \frac{8}{3}$$

3.3.3 Methods to Solve Limits

Limits An Informal Approach

The two broad areas of calculus known as differential and integral calculus are built on the foundation concept of a limit. In this section our approach to this important concept will be intuitive, concentrating on understanding what a limit is using numerical and graphical examples. In the next section, our approach will be analytical, that is, we will use algebraic methods to compare the value of a limit of a function.

1. Limit of a function informal approach Consider the function

$$f(x) = \frac{16 - x^2}{4 - x}$$

Whose domain is the set of all real numbers except -4 . Although f can not be evaluated at -4 because substituting -4 for x results in the undefined quantity $0/0$, $f(x)$ can be calculated at any number x that is very close to -4 . The two tables

x	-4.1	-4.01	-4.001
$f(x)$	8.1	8.01	8.001

x	-3.9	-3.99	-3.999
$f(x)$	7.9	7.99	7.999

(2)

show that as x approaches -4 from either the left or right, the function values $f(x)$ appear to be approaching 8, in other words, when x is near -4 , $f(x)$ is near 8. To interpret the numerical information in (1) graphically, observe that for every number $x \neq -4$, the function f can be simplified by cancellation.

$$f(x) = \frac{16 - x^2}{4 - x} = \frac{(4 + x)(4 - x)}{4 - x} = 4 + x.$$

As seen in the graph of f is essentially the graph of $y = 4 + x$ with the exception that the graph of f has a hole at the point that corresponds to $x = -4$. For x sufficiently close to -4 , represented by the two arrowheads on the x -axis, the two arrowheads on the y -axis, representing function values $f(x)$, simultaneously get closer and closer to the number 8. Indeed, in view of the numerical results in (2), the arrow heads can be made as close as we like to the number 8. We say 8 is the limit of $f(x)$ as x approaches -4 .

Informal Definition Suppose L denotes a finite number. The notion of $f(x)$ approaching L as x approaches a number a can be defined informally in the following manner.

- If $f(x)$ can be made arbitrarily close to the number L by taking x sufficiently close to but different from the number a , from both the left and right sides of a , then the limit of $f(x)$ as x approaches a is L .

Notation The discussion of the limit concept is facilitated by using a special notation. If we let the arrow symbol $\rightarrow$ represent the word approach, then the symbolism

$x \rightarrow a^-$ indicates that x approaches a number a from the left,

that is, through numbers that are less than a , and

$x \rightarrow a^+$ signifies that x approaches a from the right,

that is, through numbers that are greater than a . Finally, the notation

$x \rightarrow a$ signifies that x approaches a from both sides,

in other words, from the left and the right sides of a on a number line. In the left-hand table in (2) we are letting $x \rightarrow -4^-$ (for example, -4.001 is to the left of -4 on the number line), whereas in the right-hand table $x \rightarrow -4^+$.

One-Sided Limits In general, if a function $f(x)$ can be made arbitrarily close to a number L_1 by taking x sufficiently close to, but not equal to, a number a from the left, then we write

$$f(x) \rightarrow L_1 \text{ as } x \rightarrow a^- \quad \text{or} \quad \lim_{x \rightarrow a^-} f(x) = L_1. \quad (3)$$

Notes

Notes

The number L_1 is said to be the **left-hand limit** of $f(x)$ as x approaches a . Similarly, if $f(x)$ can be made arbitrarily close to a number L_2 by taking x sufficiently close to, but not equal to, a number a from the *right*, then L_2 is the **right-hand limit** of $f(x)$ as x approaches a and we write

$$f(x) \rightarrow L_2 \text{ as } x \rightarrow a^+ \quad \text{or} \quad \lim_{x \rightarrow a^+} f(x) = L_2. \quad (4)$$

The quantities in (3) and (4) are also referred to as **one-sided limits**.

■ **Two-Sided Limits** If both the left-hand limit $\lim_{x \rightarrow a^-} f(x)$ and the right-hand limit $\lim_{x \rightarrow a^+} f(x)$ exist and have a common value L ,

$$\lim_{x \rightarrow a^-} f(x) = L \quad \text{and} \quad \lim_{x \rightarrow a^+} f(x) = L,$$

then we say that L is the **limit** of $f(x)$ as x approaches a and write

$$\lim_{x \rightarrow a} f(x) = L. \quad (5)$$

A limit such as (5) is said to be a **two-sided limit**. See FIGURE 21.2. Since the numerical tables in (2) suggest that

$$f(x) \rightarrow 8 \text{ as } x \rightarrow -4^- \quad \text{and} \quad f(x) \rightarrow 8 \text{ as } x \rightarrow -4^+, \quad (6)$$

we can replace the two symbolic statements in (6) by the statement

$$f(x) \rightarrow 8 \text{ as } x \rightarrow -4 \quad \text{or equivalently} \quad \lim_{x \rightarrow -4} \frac{16 - x^2}{4 + x} = 8. \quad (7)$$

■ **Existence and Nonexistence** Of course a limit (one-sided or two-sided) does not have to exist. But it is important that you keep firmly in mind:

- The existence of a limit of a function f as x approaches a (from one side or from both sides), does not depend on whether f is defined at a but only on whether f is defined for x near the number a .

For example, if the function in (1) is modified in the following manner

$$f(x) = \begin{cases} \frac{16 - x^2}{4 + x}, & x \neq -4 \\ 5, & x = -4, \end{cases}$$

then $f(-4)$ is defined and $f(-4) = 5$, but still $\lim_{x \rightarrow -4} \frac{16 - x^2}{4 + x} = 8$. See FIGURE 21.3. In general, the two-sided limit $\lim_{x \rightarrow a} f(x)$ does not exist

- if either of the one-sided limits $\lim_{x \rightarrow a^-} f(x)$ or $\lim_{x \rightarrow a^+} f(x)$ fails to exist, or
- if $\lim_{x \rightarrow a^-} f(x) = L_1$ and $\lim_{x \rightarrow a^+} f(x) = L_2$, but $L_1 \neq L_2$.

EXAMPLE 1 A Limit That Exists

The graph of the function $f(x) = -x^2 + 2x + 2$ is shown in FIGURE 21.4. As seen from the graph and the accompanying tables, it seems plausible that

$$\lim_{x \rightarrow 4^-} f(x) = -6 \quad \text{and} \quad \lim_{x \rightarrow 4^+} f(x) = -6$$

and consequently $\lim_{x \rightarrow 4} f(x) = -6$.

$x \rightarrow 4^-$	3.9	3.99	3.999
$f(x)$	-5.41000	-5.94010	-5.99400

$x \rightarrow 4^+$	4.1	4.01	4.001
$f(x)$	-6.61000	-6.06010	-6.00600

Note that in Example 1 the given function is certainly defined at 4, but at no time did we substitute $x = 4$ into the function to find the value of $\lim_{x \rightarrow 4} f(x)$.

EXAMPLE 2 A Limit That Exists

The graph of the piecewise-defined function

$$f(x) = \begin{cases} x^2, & x < 2 \\ -x + 6, & x > 2 \end{cases}$$

is given in FIGURE 2.15. Notice that $f(2)$ is not defined, but that is of no consequence when considering $\lim_{x \rightarrow 2} f(x)$. From the graph and the accompanying tables,

$x \rightarrow 2^-$	1.9	1.99	1.999
$f(x)$	3.61000	3.96010	3.99600

$x \rightarrow 2^+$	2.1	2.01	2.001
$f(x)$	3.90000	3.99000	3.99900

we see that when we make x close to 2, we can make $f(x)$ arbitrarily close to 4, and so

$$\lim_{x \rightarrow 2^-} f(x) = 4 \quad \text{and} \quad \lim_{x \rightarrow 2^+} f(x) = 4.$$

That is, $\lim_{x \rightarrow 2} f(x) = 4$. ■

EXAMPLE 3 A Limit That Does Not Exist

The graph of the piecewise-defined function

$$f(x) = \begin{cases} x + 2, & x \leq 5 \\ -x + 10, & x > 5 \end{cases}$$

is given in FIGURE 2.16. From the graph and the accompanying tables, it appears that as x approaches 5 through numbers less than 5 that $\lim_{x \rightarrow 5^-} f(x) = 7$. Then as x approaches 5 through numbers greater than 5 it appears that $\lim_{x \rightarrow 5^+} f(x) = 5$. But since

$$\lim_{x \rightarrow 5^-} f(x) \neq \lim_{x \rightarrow 5^+} f(x),$$

we conclude that $\lim_{x \rightarrow 5} f(x)$ does not exist.

$x \rightarrow 5^-$	4.9	4.99	4.999
$f(x)$	6.90000	6.99000	6.99900

$x \rightarrow 5^+$	5.1	5.01	5.001
$f(x)$	4.90000	4.99000	4.99900

EXAMPLE 4 A Limit That Does Not Exist

- Recall, the greatest integer function or floor function $f(x) = \lfloor x \rfloor$ is defined to be the greatest integer that is less than or equal to x . The domain of f is the set of real numbers $(-\infty, \infty)$. From the graph in FIGURE 2.17 we see that $f(n)$ is defined for every integer n ; nonetheless, for each integer n , $\lim_{x \rightarrow n} f(x)$ does not exist. For example, as x approaches, say, the number 3, the two one-sided limits exist but have different values:

$$\lim_{x \rightarrow 3^-} f(x) = 2 \quad \text{whereas} \quad \lim_{x \rightarrow 3^+} f(x) = 3, \quad (8)$$

In general, for an integer n ,

$$\lim_{x \rightarrow n^-} f(x) = n - 1 \quad \text{whereas} \quad \lim_{x \rightarrow n^+} f(x) = n.$$

EXAMPLE 5 A Right-Hand Limit

From FIGURE 2.18 it should be clear that $f(x) = \sqrt{x} \rightarrow 0$ as $x \rightarrow 0^+$, that is

$$\lim_{x \rightarrow 0^+} \sqrt{x} = 0.$$

It would be incorrect to write $\lim_{x \rightarrow 0} \sqrt{x} = 0$ since this notation carries with it the connotation that the limits from the left and from the right exist and are equal to 0. In this case $\lim_{x \rightarrow 0} \sqrt{x}$ does not exist since $f(x) = \sqrt{x}$ is not defined for $x < 0$. ■

Notes

Notes

If $x = a$ is a vertical asymptote for the graph of $y = f(x)$, then $\lim_{x \rightarrow a} f(x)$ will always fail to exist because the function values $f(x)$ must become unbounded from at least one side of the line $x = a$.

EXAMPLE 6 A Limit That Does Not Exist

A vertical asymptote always corresponds to an infinite break in the graph of a function f . In FIGURE 2.1.9 we see that the y -axis or $x = 0$ is a vertical asymptote for the graph of $f(x) = 1/x$. The tables

$x \rightarrow 0^-$	-0.1	-0.01	-0.001
$f(x)$	-10	-100	-1000

$x \rightarrow 0^+$	0.1	0.01	0.001
$f(x)$	10	100	1000

clearly show that the function values $f(x)$ become unbounded in absolute value as we get close to 0. In other words, $f(x)$ is not approaching a real number as $x \rightarrow 0^-$ nor as $x \rightarrow 0^+$. Therefore, neither the left-hand nor the right-hand limit exists as x approaches 0. Thus we conclude that $\lim_{x \rightarrow 0} f(x)$ does not exist. ■

EXAMPLE 7 An Important Trigonometric Limit

To do the calculus of the trigonometric functions $\sin x$, $\cos x$, $\tan x$, and so on, it is important to realize that the variable x is either a real number or an angle measured in radians. With that in mind, consider the numerical values of $f(x) = (\sin x)/x$ as $x \rightarrow 0^+$ given in the table that follows.

$x \rightarrow 0^+$	0.1	0.01	0.001	0.0001
$f(x)$	0.99833416	0.99998333	0.99999983	0.99999999

It is easy to see that the same results given in the table hold as $x \rightarrow 0^-$. Because $\sin x$ is an odd function, for $x > 0$ and $-x < 0$ we have $\sin(-x) = -\sin x$ and as a consequence

$$f(-x) = \frac{\sin(-x)}{-x} = \frac{\sin x}{x} = f(x).$$

As can be seen in FIGURE 2.1.10, f is an even function. The table of numerical values as well as the graph of f strongly suggest the following result:

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1. \quad (9) \quad \blacksquare$$

The limit in (9) is a very important result and will be used in Section 3.4. Another trigonometric limit that you are asked to verify as an exercise is given by

$$\lim_{x \rightarrow 0} \frac{1 - \cos x}{x} = 0. \quad (10)$$

See Problem 43 in Exercises 2.1. Because of their importance, both (9) and (10) will be proven in Section 2.4.

An Indeterminate Form A limit of a quotient $f(x)/g(x)$, where both the numerator and the denominator approach 0 as $x \rightarrow a$, is said to have the **indeterminate form** $0/0$. The limit (7) in our initial discussion has this indeterminate form. Many important limits, such as (9) and (10), and the limit

Which form the backbone of differential calculus, also have the indeterminate form $0/0$.

EXAMPLE 8 An Indeterminate Form

The limit $\lim_{x \rightarrow 0} |x|/x$ has the indeterminate form $0/0$, but unlike (7), (9), and (10) this limit fails to exist. To see why, let us examine the graph of the function $f(x) = |x|/x$. For $x \neq 0$, $|x| = \begin{cases} x, & x > 0 \\ -x, & x < 0 \end{cases}$ and so we recognize f as the piecewise-defined function

$$f(x) = \frac{|x|}{x} = \begin{cases} 1, & x > 0 \\ -1, & x < 0. \end{cases} \quad (11)$$

From (11) and the graph of f in FIGURE 2.1.11 it should be apparent that both the left-hand and right-hand limits of f exist and

$$\lim_{x \rightarrow 0^-} \frac{|x|}{x} = -1 \quad \text{and} \quad \lim_{x \rightarrow 0^+} \frac{|x|}{x} = 1.$$

Because these one-sided limits are different, we conclude that $\lim_{x \rightarrow 0} |x|/x$ does not exist. ■

Limit Theorems

Introduction: the intention of the informal discussion in Section 2.1 was to give you an intuitive grasp of when a limit does or does not exist. However, it is neither desirable nor practical, in every instance, to reach a conclusion about the existence of a limit based on a graph or on a table of numerical values. We must be able to evaluate a limit, or discern its non-existence, in a somewhat mechanical fashion. The theorems that we shall consider in this section establish such a means. The proofs of some of these results are given in the appendix.

The first theorem gives two basic results that will be used throughout the discussion of this section.

Theorem 2.2.1 Two Fundamental Limits

- (i) $\lim_{x \rightarrow a} c = c$, where c is a constant
- (ii) $\lim_{x \rightarrow a} x = a$

Although both parts of Theorem 2.2.1 require a formal proof, Theorem 2.2.1(ii) is almost tautological when stated in words:

- The limit of x as x is approaching a is a .

See the Appendix for a proof of Theorem 2.2.1(i).

EXAMPLE 1 Using Theorem 2.2.1

- (a) From Theorem 2.2.1(i),

$$\lim_{x \rightarrow 2} 10 = 10 \quad \text{and} \quad \lim_{x \rightarrow 0} \pi = \pi.$$

- (b) From Theorem 2.2.1(ii),

$$\lim_{x \rightarrow 2} x = 2 \quad \text{and} \quad \lim_{x \rightarrow 0} x = 0. \quad \blacksquare$$

The limit of a constant multiple of a function f is the constant times the limit of f as x approaches a number a .

Notes

Theorem 2.2.2 Limit of a Constant Multiple

If c is a constant, then

$$\lim_{x \rightarrow a} c f(x) = c \lim_{x \rightarrow a} f(x).$$

We can now start using theorems in conjunction with each other.

EXAMPLE 2 Using Theorems 2.2.1 and 2.2.2

From Theorems 2.2.1 (ii) and 2.2.2,

$$(a) \lim_{x \rightarrow 8} 5x = 5 \lim_{x \rightarrow 8} x = 5 \cdot 8 = 40$$

$$(b) \lim_{x \rightarrow -2} \left(-\frac{1}{2}x\right) = -\frac{1}{2} \lim_{x \rightarrow -2} x = \left(-\frac{1}{2}\right) \cdot (-2) = 3.$$

The next theorem is particularly important because it gives us a way of computing limits in an algebraic manner.

Theorem 2.2.3 Limit of a Sum, Product, and Quotient

Suppose a is a real number and $\lim_{x \rightarrow a} f(x)$ and $\lim_{x \rightarrow a} g(x)$ exist. If $\lim_{x \rightarrow a} f(x) = L_1$ and $\lim_{x \rightarrow a} g(x) = L_2$, then

$$(i) \lim_{x \rightarrow a} [f(x) \pm g(x)] = \lim_{x \rightarrow a} f(x) \pm \lim_{x \rightarrow a} g(x) = L_1 \pm L_2,$$

$$(ii) \lim_{x \rightarrow a} [f(x)g(x)] = \left(\lim_{x \rightarrow a} f(x)\right)\left(\lim_{x \rightarrow a} g(x)\right) = L_1L_2, \text{ and}$$

$$(iii) \lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{\lim_{x \rightarrow a} f(x)}{\lim_{x \rightarrow a} g(x)} = \frac{L_1}{L_2}, L_2 \neq 0.$$

Theorem 2.2.3 can be stated in words:

- If both limits exist, then
 - (i) the limit of a sum is the sum of the limits,
 - (ii) the limit of a product is the product of the limits, and
 - (iii) the limit of a quotient is the quotient of the limits provided the limit of the denominator is not zero.

Note: If all limits exist, then Theorem 2.2.3 is also applicable to one-sided limits, that is, the symbolism $x \rightarrow a$ in Theorem 2.2.3 can be replaced by either $x \rightarrow a^-$ or $x \rightarrow a^+$. Moreover, Theorem 2.2.3 extends to differences, sums, products, and quotients that involve more than two functions. See the Appendix for a proof of Theorem 2.2.3.

EXAMPLE 3 Using Theorem 2.2.3

Evaluate $\lim_{x \rightarrow 5} (10x + 7)$.

Solution From Theorems 2.2.1 and 2.2.2, we know that $\lim_{x \rightarrow 5} 7$ and $\lim_{x \rightarrow 5} 10x$ exist. Hence, from Theorem 2.2.3(i),

$$\begin{aligned}
 \lim_{x \rightarrow 5} (10x + 7) &= \lim_{x \rightarrow 5} 10x + \lim_{x \rightarrow 5} 7 \\
 &= 10 \lim_{x \rightarrow 5} x + \lim_{x \rightarrow 5} 7 \\
 &= 10 \cdot 5 + 7 = 57.
 \end{aligned}$$

Limit of a Power Theorem 2.2.3(ii) can be used to calculate the limit of a positive integer power of a function. For example, if $\lim_{x \rightarrow a} f(x) = L$, then from Theorem 2.2.3(ii) with $g(x) = f(x)$,

$$\lim_{x \rightarrow a} [f(x)]^2 = \lim_{x \rightarrow a} [f(x) \cdot f(x)] = \left(\lim_{x \rightarrow a} f(x) \right) \left(\lim_{x \rightarrow a} f(x) \right) = L^2.$$

By the same reasoning we can apply Theorem 2.2.3(ii) to the general case where $f(x)$ is a factor n times. This result is stated as the next theorem.

Theorem 2.2.4 Limit of a Power

Let $\lim_{x \rightarrow a} f(x) = L$ and n be a positive integer. Then

$$\lim_{x \rightarrow a} [f(x)]^n = \left[\lim_{x \rightarrow a} f(x) \right]^n = L^n.$$

For the special case $f(x) = x$, the result given in Theorem 2.2.4 yields

$$\lim_{x \rightarrow a} x^n = a^n. \quad (1)$$

EXAMPLE 4 Using (1) and Theorem 2.2.3

Evaluate

(a) $\lim_{x \rightarrow 10} x^3$ (b) $\lim_{x \rightarrow 4} \frac{5}{x^2}.$

Solution

(a) From (1),

$$\lim_{x \rightarrow 10} x^3 = 10^3 = 1000.$$

(b) From Theorem 2.2.1 and (1) we know that $\lim_{x \rightarrow 4} 5 = 5$ and $\lim_{x \rightarrow 4} x^2 = 16 \neq 0$. Therefore by Theorem 2.2.3(iii),

$$\lim_{x \rightarrow 4} \frac{5}{x^2} = \frac{\lim_{x \rightarrow 4} 5}{\lim_{x \rightarrow 4} x^2} = \frac{5}{4^2} = \frac{5}{16}.$$

EXAMPLE 5 Using Theorem 2.2.3

Evaluate $\lim_{x \rightarrow 3} (x^2 - 5x + 6).$

Solution In view of Theorem 2.2.1, Theorem 2.2.2, and (1) all limits exist. Therefore by Theorem 2.2.3(i),

$$\lim_{x \rightarrow 3} (x^2 - 5x + 6) = \lim_{x \rightarrow 3} x^2 - \lim_{x \rightarrow 3} 5x + \lim_{x \rightarrow 3} 6 = 3^2 - 5 \cdot 3 + 6 = 0.$$

Notes

EXAMPLE 6 Using Theorems 2.2.3 and 2.2.4

Evaluate $\lim_{x \rightarrow 1} (3x - 1)^{10}$.

Solution First, we see from Theorem 2.2.3(i) that

$$\lim_{x \rightarrow 1} (3x - 1) = \lim_{x \rightarrow 1} 3x - \lim_{x \rightarrow 1} 1 = 2.$$

It then follows from Theorem 2.2.4 that

$$\lim_{x \rightarrow 1} (3x - 1)^{10} = \left[\lim_{x \rightarrow 1} (3x - 1) \right]^{10} = 2^{10} = 1024. \quad \blacksquare$$

■ **Limit of a Polynomial Function** Some limits can be evaluated by *direct substitution*. We can use (1) and Theorem 2.2.3(i) to compute the limit of a general polynomial function. If

$$f(x) = c_n x^n + c_{n-1} x^{n-1} + \cdots + c_1 x + c_0$$

is a polynomial function, then

$$\begin{aligned} \lim_{x \rightarrow a} f(x) &= \lim_{x \rightarrow a} (c_n x^n + c_{n-1} x^{n-1} + \cdots + c_1 x + c_0) \\ &= \lim_{x \rightarrow a} c_n x^n + \lim_{x \rightarrow a} c_{n-1} x^{n-1} + \cdots + \lim_{x \rightarrow a} c_1 x + \lim_{x \rightarrow a} c_0 \\ &= c_n a^n + c_{n-1} a^{n-1} + \cdots + c_1 a + c_0 \quad \leftarrow f \text{ is defined at } x = a \text{ and this limit is } f(a) \end{aligned}$$

In other words, to evaluate a limit of a polynomial function f as x approaches a real number a , we need only evaluate the function at $x = a$:

$$\lim_{x \rightarrow a} f(x) = f(a). \quad (2)$$

A reexamination of Example 5 shows that $\lim_{x \rightarrow 3} f(x)$, where $f(x) = x^2 - 5x + 6$, is given by $f(3) = 0$.

Because a rational function f is a quotient of two polynomials $p(x)$ and $q(x)$, it follows from (2) and Theorem 2.2.3(ii) that a limit of a rational function $f(x) = p(x)/q(x)$ can also be found by evaluating f at $x = a$:

$$\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} \frac{p(x)}{q(x)} = \frac{p(a)}{q(a)}. \quad (3)$$

Of course we must add to (3) the all-important requirement that the limit of the denominator is not 0, that is, $q(a) \neq 0$.

EXAMPLE 7 Using (2) and (3)

Evaluate $\lim_{x \rightarrow -1} \frac{3x - 4}{8x^2 + 2x - 2}$.

Solution $f(x) = \frac{3x - 4}{8x^2 + 2x - 2}$ is a rational function and so if we identify the polynomials $p(x) = 3x - 4$ and $q(x) = 8x^2 + 2x - 2$, then from (2),

$$\lim_{x \rightarrow -1} p(x) = p(-1) = -7 \quad \text{and} \quad \lim_{x \rightarrow -1} q(x) = q(-1) = 4.$$

Since $q(-1) \neq 0$ it follows from (3) that

$$\lim_{x \rightarrow -1} \frac{3x - 4}{8x^2 + 2x - 2} = \frac{p(-1)}{q(-1)} = \frac{-7}{4} = -\frac{7}{4}. \quad \blacksquare$$

You should not get the impression that we can *always* find a limit of a function by substituting the number a *directly into the function*.

EXAMPLE 8 Using Theorem 2.2.3

Evaluate $\lim_{x \rightarrow 1} \frac{x-1}{x^2+x-2}$.

Solution The function in this limit is rational, but if we substitute $x = 1$ into the function we see that this limit has the indeterminate form $0/0$. However, by simplifying *first*, we can then apply Theorem 2.2.3(iii):

$$\begin{aligned}\lim_{x \rightarrow 1} \frac{x-1}{x^2+x-2} &= \lim_{x \rightarrow 1} \frac{x-1}{(x-1)(x+2)} \quad \leftarrow \text{cancellation is valid provided that } x \neq 1 \\ &= \lim_{x \rightarrow 1} \frac{1}{x+2} \\ &= \frac{\lim_{x \rightarrow 1} 1}{\lim_{x \rightarrow 1} (x+2)} = \frac{1}{3}.\end{aligned}$$

Sometimes you can tell at a glance when a limit does not exist.

Theorem 2.2.5 A Limit That Does Not Exist

Let $\lim_{x \rightarrow a} f(x) = L_1 \neq 0$ and $\lim_{x \rightarrow a} g(x) = 0$. Then

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)}$$

does not exist.

PROOF We will give an indirect proof of this result based on Theorem 2.2.3. Suppose $\lim_{x \rightarrow a} f(x) = L_1 \neq 0$ and $\lim_{x \rightarrow a} g(x) = 0$ and suppose further that $\lim_{x \rightarrow a} (f(x)/g(x))$ exists and equals L_2 . Then

$$\begin{aligned}L_1 &= \lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} \left(g(x) \cdot \frac{f(x)}{g(x)} \right), \quad g(x) \neq 0, \\ &= \left(\lim_{x \rightarrow a} g(x) \right) \left(\lim_{x \rightarrow a} \frac{f(x)}{g(x)} \right) = 0 \cdot L_2 = 0.\end{aligned}$$

By contradicting the assumption that $L_1 \neq 0$, we have proved the theorem. ■

EXAMPLE 9 Using Theorems 2.2.3 and 2.2.5

Evaluate

(a) $\lim_{x \rightarrow 5} \frac{x}{x-5}$

(b) $\lim_{x \rightarrow 5} \frac{x^2 - 10x + 25}{x^2 - 4x - 5}$

(c) $\lim_{x \rightarrow 5} \frac{x-5}{x^2 - 10x + 25}$

Solution Each function in the three parts of the example is rational.

(a) Since the limit of the numerator x is 5, but the limit of the denominator $x - 5$ is 0, we conclude from Theorem 2.2.5 that the limit does not exist.

(b) Substituting $x = 5$ makes both the numerator and denominator 0, and so the limit has the indeterminate form $0/0$. By the Factor Theorem of algebra, $x - 5$ is a factor of both the numerator and denominator. Hence,

Notes

$$\begin{aligned}\lim_{x \rightarrow 5} \frac{x^2 - 10x + 25}{x^2 - 4x - 5} &= \lim_{x \rightarrow 5} \frac{(x-5)^2}{(x-5)(x+1)} \quad \leftarrow \text{cancel the factor } x-5 \\ &= \lim_{x \rightarrow 5} \frac{x-5}{x+1} \\ &= \frac{0}{6} = 0. \quad \leftarrow \text{limit exists}\end{aligned}$$

(c) Again, the limit has the indeterminate form $0/0$. After factoring the denominator and canceling the factors we see from the algebra

$$\begin{aligned}\lim_{x \rightarrow 5} \frac{x-5}{x^2 - 10x + 25} &= \lim_{x \rightarrow 5} \frac{x-5}{(x-5)^2} \\ &= \lim_{x \rightarrow 5} \frac{1}{x-5}\end{aligned}$$

that the limit does not exist since the limit of the numerator in the last expression is now 1 but the limit of the denominator is 0. ■

■ **Limit of a Root** The limit of the n th root of a function is the n th root of the limit whenever the limit exists and has a real n th root. The next theorem summarizes this fact.

Theorem 2.2.6 Limit of a Root

Let $\lim_{x \rightarrow a} f(x) = L$ and n be a positive integer. Then

$$\lim_{x \rightarrow a} \sqrt[n]{f(x)} = \sqrt[n]{\lim_{x \rightarrow a} f(x)} = \sqrt[n]{L},$$

provided that $L \geq 0$ when n is even.

An immediate special case of Theorem 2.2.6 is

$$\lim_{x \rightarrow a} \sqrt[n]{x} = \sqrt[n]{a}, \quad (4)$$

provided $a \geq 0$ when n is even. For example, $\lim_{x \rightarrow 9} \sqrt{x} = [\lim_{x \rightarrow 9} x]^{1/2} = 9^{1/2} = 3$.

EXAMPLE 10 Using (4) and Theorem 2.2.3

Evaluate $\lim_{x \rightarrow -8} \frac{x - \sqrt[3]{x}}{2x + 10}$.

Solution Since $\lim_{x \rightarrow -8} (2x + 10) = -6 \neq 0$, we see from Theorem 2.2.3(iii) and (4) that

$$\lim_{x \rightarrow -8} \frac{x - \sqrt[3]{x}}{2x + 10} = \frac{\lim_{x \rightarrow -8} x - [\lim_{x \rightarrow -8} x]^{1/3}}{\lim_{x \rightarrow -8} (2x + 10)} = \frac{-8 - (-8)^{1/3}}{-6} = \frac{-6}{-6} = 1. \quad \blacksquare$$

When a limit of an algebraic function involving radicals has the indeterminate form $0/0$, rationalization of the numerator or the denominator may be something to try.

EXAMPLE 11 Rationalization of a Numerator

Evaluate $\lim_{x \rightarrow 0} \frac{\sqrt{x^2 + 4} - 2}{x^2}$.

Solution Because $\lim_{x \rightarrow 0} \sqrt{x^2 + 4} = \sqrt{\lim_{x \rightarrow 0} (x^2 + 4)} = 2$ we see by inspection that the given limit has the indeterminate form $0/0$. However, by rationalization of the numerator we obtain

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{\sqrt{x^2 + 4} - 2}{x^2} &= \lim_{x \rightarrow 0} \frac{\sqrt{x^2 + 4} - 2}{x^2} \cdot \frac{\sqrt{x^2 + 4} + 2}{\sqrt{x^2 + 4} + 2} \\ &= \lim_{x \rightarrow 0} \frac{(x^2 + 4) - 4}{x^2(\sqrt{x^2 + 4} + 2)} \\ &= \lim_{x \rightarrow 0} \frac{x^2}{x^2(\sqrt{x^2 + 4} + 2)} \quad \leftarrow \text{cancel } x^2 \\ &= \lim_{x \rightarrow 0} \frac{1}{\sqrt{x^2 + 4} + 2} \quad \leftarrow \text{this limit is no longer } 0/0 \end{aligned}$$

We are now in a position to use Theorems 2.2.3 and 2.2.6:

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{\sqrt{x^2 + 4} - 2}{x^2} &= \lim_{x \rightarrow 0} \frac{1}{\sqrt{x^2 + 4} + 2} \\ &= \frac{\lim_{x \rightarrow 0} 1}{\sqrt{\lim_{x \rightarrow 0} (x^2 + 4)} + \lim_{x \rightarrow 0} 2} \\ &= \frac{1}{2 + 2} = \frac{1}{4}. \quad \blacksquare \end{aligned}$$

In case anyone is wondering whether there can be more than one limit of a function $f(x)$ as $x \rightarrow a$, we state the last theorem for the record.

Theorem 2.2.7 Existence Implies Uniqueness

If $\lim_{x \rightarrow a} f(x)$ exists, then it is unique.

Limits A Formal Approach

Introduction: In the discussion that follows we will consider an alternative approach to the notion of a limit that is based on analytical concepts rather than on intuitive concepts. A proof of the existence of a limit can never be based on one's ability to sketch graphs or on tables of numerical values. Although a good intuitive understanding of $\lim_{x \rightarrow a} f(x)$ is sufficient for proceeding with the study of the calculus in this text, an intuitive understanding is admittedly too vague to be any use in proving theorems. To give a rigorous demonstration of the existence of a limit, or to prove the important theorems of Section 2.2, we must start with a precise definition of a limit.

■ Limit of a Function Let us try to prove that $\lim_{x \rightarrow 2} (2x + 6) = 10$ by elaborating on the following idea: "If $f(x) = 2x + 6$ can be made arbitrarily close to 10 by taking x sufficiently close to 2, from either side but different from 2, then $\lim_{x \rightarrow 2} f(x) = 10$." We need to make the concepts of *arbitrarily close* and *sufficiently close* precise. In order to set a standard of arbitrary closeness, let us demand that the distance between the numbers $f(x)$ and 10 be less than 0.1; that is,

$$|f(x) - 10| < 0.1 \quad \text{or} \quad 9.9 < f(x) < 10.1. \quad (1)$$

Then, how close must x be to 2 to accomplish (1)? To find out, we can use ordinary algebra to rewrite the inequality

Notes

$$9.9 < 2x + 6 < 10.1$$

as $1.95 < x < 2.05$. Adding -2 across this simultaneous inequality then gives

$$-0.05 < x - 2 < 0.05.$$

Using absolute values and remembering that $x \neq 2$, we can write the last inequality as $0 < |x - 2| < 0.05$. Thus, for an "arbitrary closeness to 10" of 0.1, "sufficiently close to 2" means within 0.05. In other words, if x is a number different from 2 such that its distance from 2 satisfies $|x - 2| < 0.05$, then the distance of $f(x)$ from 10 is guaranteed to satisfy $|f(x) - 10| < 0.1$. Expressed in yet another way, when x is a number different from 2 but in the open interval $(1.95, 2.05)$ on the x -axis, then $f(x)$ is in the interval $(9.9, 10.1)$ on the y -axis.

Using the same example, let us try to generalize. Suppose ϵ (the Greek letter *epsilon*) denotes an arbitrary *positive number* that is our measure of arbitrary closeness to the number 10. If we demand that

$$|f(x) - 10| < \epsilon \quad \text{or} \quad 10 - \epsilon < f(x) < 10 + \epsilon, \quad (2)$$

then from $10 - \epsilon < 2x + 6 < 10 + \epsilon$ and algebra, we find

$$2 - \frac{\epsilon}{2} < x < 2 + \frac{\epsilon}{2} \quad \text{or} \quad -\frac{\epsilon}{2} < x - 2 < \frac{\epsilon}{2}. \quad (3)$$

Again using absolute values and remembering that $x \neq 2$, we can write the last inequality in (3) as

$$0 < |x - 2| < \frac{\epsilon}{2}. \quad (4)$$

If we denote $\epsilon/2$ by the new symbol δ (the Greek letter *delta*), (2) and (4) can be written as

$$|f(x) - 10| < \epsilon \quad \text{whenever} \quad 0 < |x - 2| < \delta.$$

Thus, for a new value for ϵ , say $\epsilon = 0.001$, $\delta = \epsilon/2 = 0.0005$ tells us the corresponding closeness to 2. For any number x different from 2 in $(1.9995, 2.0005)$,* we can be sure $f(x)$ is in $(9.999, 10.001)$. See FIGURE 2.6.1.

■ **A Definition** The foregoing discussion leads us to the so-called ϵ δ definition of a limit.

Definition 2.6.1 Definition of a Limit

Suppose a function f is defined everywhere on an open interval, except possibly at a number a in the interval. Then

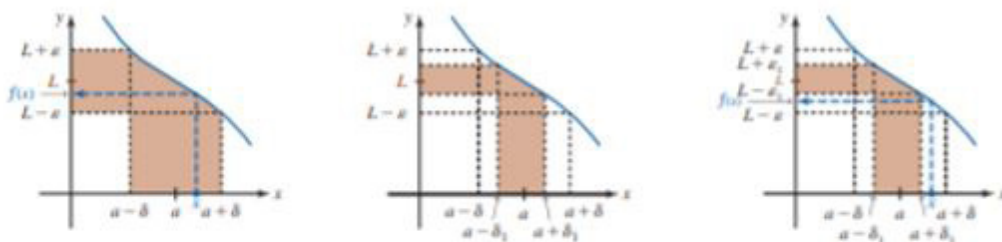
$$\lim_{x \rightarrow a} f(x) = L$$

means that for every $\epsilon > 0$, there exists a number $\delta > 0$ such that

$$|f(x) - L| < \epsilon \quad \text{whenever} \quad 0 < |x - a| < \delta.$$

*For this reason, we use $0 < |x - 2| < \delta$ rather than $|x - 2| < \delta$. Keep in mind when considering $\lim_{x \rightarrow 2} f(x)$, we do not care about f at 2.

Let $\lim_{x \rightarrow a} f(x) = L$ and suppose $\delta > 0$ is the number that "works" in the sense of Definition 2.6.1 for a given $\epsilon > 0$. As shown in FIGURE 2.6.2(a), every x in $(a - \delta, a + \delta)$, with the possible exception of a itself, will then have an image $f(x)$ in $(L - \epsilon, L + \epsilon)$. Furthermore, as in Figure 2.6.2(b), a choice $\delta_1 < \delta$ for the same ϵ also "works" in that every x not equal to a in $(a - \delta_1, a + \delta_1)$ gives $f(x)$ in $(L - \epsilon, L + \epsilon)$. However, Figure 2.6.2(c) shows that choosing a smaller ϵ_1 , $0 < \epsilon_1 < \epsilon$, will demand finding a new value of δ . Observe in Figure 2.6.2(c) that x is in $(a - \delta, a + \delta)$ but not in $(a - \delta_1, a + \delta_1)$, and so $f(x)$ is not necessarily in $(L - \epsilon_1, L + \epsilon_1)$.

(a) A δ that works for a given ϵ (b) A smaller δ_1 will also work for the same ϵ (c) A smaller ϵ_1 will require a $\delta_1 < \delta$. For x in $(a - \delta_1, a + \delta_1)$, $f(x)$ is not necessarily in $(L - \epsilon_1, L + \epsilon_1)$ FIGURE 2.6.2 $f(x)$ is in $(L - \epsilon, L + \epsilon)$ whenever x is in $(a - \delta, a + \delta)$, $x \neq a$ **EXAMPLE 1** Using Definition 2.6.1Prove that $\lim_{x \rightarrow 3} (5x + 2) = 17$.**Solution** For any arbitrary $\epsilon > 0$, regardless how small, we wish to find a δ so that

$$|(5x + 2) - 17| < \epsilon \quad \text{whenever} \quad 0 < |x - 3| < \delta.$$

To do this consider

$$|(5x + 2) - 17| = |5x - 15| = 5|x - 3|.$$

Thus, to make $|(5x + 2) - 17| = 5|x - 3| < \epsilon$, we need only make $0 < |x - 3| < \epsilon/5$; that is, choose $\delta = \epsilon/5$.**Verification** If $0 < |x - 3| < \epsilon/5$, then $5|x - 3| < \epsilon$ implies

$$|5x - 15| < \epsilon \quad \text{or} \quad |(5x + 2) - 17| < \epsilon \quad \text{or} \quad |f(x) - 17| < \epsilon. \quad \blacksquare$$

EXAMPLE 2 Using Definition 2.6.1Prove that $\lim_{x \rightarrow -4} \frac{16 - x^2}{4 + x} = 8$.**Solution** For $x \neq -4$,

$$\left| \frac{16 - x^2}{4 + x} - 8 \right| = |4 - x - 8| = |-x - 4| = |x + 4| = |x - (-4)|$$

$$\text{Thus,} \quad \left| \frac{16 - x^2}{4 + x} - 8 \right| = |x - (-4)| < \epsilon$$

whenever we have $0 < |x - (-4)| < \epsilon$; that is, choose $\delta = \epsilon$. I**EXAMPLE 3** A Limit That Does Not Exist

Consider the function

$$f(x) = \begin{cases} 0, & x \leq 1 \\ 2, & x > 1. \end{cases}$$

We recognize in FIGURE 2.6.3 that f has a jump discontinuity at 1 and so $\lim_{x \rightarrow 1} f(x)$ does not exist. However, to prove this last fact, we shall proceed indirectly. Assume that the limit exists, namely, $\lim_{x \rightarrow 1} f(x) = L$. Then from Definition 2.6.1 we know that for the choice $\epsilon = \frac{1}{2}$ there must exist a $\delta > 0$ so that

$$|f(x) - L| < \frac{1}{2} \quad \text{whenever} \quad 0 < |x - 1| < \delta.$$

Now to the right of 1, let us choose $x = 1 + \delta/2$. Since

$$0 < \left| 1 + \frac{\delta}{2} - 1 \right| = \left| \frac{\delta}{2} \right| < \delta$$

Notes

we must have

$$\left| f\left(1 + \frac{\delta}{2}\right) - L \right| = |2 - L| < \frac{1}{2}. \quad (5)$$

To the left of 1, choose $x = 1 - \delta/2$. But

$$0 < \left| 1 - \frac{\delta}{2} - 1 \right| = \left| -\frac{\delta}{2} \right| < \delta$$

$$\text{implies} \quad \left| f\left(1 - \frac{\delta}{2}\right) - L \right| = |0 - L| = |L| < \frac{1}{2}. \quad (6)$$

Solving the absolute-value inequalities (5) and (6) gives, respectively,

$$\frac{3}{2} < L < \frac{5}{2} \quad \text{and} \quad -\frac{1}{2} < L < \frac{1}{2}.$$

Since no number L can satisfy both of these inequalities, we conclude that $\lim_{x \rightarrow 1} f(x)$ does not exist. ■

In the next example we consider the limit of a quadratic function. We shall see that finding the δ in this case requires a bit more ingenuity than in Examples 1 and 2.

EXAMPLE 4 Using Definition 2.6.1

Prove that $\lim_{x \rightarrow -4} (-x^2 + 2x + 2) = -6$.

Solution For an arbitrary $\varepsilon > 0$ we must find a $\delta > 0$ so that

$$|-x^2 + 2x + 2 - (-6)| < \varepsilon \quad \text{whenever} \quad 0 < |x - 4| < \delta.$$

Now,

$$\begin{aligned} |-x^2 + 2x + 2 - (-6)| &= |(-1)(x^2 - 2x - 8)| \\ &= |(x + 2)(x - 4)| \\ &= |x + 2||x - 4|. \end{aligned} \quad (7)$$

In other words, we want to make $|x + 2||x - 4| < \varepsilon$. But since we have agreed to examine values of x near 4, let us consider only those values for which $|x - 4| < 1$. This last inequality gives $3 < x < 5$ or equivalently $5 < x + 2 < 7$. Consequently we can write $|x + 2| < 7$. Hence from (7),

$$0 < |x - 4| < 1 \quad \text{implies} \quad |-x^2 + 2x + 2 - (-6)| < 7|x - 4|.$$

If we now choose δ to be the minimum of the two numbers, 1 and $\varepsilon/7$, written $\delta = \min\{1, \varepsilon/7\}$ we have

$$0 < |x - 4| < \delta \quad \text{implies} \quad |-x^2 + 2x + 2 - (-6)| < 7|x - 4| < 7 \cdot \frac{\varepsilon}{7} = \varepsilon. \quad \blacksquare$$

The reasoning in Example 4 is subtle. Consequently it is worth a few minutes of your time to reread the discussion immediately following Definition 2.6.1, reexamine

Figure 2.3.2(b), and then think again about why $\delta = \min\{1, \varepsilon/7\}$ is the δ that “works” in the example. Remember, you can pick the ε arbitrarily: think about δ for, say, $\varepsilon = 8$, $\varepsilon = 6$, and $\varepsilon = 0.01$.

■ **One-Sided Limits** We state next the definitions of the **one-sided limits**, $\lim_{x \rightarrow a^+} f(x)$ and $\lim_{x \rightarrow a^-} f(x)$.

Definition 2.6.2 Left-Hand Limit

Suppose a function f is defined on an open interval (c, a) . Then

$$\lim_{x \rightarrow a^-} f(x) = L$$

means for every $\varepsilon > 0$ there exists a $\delta > 0$ such that

$$|f(x) - L| < \varepsilon \quad \text{whenever} \quad a - \delta < x < a.$$

Definition 2.6.3 Right-Hand Limit

Suppose a function f is defined on an open interval (a, c) . Then

$$\lim_{x \rightarrow a^+} f(x) = L$$

means for every $\varepsilon > 0$ there exists a $\delta > 0$ such that

$$|f(x) - L| < \varepsilon \quad \text{whenever} \quad a < x < a + \delta.$$

EXAMPLE 5 Using Definition 2.6.3

Prove that $\lim_{x \rightarrow 0^+} \sqrt{x} = 0$.

Solution First, we can write

$$|\sqrt{x} - 0| = |\sqrt{x}| = \sqrt{x}.$$

Then, $|\sqrt{x} - 0| < \varepsilon$ whenever $0 < x < 0 + \varepsilon^2$. In other words, we choose $\delta = \varepsilon^2$.

Verification If $0 < x < \varepsilon^2$, then $0 < \sqrt{x} < \varepsilon$ implies

$$|\sqrt{x}| < \varepsilon \quad \text{or} \quad |\sqrt{x} - 0| < \varepsilon. \quad \blacksquare$$

Limits Involving Infinity The two concepts of infinite limits

$$f(x) \rightarrow \infty \text{ (or } -\infty) \text{ as } x \rightarrow a$$

and a limit at infinity

$$f(x) \rightarrow L \text{ as } x \rightarrow \infty \text{ (or } -\infty)$$

are formalized in the next two definitions.

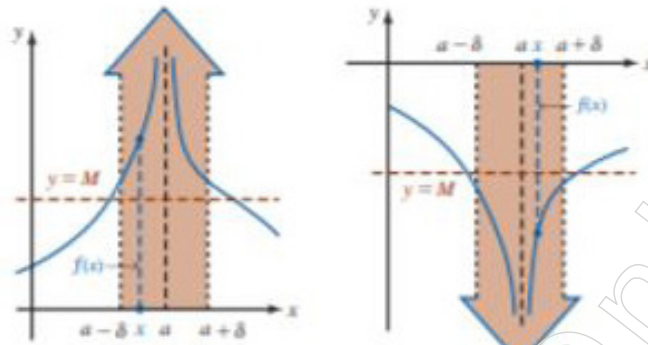
Recall, an infinite limit is a limit that does not exist as $x \rightarrow a$.

Definition 2.6.4 Infinite Limits

- (i) $\lim_{x \rightarrow a} f(x) = \infty$ means for each $M > 0$, there exists a $\delta > 0$ such that $f(x) > M$ whenever $0 < |x - a| < \delta$.
- (ii) $\lim_{x \rightarrow a} f(x) = -\infty$ means for each $M < 0$, there exists a $\delta > 0$ such that $f(x) < M$ whenever $0 < |x - a| < \delta$.

Parts (i) and (ii) of Definition 2.6.4 are illustrated in FIGURE 2.6.4(a) and Figure 2.6.4(b), respectively. Recall, if $f(x) \rightarrow \infty$ (or $-\infty$) as $x \rightarrow a$, then $x = a$ is a vertical asymptote for the graph of f . In the case when $f(x) \rightarrow \infty$ as $x \rightarrow a$, then $f(x)$ can be made larger than any arbitrary positive number (that is, $f(x) > M$) by taking x sufficiently close to a (that is, $0 < |x - a| < \delta$).

Notes



(a) For a given M , whenever $a - \delta < x < a + \delta$, $x \neq a$, then $f(x) > M$

FIGURE 2.6.4 Infinite limits as $x \rightarrow a$

(b) For a given M , whenever $a - \delta < x < a + \delta$, $x \neq a$, then $f(x) < M$

The four one-sided infinite limits

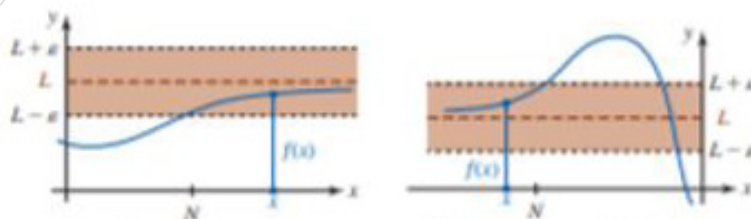
$$\begin{aligned} f(x) &\rightarrow \infty \text{ as } x \rightarrow a^-, & f(x) &\rightarrow -\infty \text{ as } x \rightarrow a^- \\ f(x) &\rightarrow \infty \text{ as } x \rightarrow a^+, & f(x) &\rightarrow -\infty \text{ as } x \rightarrow a^+ \end{aligned}$$

are defined in a manner analogous to that given in Definitions 2.6.2 and 2.6.3.

Definition 2.6.5 Limits at Infinity

- (i) $\lim_{x \rightarrow \infty} f(x) = L$ if for each $\epsilon > 0$, there exists an $N > 0$ such that $|f(x) - L| < \epsilon$ whenever $x > N$.
- (ii) $\lim_{x \rightarrow -\infty} f(x) = L$ if for each $\epsilon > 0$, there exists an $N < 0$ such that $|f(x) - L| < \epsilon$ whenever $x < N$.

Parts (i) and (ii) of Definition 2.6.5 are illustrated in FIGURE 2.6.5(a) and Figure 2.6.5(b), respectively. Recall, if $f(x) \rightarrow L$ as $x \rightarrow \infty$ (or $-\infty$), then $y = L$ is a horizontal asymptote for the graph of f . In the case when $f(x) \rightarrow L$ as $x \rightarrow \infty$, then the graph of f can be made arbitrarily close to the line $y = L$ (that is, $|f(x) - L| < \epsilon$) by taking x sufficiently far out on the positive x -axis (that is, $x > N$).



(a) For a given ϵ , $x > N$ implies $L - \epsilon < f(x) < L + \epsilon$

FIGURE 2.6.5 Limits at infinity

(b) For a given ϵ , $x < N$ implies $L - \epsilon < f(x) < L + \epsilon$

EXAMPLE 6 Using Definition 2.6.5(i)

Prove that $\lim_{x \rightarrow \infty} \frac{3x}{x+1} = 3$.

Solution By Definition 2.6.5(i), for any $\varepsilon > 0$, we must find a number $N > 0$ such that

$$\left| \frac{3x}{x+1} - 3 \right| < \varepsilon \quad \text{whenever} \quad x > N.$$

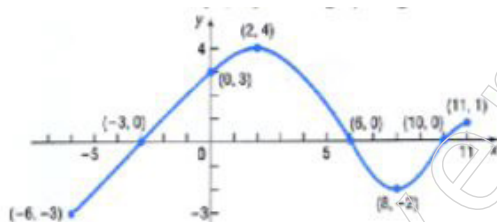
Now, by considering $x > 0$, we have

$$\left| \frac{3x}{x+1} - 3 \right| = \left| \frac{-3}{x+1} \right| = \frac{3}{x+1} < \frac{3}{x} < \varepsilon$$

whenever $x > 3/\varepsilon$. Hence, choose $N = 3/\varepsilon$. For example, if $\varepsilon = 0.01$, then $N = 3/(0.01) = 300$ will guarantee that $|f(x) - 3| < 0.01$ whenever $x > 300$. ■

Limits: Graphic Solutions**Graphical Limits**

Let $f(x)$ be a function defined on the interval $[-6, 11]$ whose graph is given as:



The limits are defined as the value that the function approaches as it goes to an x value. Using this definition, it is possible to find the value of the limits given a graph. A few examples are below:

$$\lim_{x \rightarrow -3} f(x) = 0$$

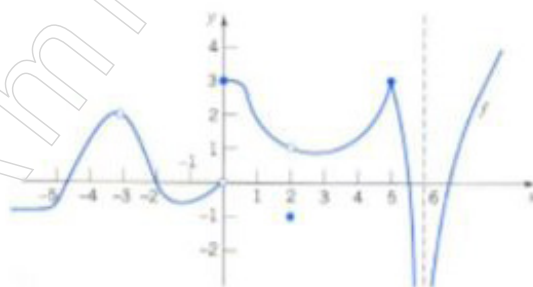
$$\lim_{x \rightarrow 2} f(x) = 4$$

$$\lim_{x \rightarrow 8} f(x) = -2$$

In general, you can see that these limits are equal to the value of the function. This is true if the function is continuous.

Continuity

Continuity of a graph is loosely defined as the ability to draw a graph without having to lift your pencil. To better understand this, see the graph below:



Let's investigate at the following points:

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$x = -3$ Discontinuous at this point the value is not defined at -3 "Removable discontinuity"	$x = 0$ discontinuous at this point The limit of the left is not equal to the limit from the right "Jump discontinuity"	$x = 2$ discontinuous at this point The limit from the left is equal to the right, but is not equal to the value of the function "Removable discontinuity"
$x = 4$ continuous at this point The limit from the left is equal to the limit from the right and equal to the value of the function	$x = 5$ Continuous at this point The limit from the left is equal to the limit from the right and equal to the value of the function	$x = 6$ Discontinuous at this point The value of the limit is equal to negative infinity and therefore not defined "Infinite discontinuity"

One-Sided Limits: General Definition

One-sided limits are differentiated as right hand limits (when the limit approaches from the right) and left-hand limits (when the limit approaches from the left) whereas ordinary limits are sometimes referred to as two-sided limits. Right-hand limits approach the specified point from positive infinity. Left-hand limits approach this point from negative infinity.

The right-handed limit:

$$\lim_{x \rightarrow a^+} f(x) = L$$

The left-handed limit:

$$\lim_{x \rightarrow a^-} f(x) = L$$

A More formal Definition of Continuity

From this information, a more formal definition can be found. Continuity, at a point a , is defined when the limit of the function from the left equals the limit from the right and this value is also equal to the value of the function. using notation, for all points 0 where

$$\lim_{x \rightarrow a^+} f(x) = \lim_{x \rightarrow a^-} f(x) = f(a),$$

the function is said to be continuous.

Summary: When does a limit not exist?

- A general limit does not exist if the left and right-hand limits aren't equal (resulting in a discontinuity in the function).
- A general limit does not exist wherever a function increases or decreases infinitely ("without bound") as it approaches a given x -value.
- A general limit does not exist in the cases of infinite oscillation when approaching a fixed point.

Limits: Numeric Solutions

Now that you know how to solve a limit graphically, you may be asking yourself:

'That's great, but what about when there isn't a graph in the problem?' That is a good question, and that is what this next section is about. There are a many better (and more accurate) ways to find the value of the limit than graphing or plugging in numbers that get closer and closer to the value of interest. These solution methods fall under three categories: substitution, factoring, and the conjugate method. But first things first, let's discuss some of the general rules for limits.

Limit Rules

Here are some of the general limit rules (with $\lim_{x \rightarrow c} f(x) = L$ and $\lim_{x \rightarrow c} g(x) = M$):

1. **Sum Rule:** The limit of the sum of two functions is the sum of their limits

$$\lim_{x \rightarrow c} (f(x) + g(x)) = \lim_{x \rightarrow c} f(x) + \lim_{x \rightarrow c} g(x) = L + M$$

2. **Difference Rule:** The limit of the difference of two functions is the difference of their limits

$$\lim_{x \rightarrow c} (f(x) - g(x)) = \lim_{x \rightarrow c} f(x) - \lim_{x \rightarrow c} g(x) = L - M$$

3. **Product Rule:** The limit of a product of two functions is the product of their limits

$$\lim_{x \rightarrow c} (f(x) \cdot g(x)) = \lim_{x \rightarrow c} f(x) \cdot \lim_{x \rightarrow c} g(x) = L \cdot M$$

4. **Constant Multiple Rule:** The limits of a constant times a function is the constant times the limit of the function

$$\lim_{x \rightarrow c} (k \cdot f(x)) = k \lim_{x \rightarrow c} f(x) = k \cdot L$$

5. **Quotient Rule:** The limit of a quotient of two functions is the quotient of their limits, provided the limit of the denominator is not zero

$$\lim_{x \rightarrow c} \left(\frac{f(x)}{g(x)} \right) = \frac{\lim_{x \rightarrow c} f(x)}{\lim_{x \rightarrow c} g(x)} = \frac{L}{M}, \quad M \neq 0$$

Limit Rule Examples Find the following limits using the above limit rules:

$$1. \lim_{x \rightarrow c} (x^3 + 4x^2 - 3) = \lim_{x \rightarrow c} (x^3) + \lim_{x \rightarrow c} (4x^2) - \lim_{x \rightarrow c} (3) \\ = c^3 + 4c^2 - 3$$

$$2. \lim_{x \rightarrow c} \frac{x^4 + x^2 - 1}{x^2 + 5} = \frac{\lim_{x \rightarrow c} (x^4 + x^2 - 1)}{\lim_{x \rightarrow c} (x^2 + 5)} \\ = \frac{\lim_{x \rightarrow c} (x^4) + \lim_{x \rightarrow c} (x^2) - \lim_{x \rightarrow c} (1)}{\lim_{x \rightarrow c} (x^2) + \lim_{x \rightarrow c} (5)} \\ = \frac{c^4 + c^2 - 1}{c^2 + 5}$$

$$3. \lim_{x \rightarrow 1} \frac{x^4 + x^2 - 1}{x^2 + 5} = \frac{\lim_{x \rightarrow 1} (x^4 + x^2 - 1)}{\lim_{x \rightarrow 1} (x^2 + 5)} \\ = \frac{\lim_{x \rightarrow 1} (x^4) + \lim_{x \rightarrow 1} (x^2) - \lim_{x \rightarrow 1} (1)}{\lim_{x \rightarrow 1} (x^2) + \lim_{x \rightarrow 1} (5)} \\ = \frac{1^4 + 1^2 - 1}{1^2 + 5} = \frac{1}{6}$$

B. Now you try some!

$$1. \lim_{x \rightarrow c} (2x + 5)$$

$$2. \lim_{t \rightarrow 6} 8(t - 5)(t - 7)$$

$$3. \lim_{x \rightarrow 2} \frac{x+2}{x^2+5x+6}$$

Limits of Rational Functions: Substitution Method

A rational function is a function that can be written as the ratio of two algebraic

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expressions. If a function is considered rational and the denominator is not zero, the limit can be found by substitution. This can be seen in the example below (which is similar to the example #3 above, but now done in one quick, convenient step):

$$\lim_{x \rightarrow -1} \frac{x^3 + 4x^2 - 3}{x^2 + 5} = \frac{(-1)^3 + 4(-1)^2 - 3}{(-1)^2 + 5} = \frac{0}{6} = 0$$

This can be defined more formally as: If $P(x)$ and $Q(x)$ are algebraic expressions and $Q(c) \neq 0$, then:

$$\lim_{x \rightarrow c} \frac{P(x)}{Q(x)} = \frac{P(c)}{Q(c)}$$

C. Now you try some!

1. $\lim_{x \rightarrow -5} \frac{x^2 + 3x - 5}{x + 7}$
2. $\lim_{x \rightarrow 2} \frac{x + 3}{x + 6}$

Factoring Method

Consider the function $f(x) = \frac{x^2 - 9}{x + 3}$. How would you find the limit of f as x approaches -3 ? If you try to use substitution to find the limit, world-ending paradoxes ensue:

$$\lim_{x \rightarrow -3} \frac{x^2 - 9}{x + 3} = \frac{(-3)^2 - 9}{(-3) + 3} = \frac{0}{0}$$

But fear not, this answer just tells us that we must use a different method to find the limit, because the function likely has a “hole” at the given. Therefore, the factoring method can be tried. To start this method, the numerator and denominator must be factored (in this case the denominator is “factored” already).

$$\lim_{x \rightarrow -3} \frac{(x + 3)(x - 3)}{x + 3}$$

The factor $(x + 3)$ can be canceled to get the much simpler limit expression of $\lim_{x \rightarrow -3} (x - 3)$ that can easily be evaluated via substitution:

$$\lim_{x \rightarrow -3} (x - 3) = -3 - 3 = -6$$

Therefore, the result of the limit can be found, with the understanding that there is a “hole” in the graph at $x = -3$. Hence, $\lim_{x \rightarrow -3} \frac{x^2 - 9}{x + 3} = -6$

D. Now you try some!

- a) $\lim_{x \rightarrow 5} \frac{2x^2 - 7x - 15}{x - 5}$
- b) $\lim_{x \rightarrow 1} \frac{x^3 - 1}{x - 1}$

Conjugate Method

The conjugate of a binomial expression (i.e. an expression with two terms, you can tell this because of the Latin root bi-meaning two) is the same expression with opposite middle signs. For example, the conjugate of $(\sqrt{x} - 5)$ is $(\sqrt{x} + 5)$. “This is really useful if you have a radical in your limit. This is because the product of two conjugates containing radicals will, itself, contain no radical in your limit. This is because the product of two conjugates containing radicals will, itself, contain no radical expressions. See below:

$$(\sqrt{x} - 5)(\sqrt{x} + 5) = \sqrt{x}^2 + 5\sqrt{x} - 5\sqrt{x} - 25 = x - 25$$

you should use the conjugate method whenever you have a limit problem containing radicals for which substitution does not work.

Example:

Evaluate $\lim_{x \rightarrow 5} \frac{\sqrt{x+11}-4}{x-5}$

First try the substitution method:

$$\lim_{x \rightarrow 5} \frac{\sqrt{x+11}-4}{x-5} = \frac{\sqrt{5+11}-4}{5-5} = \frac{0}{0}$$

Well, another hole in the universe, or at least the graph, indicating that you'll need another method to find the limit since the function probably has a hole at $x = 5$. To start, multiply both the numerator and denominator by the conjugate of the radical expression $(\sqrt{x+11}+4)$:

$$\lim_{x \rightarrow 5} \frac{\sqrt{x+11}-4}{x-5} \cdot \frac{(\sqrt{x+11}+4)}{(\sqrt{x+11}+4)}$$

$$\lim_{x \rightarrow 5} \frac{(x+11)-16}{(x-5)(\sqrt{x+11}+4)}$$

$$\lim_{x \rightarrow 5} \frac{x-5}{(x-5)(\sqrt{x+11}+4)}$$

Cancel the $x - 5$ factor in the numerator and denominator.

$$\lim_{x \rightarrow 5} \frac{1}{(\sqrt{x+11}+4)} = \frac{1}{(\sqrt{5+11}+4)} = \frac{1}{8}$$

E. Now you try some!

a) $\lim_{x \rightarrow -2} \frac{x+2}{\sqrt{x+6}-2}$

b) $\lim_{x \rightarrow -1} \frac{\sqrt{x+10}-3}{x+1}$

Formal Definition: Limits

Limits are more formally defined as

'L is the limit of $f(x)$ as x approaches a if for every number $\epsilon > 0$, there is a corresponding number $\delta > 0$ such that for all x . Using notation we write

$$\lim_{x \rightarrow a} f(x) = L \text{ IFF } 0 < |x - a| < \delta \rightarrow |f(x) - L| < \epsilon$$

From the formal definition, right-handed limit can be defined as:

$$0 < x - a < \delta \rightarrow |f(x) - L| < \epsilon$$

And written as:

$$\lim_{x \rightarrow a^+} f(x) = L$$

Whereas the left-handed limit can be defined as:

$$-\delta < x - a < 0 \rightarrow |f(x) - L| < \epsilon$$

And written as:

$$\lim_{x \rightarrow a^-} f(x) = L$$

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Example 1: Testing the Definition

Show: $\lim_{x \rightarrow 1} (5x - 3) = 2$

We have to find a suitable $\delta > 0$ so that if $x \neq 1$ and x is within distance δ , that is if:

$$0 < |x - 1| < \delta$$

Then $f(x)$ is within distance ϵ of $L = 2$, that is

$$|f(x) - 2| < \epsilon$$

$$|(5x - 3) - 2| < \epsilon$$

$$|5x - 5| < \epsilon$$

$$5|x - 1| < \epsilon$$

$$|x - 1| < \frac{\epsilon}{5}$$

Thus, we can take $\delta = \frac{\epsilon}{5}$ due to the fact that $0 < |x - 1| < \delta$, then:

$$|f(x) - 2| < \epsilon$$

$$\text{So, if } |(5x - 3) - 2| = |5x - 5| = 5|x - 1| < 5(\delta) = 5\left(\frac{\epsilon}{5}\right) = \epsilon$$

$$\text{therefore, } 0 < |x - 1| < \frac{\epsilon}{5} \rightarrow |f(x) - 2| < \epsilon \text{ and } \lim_{x \rightarrow 1} (5x - 3) = 2.$$

Limits: Advanced Topics

Previously, when we found that the result of a limit doing straight substitution yielded $0/0$ we used factoring or conjugation to be able to solve the problem. What happens when neither of those methods prove useful? You become very grateful for the 17th-century French mathematician Guillaume de L'Hopital. L'Hopital was the man that derived a method of solving these types of equations, known as indeterminate forms. This method, known as L'Hopital's Rule, is formally defined below.

Formal Definition: L'Hôpital's Rule

If the limit $\lim_{x \rightarrow c} \frac{f(x)}{g(x)}$ results in one of the following forms:

$$\frac{0}{0}, \quad \pm \frac{\infty}{\infty}, \quad 0 \pm \pm \infty, \quad \infty - \infty, \quad 0^0, \quad 1^\infty, \quad \infty^0$$

And $\lim_{x \rightarrow c} \frac{f'(x)}{g'(x)}$ exists and $g'(x) \neq 0$, then:

$$\lim_{x \rightarrow c} \frac{f(x)}{g(x)} = \lim_{x \rightarrow c} \frac{f'(x)}{g'(x)}$$

Example 1: Indeterminate form of $\frac{0}{0}$

Find the limit $\lim_{x \rightarrow 0} \frac{e^x - 1}{x}$

$$\lim_{x \rightarrow 0} \frac{e^x - 1}{x} = \frac{0}{0} = \lim_{x \rightarrow 0} \frac{e^x}{1}$$

Using L'Hôpital's Rule:

$$\lim_{x \rightarrow 0} \frac{e^x - 1}{x} = \lim_{x \rightarrow 0} \frac{e^x}{1} = \frac{1}{1} = 1$$

Example 2: Indeterminate form of $\frac{\infty}{\infty}$

Find the limit $\lim_{x \rightarrow \infty} \frac{x^2}{2^x}$

$$\lim_{x \rightarrow \infty} \frac{x^2}{2^x} = \frac{\infty}{\infty}$$

Using L'Hôpital's Rule:

$$\lim_{x \rightarrow \infty} \frac{x^2}{2^x} = \lim_{x \rightarrow \infty} \frac{2x}{2^x \ln 2} = \frac{2}{\ln 2} \lim_{x \rightarrow \infty} \frac{x}{2^x} = \frac{\infty}{\infty}$$

Using L'Hôpital's Rule again:

$$\frac{2}{\ln 2} \lim_{x \rightarrow \infty} \frac{x}{2^x} = \frac{2}{\ln 2} \lim_{x \rightarrow \infty} \frac{1}{2^x \ln 2} = \frac{2}{(\ln 2)^2} \lim_{x \rightarrow \infty} \frac{1}{2^x} = \frac{2}{(\ln 2)^2} \cdot 0 = 0$$

Example 3: indeterminate form of $\infty - \infty$

Find the limit $\lim_{x \rightarrow 2} \left(\frac{4}{x^2 - 4} - \frac{1}{x - 2} \right)$

$$\lim_{x \rightarrow 2} \left(\frac{4}{x^2 - 4} - \frac{1}{x - 2} \right) = \infty - \infty$$

Using L'Hôpital's Rule:

$$\lim_{x \rightarrow 2} \left(\frac{4}{x^2 - 4} - \frac{1}{x - 2} \right) = \lim_{x \rightarrow 2} \frac{4 - (x + 2)}{(x - 2)(x + 2)} = \lim_{x \rightarrow 2} \frac{2 - x}{x^2 - 4} = \lim_{x \rightarrow 2} \frac{-1}{2x} = -\frac{1}{4}$$

Example 4: indeterminate form of 1^∞

Find the limit $\lim_{x \rightarrow 1} x^{\frac{1}{1-x}}$

$$\lim_{x \rightarrow 1} x^{\frac{1}{1-x}} = 1^\infty$$

Let $y = x^{\frac{1}{1-x}}$. Then $\ln y = \ln x^{\frac{1}{1-x}} = \frac{\ln x}{1-x}$

Using L'Hôpital's Rule:

$$\lim_{x \rightarrow 1} \ln y = \lim_{x \rightarrow 1} \frac{\ln x}{1-x} = \lim_{x \rightarrow 1} \frac{\frac{1}{x}}{-1} = -\lim_{x \rightarrow 1} \frac{1}{x} = -1$$

Therefore

$$\lim_{x \rightarrow 1} \ln y = -1$$

$$\lim_{x \rightarrow 1} x^{\frac{1}{1-x}} = \lim_{x \rightarrow 1} y = e^{-1} = \frac{1}{e}$$

Example 5: indeterminate form of $0 \cdot \infty$

Find the limit $\lim_{x \rightarrow 0^+} (x \cdot \ln x)$

$$\lim_{x \rightarrow 0^+} x \cdot \ln x = 0 \cdot \infty$$

Using L'Hôpital's Rule:

$$\lim_{x \rightarrow 0^+} x \cdot \ln x = \lim_{x \rightarrow 0^+} \frac{\ln x}{\frac{1}{x}} = \lim_{x \rightarrow 0^+} \frac{\frac{1}{x}}{-\frac{1}{x^2}} = \lim_{x \rightarrow 0^+} -x = 0$$

F. Now you try some!

1. $\lim_{x \rightarrow 0} \frac{\sin 3x}{4x}$

2. $\lim_{x \rightarrow 1} \frac{5x^4 + 4x^2 - 1}{10 - x - 9x^2}$

3. $\lim_{x \rightarrow \infty} \frac{e^x}{x^3}$

4. $\lim_{x \rightarrow \infty} x \cdot \ln x$

5. $\lim_{x \rightarrow 0} x^{\frac{1}{x}}$

6. Why does the $\lim_{x \rightarrow -1} \frac{\sqrt{x+4}-3}{x+1}$ not equal $\frac{1}{2\sqrt{3}}$?

Note the trick that is needed here – what happens if you use L'Hôpital's Rule without making this initial change?

7. Early in their presentation of limits, most teachers present the following two limit facts as special cases: $\lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} = 1$ and $\lim_{\theta \rightarrow 0} \frac{\cos \theta - 1}{\theta} = 0$. Verify each by using L'Hôpital's Rule.

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Limits to Infinity (Horizontal Asymptotes)

What happens to a function as it goes further and further out to the left and right? well, that depends on the function. But half of the answer can be discovered by allowing the independent variable to take on increasingly large, positive values and keeping an eye on the output (the graph)- this investigates what is happening as we go further and further to the right. The other half is discovered by allowing the independent variable to take on increasingly large, negative values and, again, keeping an eye on the output- this investigates what is happening as we go further and further to the left.

Here are some basic facts and some generalization that will be sufficient to evaluate most 'limits to infinity'.

Consider the function as an algebraic fraction, and consider the ratio of the leading terms. Let the algebraic expression in the numerator be expressed as $n(x)$ and the algebraic expression found in the denominator be expressed as $d(x)$, then

- if the degree of the numerators is lower than the degree of the denominator, then $\lim_{x \rightarrow \infty} \frac{n(x)}{d(x)} = 0$. In general, whenever the denominator grows faster than the numerator, the limit will go to zero. Thus, in these cases, as the graph extends far to both the left and the right, the output (i.e., the graph) gets closer and closer to zero.

Here is a list of functions in order of their rate of growth – quickest to slowest:

$x!, \dots, 4^x, 3^x, 2^x, \dots, x^4, x^3, x^2, x \log x, x, \log x, \dots, 3, 2, 1$

Examples:

$$a. \lim_{x \rightarrow \infty} \frac{5x^3 - 2x^2 + 7x - 13}{12 - 2x + x^4} = \lim_{x \rightarrow \infty} \frac{5x^3}{x^4} = \lim_{x \rightarrow \infty} \frac{5}{x} = 0$$

$$b. \lim_{x \rightarrow \infty} \frac{6x \log x + 5x - 2}{2x^2 + 2x - 14 \log x} = \lim_{x \rightarrow \infty} \frac{6x \log x}{2x^2} = 0$$

- If the degree of the numerator is higher than the degree of the denominator, then $\lim_{x \rightarrow \infty} \frac{n(x)}{d(x)} = \infty$ or $-\infty$. In general, whenever the numerator grows faster than the denominator, the limit will go to positive or negative infinity. Thus, in these cases, as the graph extends far to both the left and the right, the output (i.e., the graph) increases or decreases without bound. In these cases, each side needs to be considered separately.

Examples:

$$a. \lim_{x \rightarrow \infty} \frac{3x^5 + x^2 - 5}{6 + x + 7x^2} = \lim_{x \rightarrow \infty} \frac{3x^5}{7x^2} = \lim_{x \rightarrow \infty} \frac{3}{7}x^3 = \infty$$

$$\text{whereas } \lim_{x \rightarrow -\infty} \frac{3x^5 + x^2 - 5}{6 + x + 7x^2} = \lim_{x \rightarrow -\infty} \frac{3}{7}x^3 = -\infty$$

$$b. \lim_{x \rightarrow \infty} \frac{x! - \log x + 1}{2^x + 3x^5 - 5 \log x} = \lim_{x \rightarrow \infty} \frac{x!}{2^x} = +\infty \quad [\text{Note that } \lim_{x \rightarrow -\infty} \frac{x! - \log x + 1}{2^x + 3x^5 - 5 \log x} \text{ is meaningless since } x! \text{ is not defined for negative values.}]$$

$$c. \lim_{x \rightarrow \infty} 2x^4 - 5x + 3 = +\infty \quad (\text{Note that } \lim_{x \rightarrow -\infty} 2x^4 - 5x + 3 = +\infty \text{ as well [why?]})$$

- If the degrees are equal, then $\lim_{x \rightarrow \infty} \frac{n(x)}{d(x)}$ is equal to the leading coefficient of $n(x)$ over the leading coefficient of $d(x)$.

Examples:

$$\text{a. } \lim_{x \rightarrow \infty} \frac{3x^3 - 2x^2 + x - 4}{4 + 2x - 5x^3} = \lim_{x \rightarrow \infty} \frac{3x^3}{-5x^3} = \lim_{x \rightarrow \infty} \frac{3}{-5} = -\frac{3}{5}$$

$$\text{b. } \lim_{x \rightarrow \infty} \frac{\frac{1}{2}2^x}{\frac{3}{4}2^x - \log x + 4} = \frac{1/2}{3/4} = \frac{2}{3}$$

3.3.4 Continuous and Discontinuous functions – Introduction and Application

Continuity of a Function at a Point

A function $f(x)$ is said to be continuous at an interior point $x=a$ of its domain if $\lim_{x \rightarrow a} f(x) = f(a)$. In other words, a function $f(x)$ is said to be continuous at a point $x=a$ provided left hand limit, right hand limit and value of the function are equal:

A function $f(x)$ is continuous at a point $x=a$ if

$$\lim_{h \rightarrow 0} f(a-h) = \lim_{h \rightarrow 0} f(a+h) = f(a)$$

Continuity on a function on an open interval

A function $f(x)$ is said to be continuous on an open interval (a,b) if it is continuous at each point of (a,b) .

Continuity on a function on a closed interval

A function $f(x)$ is said to be continuous on a closed interval $[a,b]$ if

it is continuous at each point of (a,b) .

$f(x)$ is continuous from right at $x=a$, i.e.,

$$\lim_{h \rightarrow 0} f(a+h) = f(a)$$

$f(x)$ is continuous from left at $x=b$, i.e.,

$$\lim_{h \rightarrow 0} f(b-h) = f(b)$$

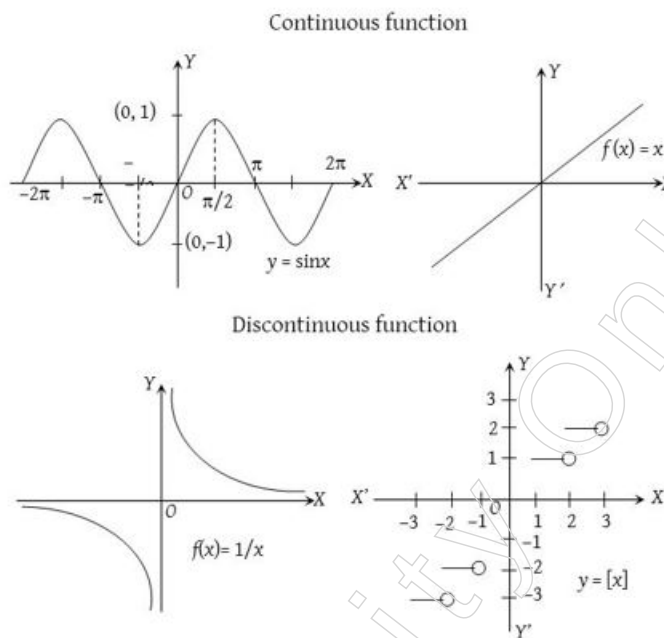
$f(x)$ is continuous at each point of the open interval (a,b) .

Discontinuity of a function

A function $f(x)$, which is not continuous at a point $x=a$, is said to be discontinuous at that point.

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Ex. 3.13: Check whether the function $f(x) = \begin{cases} 4x^2, & \text{if } x \leq 2 \\ 3, & \text{if } x > 2 \end{cases}$ is continuous at $x = 2$, or not?

Solution: Given, $f(x) = \begin{cases} 4x^2, & \text{if } x \leq 2 \\ 3, & \text{if } x > 2 \end{cases}$

At $x = 2$,

$$\begin{aligned} LHL &= \lim_{x \rightarrow 2^-} f(x) \\ &= \lim_{x \rightarrow 2^-} (4x^2) \\ &= 16 \end{aligned}$$

At $x = 2$,

$$\begin{aligned} RHL &= \lim_{x \rightarrow 2^+} f(x) \\ &= 3 \end{aligned}$$

$$f(2) = 16$$

$$\therefore LHL = f(2) \neq RHL$$

$\therefore$ function is not continuous at $x = 2$.

Ex. 3.14: Check whether the function $f(x) = \begin{cases} x+1, & \text{if } x \geq 1 \\ x^2+1, & \text{if } x < 1 \end{cases}$ is continuous at $x = 1$, or not?

Solution: Given, $f(x) = \begin{cases} x+1, & \text{if } x \geq 1 \\ x^2+1, & \text{if } x < 1 \end{cases}$

At $x = 1$,

$$\begin{aligned} RHL &= \lim_{x \rightarrow 1^+} f(x) \\ &= \lim_{x \rightarrow 1^+} (x+1) \\ &= 2 \end{aligned}$$

At $x = 1$,

$$\begin{aligned} LHL \lim_{x \rightarrow 1^-} f(x) \\ &= \lim_{x \rightarrow 1^-} (x + 1) \\ &= 1 + 1 \end{aligned}$$

$$f(1) = x + 1$$

$$= 2$$

$$\therefore LHL = RHL = f(1)$$

$\therefore$ function is continuous at $x = 1$.

Ex. 3.15: If function $f(x) = \begin{cases} kx^2, & \text{if } x \leq 2 \\ 3, & \text{if } x > 2 \end{cases}$ is continuous at $x = 2$, then find the value of k ?

Solution: Given, $f(x) = \begin{cases} kx^2, & \text{if } x \leq 2 \\ 3, & \text{if } x > 2 \end{cases}$

At $x = 2$,

$$\begin{aligned} LHL &= \lim_{x \rightarrow 2^-} f(x) \\ &= \lim_{x \rightarrow 2^-} (kx^2) \\ &= 4k \end{aligned}$$

At $x = 2$,

$$\begin{aligned} RHL &= \lim_{x \rightarrow 2^+} f(x) \\ &= 3 \end{aligned}$$

$$f(2) = 4k$$

If function is continuous then $LHL = RHL = f(x)$.

So,

$$4k = 3$$

$$k = \frac{3}{4}$$

Check your Understanding

1. The two broad areas of calculus known as differential and integral calculus are built on the foundation concept of a _____.
2. A _____ of the existence of a limit can never be based on one's ability to sketch graphs or on tables of numerical values.
3. _____ of a graph is loosely defined as the ability to draw a graph without having to lift your pencil.
4. Left hand limits approach the specified point from positive infinity. True / False
5. A _____ function is a function that can be written as the ratio of two algebraic expressions.

Notes

6. The _____ of a binomial expression is the same expression with opposite middle signs.

Multiple Choice

- The value of $\lim_{x \rightarrow 1} \frac{(1+x)^{-1} - 1}{(1+x)^{-1} - 1}$ is
 - 20
 - 21
 - 15
 - 17
- The value of $\lim_{x \rightarrow \sqrt{2}} \frac{x^4 - 4}{x^2 + 3\sqrt{2}x - 8}$ is
 - $-\frac{8}{5}$
 - $\frac{7}{5}$
 - $\frac{8}{5}$
 - None of these
- The value of $\lim_{x \rightarrow 1} \frac{x^7 - 2x^5 + 1}{x^3 - 3x^2 + 2}$ is
 - 0
 - 1
 - 1
 - 3
- The value of $\lim_{x \rightarrow 2} \frac{x^2 - 4}{\sqrt{3x - 2} - \sqrt{x + 2}}$ is
 - 8
 - 8
 - 7
 - 5
- Find n, if $\lim_{x \rightarrow 2} \frac{x^n - 2^n}{x - 2} = 80, n \in N$.
 - 5
 - 6
 - 4
 - 7

Summary

- The two broad areas of calculus known as differential and integral calculus are built on the foundation concept of a limit.
- The limit of the n^{th} root of a function is the n^{th} root of the limit whenever the limit exists and has a real n^{th} root.

Notes

- A proof of the existence of a limit can never be based on one's ability to sketch graphs or on tables of numerical values.
- Continuity of a graph is loosely defined as the ability to draw a graph without having to lift your pencil.
- One sided limits are differentiated as right-hand limits and left-hand limits whereas ordinary limits are sometimes referred to as two-sided limits. Right hand limits approach the specified point from positive infinity. Left hand limits approach this point from negative infinity.
- A general limit does not exist wherever a function increases or decreases infinitely as it approaches a given x-value.
- A rational function is a function that can be written as the ratio of two algebraic expressions. If a function is considered rational and the denominator is not zero, the limit can be found by substitution.
- The conjugate of a binomial expression is the same expression with opposite middle signs.
- A function $f(x)$ is said to be continuous at a point $x=a$ provided left hand limit, right hand limit and value of the function are equal.
- A function $f(x)$ is said to be continuous on an open interval (a,b) if it is continuous at each point of (a,b) .
- A function $f(x)$, which is not continuous at a point $x=a$, is said to be discontinuous at that point.

Activity

1. Find the point(s) of discontinuity for the function $f(x) = x^2 + 1 + |2x - 1|$
2. Consider the function:

$$f(x) = \begin{cases} \ln x & \text{if } 0 < x < 1 \\ ax^2 + b & \text{if } 1 \leq x < \infty \end{cases}$$

3. If $f(2) = 3$, determine the values of **a** and **b** for which $f(x)$ is continuous.
4. Given the function:

$$f(x) = \begin{cases} \frac{x^2-9}{x-3} & \text{if } x \neq 3 \\ a & \text{if } x = 3 \end{cases}$$

5. Determine the value of **a** for which the function is continuous at **$x = 3$** .

Questions and Exercises

1. Describe a limit of a function.
2. What are the theorems of limits?
3. List the five limit rules.
4. What is the factoring method?
5. Why is it beneficial to have conjugates?

Notes

Glossary

- **Continuity of a graph:** is loosely defined as the ability to draw a graph without having to lift your pencil.
- **Rational function:** is a function that can be written as the ratio of two algebraic expressions.
- Further Reading
- Boelkins, Matthew. Active Calculus 2018: Single Variable. CreateSpace Independent Publishing Platform; 2018th edition (August 13, 2018)
- Edwards, Bruce H.; Larson, Ron. Calculus: Early Transcendental Functions Cengage Learning; 7th edition (January 1, 2018)
- Stewart, James. Calculus 8th Edition. Cengage Learning; 8th edition (May 19, 2015)

Check your Understanding: Answers

1. Limit
2. Proof
3. Continuity
4. False
5. Rational
6. Conjugate

Multiple Choice

1. B
2. C
3. B
4. A
5. A

Module - 4: Data Analysis

Notes

Structure:

- 4.1 Data interpretation
 - 1.1.1 Data and Statistical Data, Frequency Distribution
 - 1.1.2 Graphical Representation
 - 1.1.3 Histogram
 - 1.1.4 Frequency Polygon and Frequency Curve
 - 1.1.5 Ogive- Part 1
 - 1.1.6 Ogive- Part 2
- 4.2 Descriptive Measures
 - 2.1.1 Measure of the Central Tendency - I
 - 2.1.2 Measure of the Central Tendency - II
 - 2.1.3 Measure of Dispersion
 - 2.1.4 Kurtosis, skewness

Notes

Unit - 4.1: Data Interpretation

Objectives:

At the end of this unit, you will be able to understand:

- Data and Statistical Data, Frequency Distribution
- Graphical Representation
- Histogram
- Frequency Polygon and Frequency Curve
- Ogive Part 1
- Ogive Part 2

4.1.1 Data and Statistical Data, Frequency Distribution

Introduction

The word Statistics is derived from the Italian word 'Stato' which means 'state'; and the word 'Statista' refers to a person who is involved with the affairs of state. Thus, statistics originally was meant for collection of facts useful for affairs of the state, like the taxes, land records, population demography, etc. There is an evidence of use of some of the principles of statistics by ancient Indian civilization as well. Some of the techniques have found their mention in Vedic Mathematics. However, the modern statistical methods spread from Italy to France, Holland and Germany in 16th century.

Definitions of Statistics

The definitions of statistics are as follows: "Statistics are the classified facts representing the conditions of the people in the state. Specially those facts which can be stated in number or in table of numbers or in any tabular or classified arrangement." – Webster

"Statistics may be defined as the science of collection, presentation, analysis and interpretation of data." – Croxton and Cowden "By statistics we mean quantitative data affected to a marked extent by multiplicity of causes." –Yule and Kendall

Functions of Statistics

1. **Condensation:** Statistics compresses a mass of figures to small meaningful information, for example, average sales, BSE index, the growth rate etc. It is impossible to get a precise idea about the profitability of a business from a mere record of income and expenditure transactions. The information of Return On Investment (ROI), Earnings Per Share (EPS), profit margins, etc., however, can be easily remembered, understood and thus used in decision-making.
2. **Forecast:** Statistics helps in forecasting by analyzing trends, which are essential for planning and decision-making. Predictions based on the gut feeling or hunch can be harmful for the business. For example, to decide the refining capacity for a petrochemical plant, it is required to predict the demand of petrochemical product

mix, supply of crude oil, the cost of crude, substitution products, etc., for next 10 to 20 years, before committing an investment.

3. **Testing of hypotheses:** Hypotheses are the statements about population parameters based on past knowledge or information. It must be checked for its validity in the light of current information. Inductive inference about the population based on the sample estimates involves an element of risk. However, sampling keeps the decision-making costs low. Statistics provides quantitative base for testing our beliefs about the population.
4. **Relationship between Facts:** Statistical methods are used to investigate the cause and effect relationship between two or more facts. The relationship between demand and supply, money-supply and price level can be best understood with the help of statistical methods.
5. **Expectation:** Statistics provides the basic building block for framing suitable policies. For example how much raw material should be imported, how much capacity should be installed, or manpower recruited, etc., depends upon the expected value of outcome of our present decisions.

Limitations of Statistics

Statistical techniques, because of their flexibility have become popular and are used in numerous fields. But statistics is not a cure-all technique and has few limitations. It cannot be applied to all kinds of situations and cannot be made to answer all queries. The major limitations are:

1. Statistics deals with only those problems, which can be expressed in quantitative terms and amenable to mathematical and numerical analysis. These are not suitable for qualitative data such as customer loyalty, employee integrity, emotional bonding, motivation etc.
2. Statistics deals only with the collection of data and no importance is attached to an individual item.
3. Statistical results are only an approximation and not mathematically correct. There is always a possibility of random error.
4. Statistics, if used wrongly, can lead to misleading conclusions, and therefore, should be used only after complete understanding of the process and the conceptual base.
5. Statistics laws are not exact laws and are liable to be misused.
6. The greatest limitation is that the statistical data can be used properly only by a professional. A person having thorough knowledge of the methods of statistics and proper training can only come to conclusions.
7. If statistical data are not uniform and homogenous, then the study of the problem is not possible. Homogeneity of data is essential for a proper study.
8. Statistical methods are not the only method for studying a problem. There are other methods as well, and a problem can be studied in various ways.

Notes

Data Collection

The collection and analysis of data constitute the primary stages of execution of any statistical investigation. The procedure for collection of data depends upon various considerations such as the scope, objective, nature of investigation, etc. Availability of resources such as time, money, manpower, etc., also affect the procedure choice. Data may be collected either from a primary or from a secondary source, which are described below.

Types of Data – Primary and Secondary

The data used in statistical study is termed as either 'primary' or 'secondary' depending upon whether it was collected specifically for the undertaken study or for some other purpose.

When the data used in a statistical study is collected under the control and supervision of the investigator, such type of data is referred to as 'primary data'. Primary data is collected afresh and for the first time, and thus, happen to be original in character. On the other hand, when the data is not collected for this purpose, but is derived from other sources then such data is referred to as 'secondary data'. Often, secondary data is collected by some other organization to satisfy their needs, but it is used by someone else for entirely different reasons.

The difference between primary and secondary data is only in terms of degree. For example, data, which are primary in the hands of one, becomes secondary in hands of other. Suppose an investigator wants to study the working conditions of labourers in an industry. If the investigator or their agent collects the data directly, then it is called a 'primary data'. But if subsequently someone else uses this collected data for some other purpose, then this data becomes a 'secondary data'.

Types of Statistics

The study of statistics can be categorized into two main branches. These branches are descriptive statistics and inferential statistics.

Descriptive statistics is used to sum up and graph the data for a category picked. This method helps to understand a particular collection of observations. A sample is defined on descriptive statistics. There is no confusion in concise numbers, since you just identify the individuals or things which are calculated.

Descriptive statistics give information that describes the data in some manner. For example, suppose a pet shop sells cats, dogs, birds and fish. If 100 pets are sold, and 35 out of the 100 were dogs, then one description of the data on the pets sold would be that 35% were dogs.

Inferential statistics are techniques that allow us to use certain samples to generalize the populations from which the samples were taken. Hence, it is crucial that the sample represents the population accurately. The method to get this done is called sampling. Since the inferential statistics aim at drawing conclusions from a sample and generalizing them to a population, we need to be sure that our sample accurately represents the population. This requirement affects our process. At a broad level, we must do the following:

- Define the population we are studying.
- Draw a representative sample from that population.
- Use analyses that incorporate the sampling error.

Notes

Methods of Collecting Data

Methods of collecting primary data

Generally, for managerial decision-making, it is necessary to analyze information regarding a large number of characteristics. Collection of primary data can be time consuming, expensive, and hence requires a great deal of deliberation. According to the nature of information required, one of the following methods or their combination can be selected.

1. **Observation Method:** In this method, the investigator collects the data through personal observations. This method is very useful if data is created in the system through capturing transactions. Computerized transaction processing can be modified to generate necessary data or information. An investigator well versed with the system or a part of the system is ideally suited for collecting this kind of data. Since the investigator is solely involved in collecting the data, their training, knowledge and skills play an important role as far as the quality of the data is concerned. Sometimes, the audio/video aids can also be used to record the observations.
2. **Indirect Investigation:** In this case, the information collected by oral or written interrogation forms the primary data. Usually enquiry commissions, board of investigations, investigation teams and committees collect data in this manner. Quality of the data largely depends upon the person interviewed, their motives, memory, overall cooperation, and the interviewer's repute with the person being interviewed.
3. **Questionnaire with Personal Interview:** This is the most common and popular method for data collection. In this method, individuals are personally interviewed and answers are recorded to collect the data. Questionnaire is structured and followed in specific sequence. Occasionally, a part of the questionnaire may be unstructured to motivate the interviewee to give additional information or information on intimate matters. Accuracy of the data depends on the ability, sincerity and the tactfulness of the interviewer to conduct the interview in friendly and professional environment.
4. **Mailed Questionnaire:** In this method, the structured questionnaire is mailed to selected people with a request to fill it and return. Along with the questions, the supplementary information clarifying terms, explaining process, etc., is also attached. In a few cases, inducements for filling and returning the questionnaire are also given. To develop a report, the covering of a letter with a questionnaire is necessary to explain the reason for the data collection and, if any, to alleviate the respondent's fears. The respondents are believed to be literate and be able to answer the questions without any confusion. This is a less expensive and faster method to collect large volume of data, over a wide geographic area, in a standard form, and at the convenience of the respondent. Hence this method is most popular and extensively used. However, this method needs a guard against two drawbacks viz. The absence of an interviewer, which results in a large proportion of the non-response and the possibility of reducing the reliability of the replies if the respondent

Notes

is not sufficiently motivated. These shortcomings can be overcome by increasing sample size and designing the questionnaire comprehensively.

5. **Telephonic Interview:** This method is less expensive but has limited in scope, as the respondent must possess a telephone and has it listed. Further, the respondent must be available and in the frame of mind to provide correct answers. This method is comparatively less reliable for public surveys. However, for industrial survey, in developed regions, and with known customers, this method is best suited. There is a limit to the number of questions that the interviewee could answer in three to four minutes. The method is efficient if there are just three to five yes/no type questions and two to three short questions.
6. **Internet Surveys:** Of late, Internet surveys have become popular. These are less expensive, fast and can be interactive. However, its scope is limited to those who have regular Internet access. With rapid growth in technology and Internet connectivity it would be one of the main methods of collecting primary data. With its interactivity and multimedia facilities it also combines the advantages of other methods.

Methods of Collecting Secondary Data

Secondary data is one that has been collected or analyzed by some other agency for another purpose. Sources of secondary data are -

1. Publications of central, state and local governments. This is an important and reliable source to get unbiased data.
2. Publications of foreign governments or of international bodies. Although it is a good source, context under which it is collected needs to be verified before using this data. For international situations this data could be very useful and authentic.
3. Journals of trade, commerce, economics, scientific, engineering, medicine, etc. This data could be very reliable for a specific purpose.
4. Other published sources like books, magazines, reports, newspapers, etc.
5. Unpublished data, based on internal records and documents of an organization can provide most authentic and much cheaper information provided we could identify the source.
6. Diaries, letters, mailers can also provide secondary data. The problem with the unpublished data is that it's difficult to locate and get access.

Applications of Statistics

Data is a collection of any number of related observations. We can collect the number of telephones installed in a given day by several workers or the numbers of telephones installed per day over a period of several days by one worker and call the results our data. A collection of data is called a data set and a single observation is called as a data point. Data is used everywhere in day to day life. It is applicable in very large number of fields such as economics, management, sociology, anthropology, agriculture, medicine, psychology, education. All the fields lean heavily on data and its analysis. The application of data is so vast and ever expanding that it is very difficult to define. Its use has permeated almost in every facet of our lives.

Application of Statistics in Business Decision

Statistics is not restricted to only information about the State, but it also extends to almost every realm of the business. Statistics is about scientific methods to gather, organize, summarize and analyze data. More important still is to draw valid conclusions and make effective decisions based on such analysis. To a large degree, company performance depends on the preciseness and accuracy of the forecast. Statistics is an indispensable instrument for manufacturing control and market research. Statistical tools are extensively used in business for time and motion study, consumer behaviour study, investment decisions, credit ratings, performance measurements and compensations, inventory management, accounting, quality control, distribution channel design, etc. For managers, therefore, understanding statistical concepts and knowledge about using statistical tools is essential. With an increase in a company's size and market uncertainty due to reduced competition, the need for statistical knowledge and statistical analysis of various business circumstances has greatly increased. Prior to this, when the size of business used to be small without much complexities, a single person, usually owner or manager of the firm, used to take all decisions regarding the business. Example: A manager used to decide, from where the necessary raw materials and other factors of production were to be acquired, how much of output will be produced, where it will be sold, etc. This type of decision making was usually based on experience and expectations of this single individual and as such had no scientific basis.

Classification of Data

Classification refers to the grouping of data into homogeneous classes and categories. It is the process of arranging things in groups or classes as per their resemblances and affinities.

Rules of Classification - The principal rules of classifying data are:

1. To condense the data mass in such a way that salient features can be readily noticed; for example, household incomes can be grouped as higher income group, middle-income group and lower income group based on certain criterion.
2. To facilitate comparison between attributes of variables; for example, comparison between education and income, income and expenditure on consumer durables, etc.
3. To prepare the data for tabulation.
4. To highlight the significant features of the data; for example, data is concentrated on one side, or one particular value may be dominant.
5. To enable grasping of data.
6. To study the relationship formed.

Bases of Classification

Some common types of bases of classification are:

1. Geographical classification: In this type, data is classified according to area or region. For example, State wise industrial production, city wise consumer behaviour, area wise sales figures, etc.

Notes

2. Chronological classification: In this type, data is classified according to the time of its occurrence; for example, monthly sales, daily demand, yearly production, etc.
3. Qualitative classification: When the data is classified according to some attributes, which are not capable of measurement, it is known as qualitative classification. In dichotomous classification, an attribute is divided into two classes, one possessing the attribute and other not possessing it; for example, smoker, non-smoker, employed, unemployed, etc. In many-fold classification, attribute is divided so as to form several classes like education level, religion, mother tongue, etc.
4. Classification of data according to characteristics: It refers to the classification of data according to some characteristics which can be measured; for example, age, salary, height, etc. Quantitative data may be further classified into two types namely discrete and continuous. In case of discrete type, values of the variables taken are countable (could be infinitely large also for example, integers). Examples of these are number of accidents, number of defectives, etc. In case of continuous quantities, data can take any real values; for example, weight, height, distance, volume, etc.

Frequency Distribution

Classification of data shows the different values of a variable and their respective frequency of occurrence is called a frequency distribution of the values.

There are two kinds of frequency distributions, namely, discrete frequency distribution (or simple, or ungrouped frequency distribution), and continuous frequency distribution (or condensed or grouped frequency distribution).

a. Discrete Frequency Distribution

The process of preparing discrete frequency distribution is simple. First, all the possible values of variables are arranged in ascending order in a column. Then another column of 'Tally' mark is prepared to count the number of times a particular value of the variable is repeated. To facilitate counting, a block of five 'Tally' marks is prepared. The last column contains frequency. To illustrate this let us consider one example.

Example:

Construct frequency distribution table for the following data of number of family members in 30 families:

4	3	2	3	4	5	5	7	3	2
3	4	2	1	1	6	3	4	5	4
2	7	3	4	5	6	2	1	5	3

Number of Family Members	'Tally Marks'	Frequency
1		3
2		5
3		7
4		6

5		5
6		2
7		2
		Total N = 30

Notes

b. Continuous Frequency Distribution

For continuous data a 'grouped frequency distribution' is necessary. For discrete data, discrete frequency distribution is better than array, but this does not condense the data. 'Grouped frequency distribution' is useful for condensing discrete data by putting them into smaller groups or classes called class-intervals. Some important terms used in case of continuous frequency distribution are as follows:

1. **Class limits:** Class limits denote the lowest and highest value which can be included in the class. The two boundaries of class are known as the lower limit and upper limit of the class. For example, 10-18, 20-28, where 10 and 18 are limits of the first class; 20 and 28 are limits of second class, etc.
2. **Class intervals:** The class interval represents the width, the span or the size of a class. The width may be determined by subtracting the lower limit of one class from the lower limit of the following class. For example, classes 10-15, 15-20, etc have class interval $20 - 15 = 5$.
3. **Class frequency:** The number of observations falling within a particular class is known as its class frequency. Total frequency indicates the total number of observations $N = \sum f$.
4. **Class mark or class mid-point:** Mid-point of a class is defined as the sum of two successive lower limits divided by two. Thus class mark is the value lying halfway between lower and upper class limits. For example, classes 10-20, 20-30, etc have class marks 15, 25 etc.
5. **Types of class intervals:** There are many different ways in which limits of class intervals can be shown.
6. **Exclusive method:** In this method, the class intervals are so arranged that upper limit of one class is the lower limit of next class. This method always presumes that the upper limit is excluded from the class, for example, with class limits 20-25, 25-30 observation with value 25 is included in class 25-30.
7. **Inclusive method:** In this method, the upper limit of the class is included in the same class itself. In such case there is no overlap of upper limit of former class and lower limit of successive class. For example, with class limits 20-29.5, 30-39.5, 40-49.5, etc. there is no ambiguity but values from 29.5 to 30 or 39.5 to 40 etc. are not allowed.
8. **Open end:** In an open-end distribution, the lower limit of the very first class or upper limit of the last class is not given. For example, while stating the distribution of monthly salary of managers in rupees, one may specify class limits as, below 10000, 10000-15000, 15000-20000, 20000-25000, above 25000. Similarly, while recording weights of college students in kg as grouped data the class intervals could be less than 40, 40 to 50, 50 to 60, 60 to 70, 70 to 80, greater than 80.

Notes

9. **Unequal class interval:** The method is also used to limit the class intervals where the width of the classes is not equal for all classes. This method is of practical use when there are large gaps in the data, or distribution of the data is uneven. It is used for explaining, visualizing and plotting data with unequal class interval. However, we must adjust formulae for calculations accordingly.

Cumulative and Relative Frequency

In many situations rather than listing the actual frequency opposite each class, it may be appropriate to list either cumulative frequencies or relative frequencies or both.

Cumulative Frequencies

The cumulative frequency of a given class interval thus, represents the total of all the previous class frequencies including the class against which it is written.

Relative Frequencies

Relative frequency is obtained by dividing the frequency of each class by the total number of observations i.e. the total frequency.

- If the relative frequency is multiplied by 100, we get the percentage frequency.
- There are two important advantages in looking at relative frequencies (percentages) instead of the absolute frequencies in a frequency distribution. These are:
- Relative frequencies facilitate the comparison of two or more than sets of data.
- Relative frequencies constitute the basis of understanding the probability concept.

Example: Age of 50 employees is given. Find cumulative frequency, relative frequency and percentage frequency.

22	21	37	33	28	42	56	33	32	59
40	47	29	65	45	48	55	43	42	40
37	39	56	54	38	49	60	37	28	27
32	33	47	36	35	42	43	55	53	48
29	30	32	37	43	54	55	47	38	62

Class Interval	Class Frequency	Cumulative Frequency	Relative Frequency	Percentage Frequency
20-30	7	$(0+7) = 7$	$7/50 = 0.14$	14
30-40	16	$(7+16) = 23$	$16/50 = 0.32$	32
40-50	15	$(23+15) = 38$	$15/50 = 0.30$	30
50-60	9	$(38+9) = 47$	$9/50 = 0.18$	18
60-70	3	$(47+3) = 50$	$3/50 = 0.06$	6
	$N = f = 50$		Total = 1	Total = 100

A frequency distribution is constructed to satisfy three objectives: (i) to facilitate the analysis of data, (ii) to estimate frequencies of the unknown population distribution from the distribution of sample data, and (iii) to facilitate the computation of various statistical measures.

4.1.2 Graphical Representation

Methods of Re-presentation of Data

One of the most convincing and appealing ways in which statistical results may be represented is through graphs and diagrams.

Diagrams and graphs are extremely used because of the following reasons:

- (i) Diagrams and Graphs attract to the eye.
- (ii) They have more memorizing effect.
- (iii) It facilitates for easy comparison of data from one period to another.
- (iv) Diagram and graphs give bird's eye view of entire data; therefore, it conveys meaning very quickly.

a. Bar Diagram

In a bar diagram, only the length of the bar is taken into account but not the width. In other words bar is a thick line whose width is merely shown, but length of the bar is taken into account and is called one-dimensional diagram.

Simple Bar Diagram

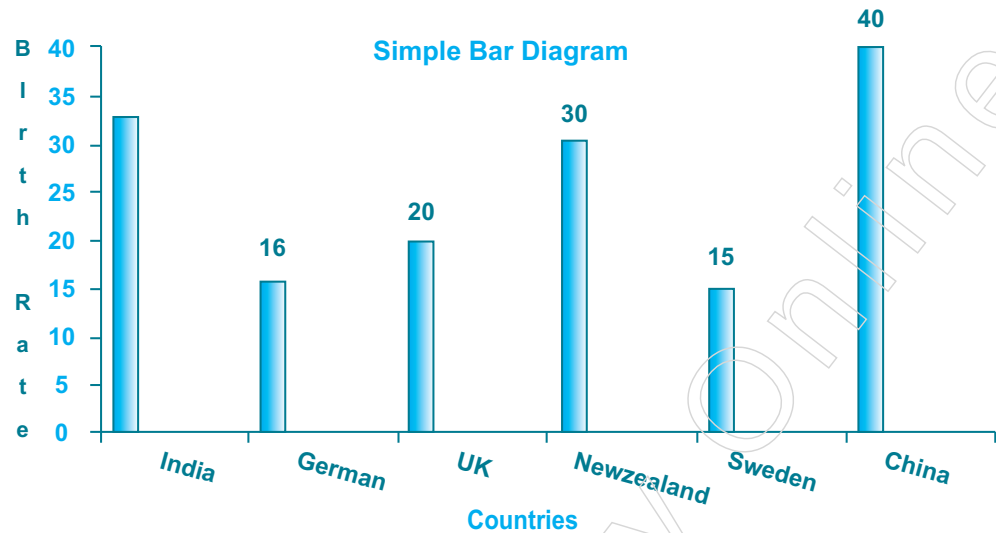
It represents only one variable. Since these are of the same width and vary only in lengths (heights), it becomes very easy for a comparative study. Simple bar diagrams are very popular in practice. A bar chart can be either vertical or horizontal; for example sales, production, population figures etc. for various years may be shown by simple bar charts

Illustration - 1

The following table gives the birth rate per thousand of different countries over a certain period of time.

Country	India	Germany	U. K.	New Zealand	Sweden	China
Birth Rate	33	16	20	30	15	40

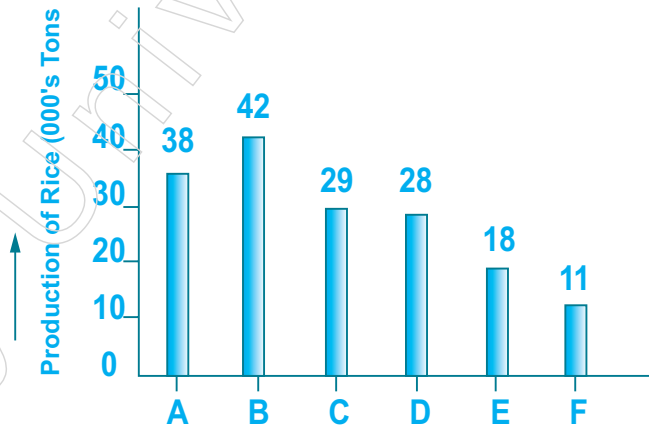
Notes



Comparing the size of bars, China's birth rate is highest, next is India whereas Germany and Sweden equal in the lowest positions.

Illustration 2 - Represent the data by using a simple bar diagram.

Countries:	A	B	C	D	E	F
Production of Rice (000's tons):	38	42	29	28	18	11



Sub-divided BarDiagram

In a subdivided bar diagram, each bar representing the magnitude of given value is further subdivided into various components. Each component occupies a part of the bar proportional to its share in total.

Illustration -

Present the following data in a sub-divided bar diagram.

Notes

Year/Faculty	Science	Humanities	Commerce
2014-2015	240	560	220
2015-2016	280	610	280

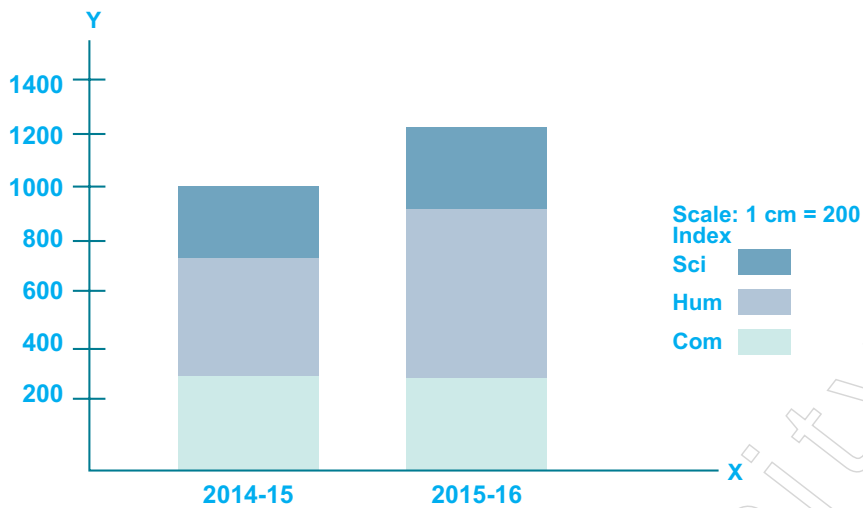
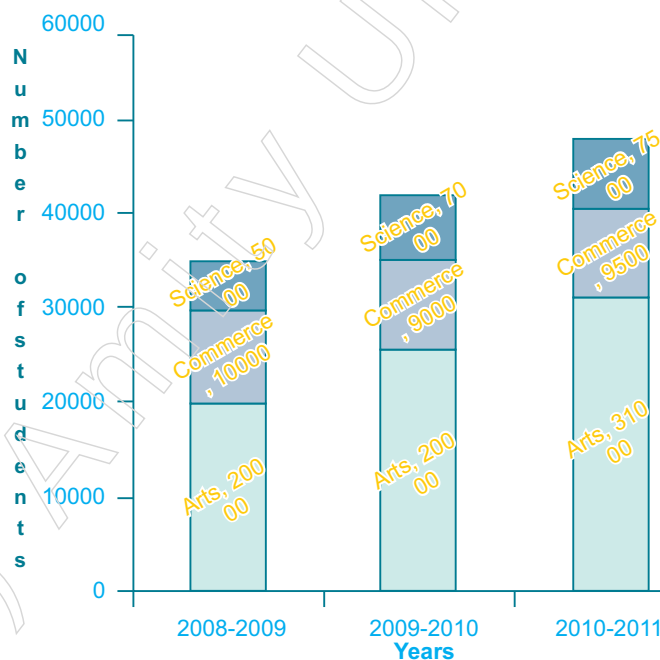


Illustration – 2

The Number of Students in University X during 2008 to 2011 areas follows. Represent the data by a similar diagram.

Year	Arts	Commerce	Science	Total
2008 - 2009	20,000	10,000	5,000	35,000
2009 - 2010	26,000	9,000	7,000	42,000
2010 - 2011	31,000	9,500	7,500	48,000



Notes

Multiple Bar Diagram

In a multiple bar diagram, two or more sets of related data are represented and the components are shown as separate adjoining bars. The height of each bar represents the actual value of the component. The components are shown by different shades or colours.

Illustration 1 - Construct a suitable bar diagram for the following data of number of students in two different colleges in different faculties.

College	Arts	Science	Commerce	Total
A	1200	800	600	2600
B	700	500	600	1800

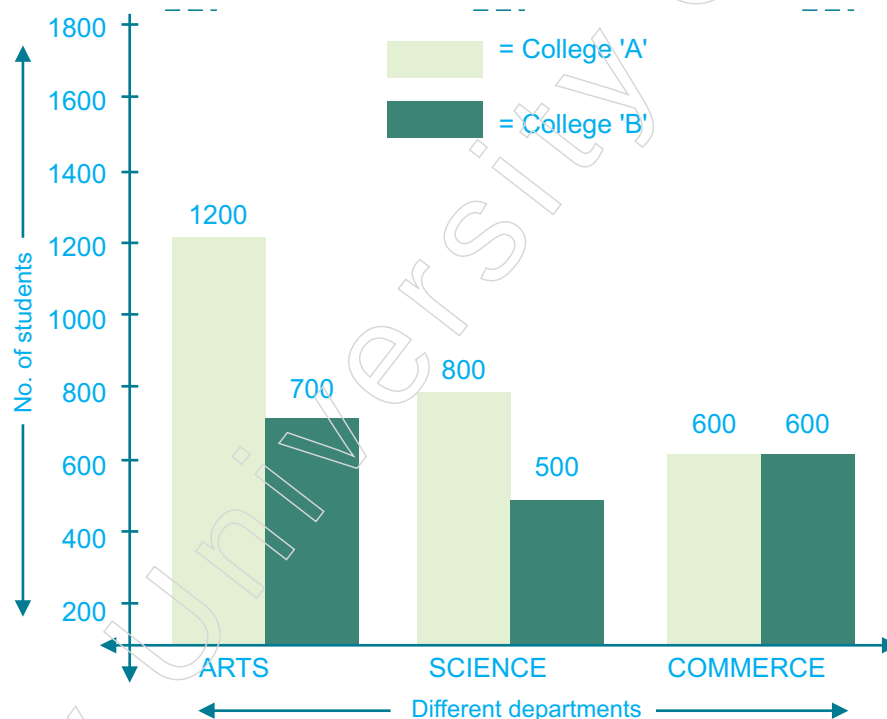


Fig: A multiple bar diagram showing numbers of students in two different colleges in different departments.

Illustration 2

Read the following data of results of III semester. B.B.A. examination of Mangalore University held in May 2006, 2007 and 2008 in a multiple bar diagram

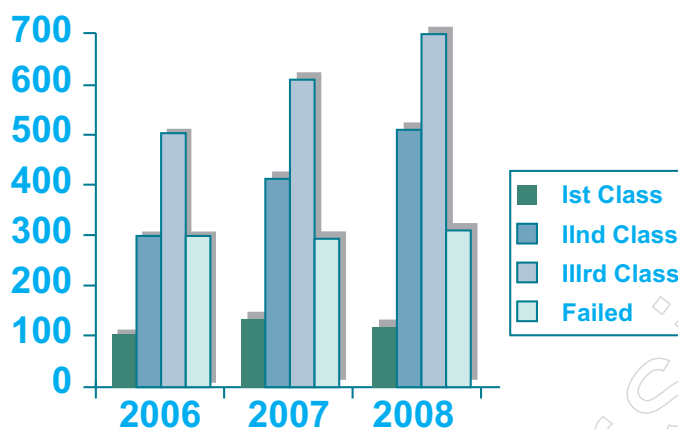
Year	Class I	Class II	Class III	Failed
2006	100	300	500	300
2007	120	400	600	280
2008	100	500	700	300

Percentage bar Diagram

In percentage bar diagram the length of the entire bar kept equal to 100 (Hundred). Various segment of each bar may change and represent percentage on an aggregate.

Illustration 1

Year	Men	Women	Children
1995	45%	35%	20%
1996	44%	34%	22%
1997	48%	36%	16%



- axis on the graph paper. Make sure to write the title above the table so that it determines the purpose of the graph.
- For instance, if one of the factors is time, it goes on the horizontal axis, referred to as the x-axis. The other factor would subsequently go on the vertical axis, which is known as the y-axis. Both the axes are to be labeled as per their respective factors. For example, The x axis can be labeled as time or day.
- Afterward, with the help of the already given data, the exact values on the graph can be pointed. Once the points are joined, a clear inference about the trend can be made.

Pie Chart

A pie chart or a circle chart is a circular statistical graphic that is divided into slices to illustrate a numerical proportion. In a pie chart, the arc length of each slice is proportional to the quantity it represents. While it is named for its resemblance to a pie which has been sliced, there are variations on the way it can be presented..In a pie chart, categories of data are represented by wedges in the circle and are proportional in size to the percent of individuals in each category.

Pie charts are very widely used in the business world and the mass media. Pie charts are generally used to show percentage or proportional data and usually the percentage represented by each category is provided next to the corresponding slice of pie. Pie charts are good for displaying data for around six categories or fewer.

Example:

Show the following data of expenditure of an average working class family by a suitable diagram

Notes

Notes

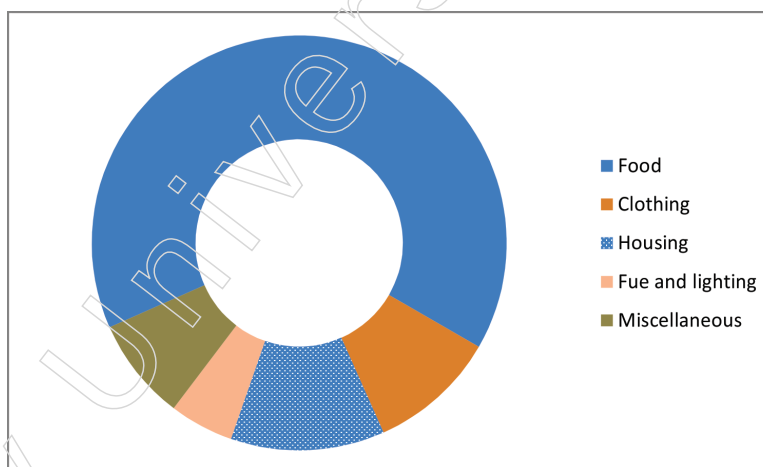
Item of Expenditure	Percent of Total Expenditure
Food	65
Clothing	10
Housing	12
Fuel and Lighting	5
Miscellaneous	8

Solution:

1. Food = $65/100 \times 360 = 234$
2. Clothing = $10/100 \times 360 = 36$
3. Housing = $12/100 \times 360 = 43.2$
4. Fuel and Lighting = $5/100 \times 360 = 18$
5. Miscellaneous = $8/100 \times 360 = 28.8$

The angles of different sectors are calculated as shown below:

Food Pie Chart



4.1.3 Histogram

A histogram consists of contiguous boxes and has both horizontal axis and a vertical axis. The horizontal axis is labeled with what the data represents (for instance, distance from your home to school). The vertical axis is labeled either Frequency or relative frequency. The graph will have the same shape with either label. The histogram (like the stemplot) can give you the shape of the data, the center, and the spread of the data. (The next section tells you how to calculate the center and the spread.)

The relative frequency is equal to the frequency for an observed value of the data divided by the total number of data values in the sample. (In the chapter on Sampling and Data (Section 1.1), we defined frequency as the number of times an answer occurs.)

$$RF = f/n$$

Notes

Where f is the frequency n is the total number of data values (or the sum of the individual frequencies), and RF is the relative frequency.

Example – If 3 students in Mathematics class of 40 students received from 90% to 100%, then,

$$f = 3, n = 40 \text{ and}$$

$$RF = f/n$$

$$= 3/40$$

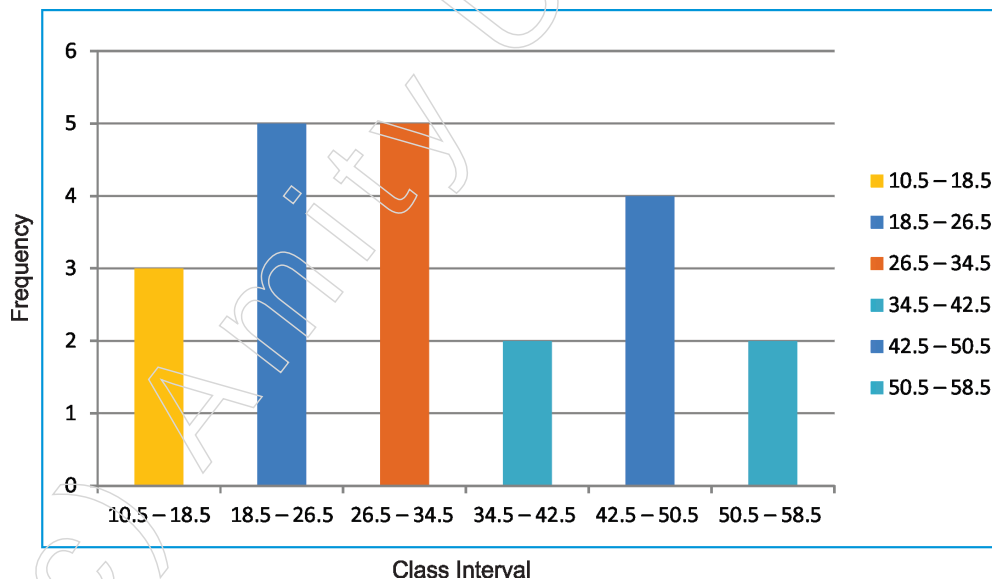
$$= 0.075$$

Seven and a half percent of the students received 90% to 100%. Ninety percent to 100% are quantitative measures.

Example:

Formulate the Histogram from the following data –

Class Interval	Frequency
10.5 – 18.5	3
18.5 – 26.5	5
26.5 – 34.5	5
34.5 – 42.5	2
42.5 – 50.5	4
50.5 – 58.5	2

Solution:**Histogram**

Notes

4.1.5 Ogive – Part – 1

Ogives

When frequencies are added, they are called the cumulative frequencies. The curve obtained by plotting cumulating frequencies is called a cumulative frequency curve or an ogive (pronounced as ojive).

To construct an Ogive: (i) Add up the progressive totals of frequencies, class by class, to get the cumulative frequencies. (ii) Plot classes on the horizontal (x-axis) and cumulative frequencies on the vertical (y-axis).

Less than Ogive: To plot a less than ogive, data is arranged in ascending order of magnitude and frequencies are cumulated from the top i.e. adding. Cumulative frequencies are plotted against the upper class limits. Ogives under this method, gives a positive curve

Greater than Ogive: To plot a greater than ogive, the data is arranged in the ascending order of magnitude and frequencies are cumulated from the bottom or subtracted from the total from the top. Cumulative frequencies are plotted against the lower class limits. Ogives under this method, gives negative curve

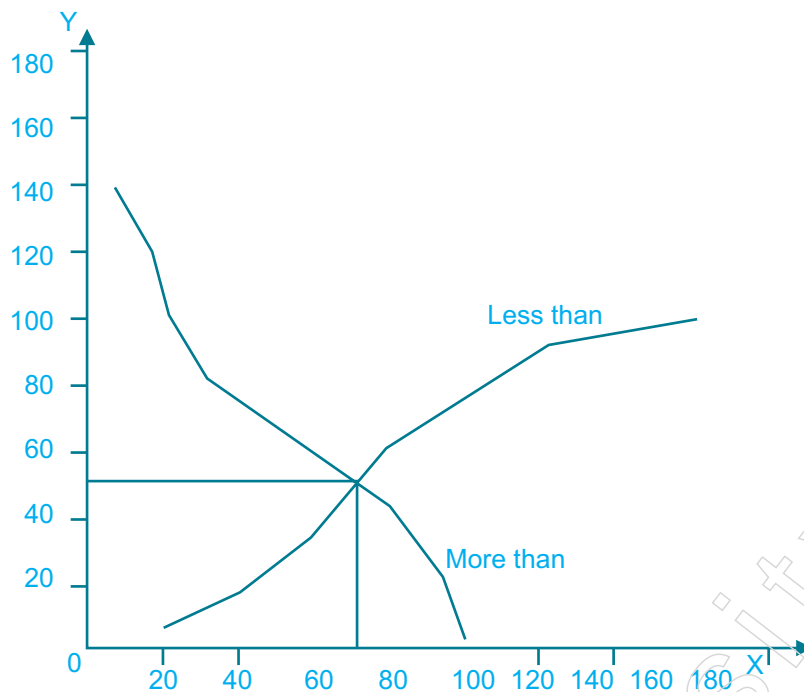
Uses: Certain values like median, quartiles, quartile deviation, co-efficient of skewness etc. can be located using ogives. Ogives are helpful in the comparison of the two distributions.

Illustration 1 –

Draw less than and more than ogive curves for the following frequency distribution and obtain median graphically. Verify the result.

CI	0-20	20-40	40-60	60-80	80-100	100-120	120-140	140-160
f	5	12	18	25	15	12	8	5

Size	lcf	Mcf	Size
20	5	100	0
40	17	95	20
60	35	83	40
80	60	65	60
100	75	40	80
120	87	25	100
140	95	13	120
160	100	5	140



Notes

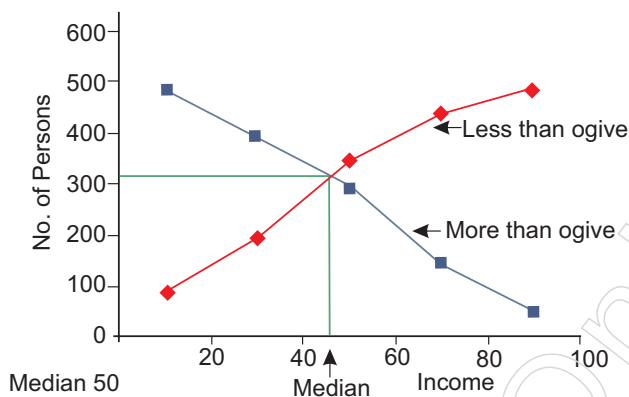
Example: Find the median from the following series. Also draw less than ogive, more than ogive and locate median on a graph.

Income (₹)	No. of Persons
0-20	82
20-40	112
40-60	150
60-80	95
80-100	48

Solution:

C.I.	F	Class (Less than)	L.C.F.	Class (More than)	M.C.F.
0-20	82	20	82	0	487
20-40	112	40	194	20	405
40-60	150	60	344	40	293
60-80	95	80	439	60	143
80-100	48	100	487	80	48

Notes



4.1.6 Ogive – Part 2

Cumulative Frequency Curve or the Ogive

If from a cumulative frequency table, the upper limits of the class taken as x-coordinates and the cumulative frequencies as the y-coordinates and the points are plotted, then these points when joined by straight lines, we obtain less than type cumulative frequency polygon.

If more than cumulative frequency is plotted against the corresponding lower limits of each class and the points plotted are joined by straight lines, we obtain more than type cumulative frequency polygon.

However, when the points plotted are joined by a free hand smooth curve, we obtain cumulative frequency curve.

Example: A cumulative frequency polygon for the given following data

Class	Frequency	Cumulative frequency
0-10	2	2
10-20	4	6
20-30	10	16
30-40	4	20
40-50	3	23
50-60	8	31
60-70	1	32
70-80	5	37
80-90	11	48
90-100	2	50

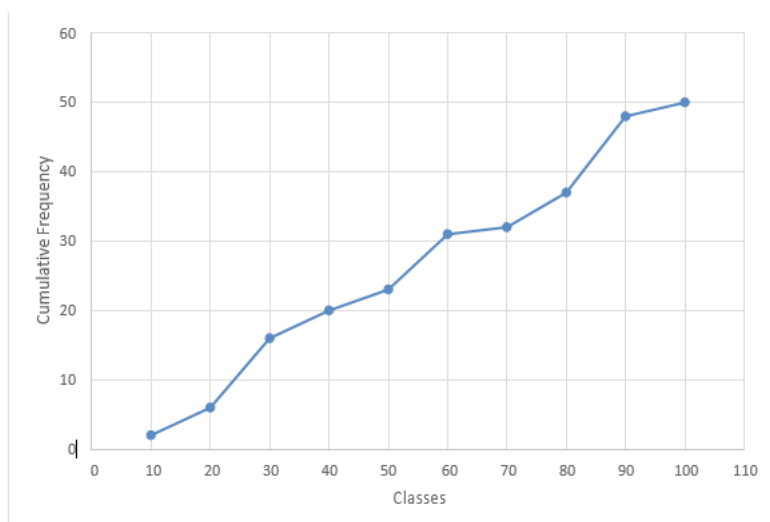


Fig. 5.1.4 Cumulative Frequency

Activity1: Construct an ogive curve for the following frequency distribution of Cotton Mills in Bombay according to the quantities of cotton consumed-

Cotton consumed in thousand candles	
No. of Mills	
0-2	5
2-4	13
4-6	12
6-8	11
8-10	8
10-12	4
12-14	1
14-16	3
16-18	1
18-20	1
over 20	2

The following table shows the frequency distribution for the number of students per teacher in 750 colleges and professional schools-

Students x	Frequency F
1	7
4	46
7	165
10	195
13	189
16	89

Notes

Notes

19	28
22	19
2	9
528	3

Check your Understanding

- _____ helps in forecasting by analysing trends, which are essential for planning and decision-making.
- _____ are the statements about populations parameters based on past knowledge or information.
- Inductive _____ about the population based on the sample estimates involves an element of risk.
- Statistics deals with only those problems which can be expressed in quantitative terms and amenable to mathematical and numerical analysis. True / False
- _____ statistics is used to sum up and graph the data for a category picked.
- _____ refers to the grouping of data into homogeneous classes and their categories.
- In a _____ graph, only the length of the bar is taken into account but not the width.
- A _____ graph is a type of chart used to show information changing over time.
- A _____ chart or a circle chart is a circular statistical graphic that is divided into slices to illustrate a numerical proportion.
- A _____ consists of contiguous boxes and has both horizontal axis and a vertical axis.
- The _____ frequency is equal to the frequency for an observed value of the data divided by the total number of data values in the sample.
- The curve obtained by plotting cumulating frequencies is called a cumulative frequency curve or an ogive. True / False

Summary

- Statistics originally was meant for collection of facts useful for affairs of the state, like the taxes, land records, population demography, etc.
- "Statistics are the classified facts representing the conditions of the people in the state. Specially those facts which can be stated in number or in table of numbers or in any tabular or classified arrangement."
- The collection and analysis of data constitute the primary stages of execution of any statistical investigation. The procedure for collection of data depends upon various considerations such as the scope, objective, nature of investigation, etc. Data may be collected either from a primary or secondary source.
- Descriptive statistics is used to sum up and graph the data for a category picked.

Notes

This method helps to understand a particular collection of observations. There is no confusion in concise numbers, since you just identify the individuals or things which are calculated.

- Inferential statistics are techniques that allow us to use certain samples to generalize the populations from which the samples were taken. Hence, it is crucial that the sample represents the populations accurately.
- Statistical tools are extensively used in business for time and motion study, consumer behaviour study, investment decisions, credit ratings, performance measurements and compensations, inventory management, accounting, quality control, distribution channel design, etc. For managers, therefore, understanding statistical concepts and knowledge about using statistical tools is essential.
- Classification refers to the grouping of data into homogeneous classes and their categories. It is the process of arranging things in groups or classes as per their resemblances and affinities.
- One of the most convincing and appealing ways in which statistical results may be represented is through graphs and diagrams.
- A line graph is a type of chart used to show information changing over time. We use multiple dots to plot line graphs connected by straight lines. It is also known as a line chart. It consists of two axes defined as the 'x' and 'y' axis.
- A pie chart or a circle chart is a circular statistical graphic that is divided into slices to illustrate a numerical proportion. Pie Charts are widely used in the business world and the mass media. They are generally used to show percentages or proportional data and usually the percentages represented by each category is provided next to the corresponding slice of pie.
- A histogram consists of contiguous boxes and has both horizontal axis and a vertical axis. The horizontal axis is labelled with what the data represents and the vertical axis is labelled either Frequency or relative frequency.
- When frequencies are added, they are called the cumulative frequencies. The curve obtained by plotting cumulating frequencies is called a cumulative frequency curve or an ogive.

Activity

Create the cumulative frequency table and draw the Ogive for the data below.

Marks	1-10	11-20	21-30	31-40	41-50	51-60	61-70	71-80
Frequency	3	8	12	14	10	6	5	2

Questions and Exercises

1. Describe the functions of statistics.
2. What are the two forms of collected data? Give examples of each.
3. List the different methods of collecting primary data.
4. What are the rules of classification?
5. What type of data is best represented by a pie chart?

Notes

6. How does one construct an ogive?

Glossary

- **Classification:** the grouping of data into homogenous classes and categories.
- **Line Graph:** is a type of chart used to show information changing over time.
- **Primary Data:** when the data used in a statistical study is collected under the control and supervision of the investigator.
- **Relative frequency:** is equal to the frequency for an observed value of the data divided by the total number of data values in the sample.
- **Secondary Data:** when data is derived from other sources that have not been collected for the specific purpose of research.
- Further Reading
- Bekes, Gabor. Data Analysis for Business, Economics, and Policy. Cambridge University Press (May 6, 2021)
- Huberman, A. Michael; Miles, Matthew B.; Saldana, Johnny. Qualitative Data Analysis: A Methods Sourcebook
- Knaflic, Cole Nussbaumer. Storytelling with Data: A Data Visualization Guide for Business Professionals. Wiley; 1st edition (November 2, 2015)

Check your Understanding: Answers

1. Statistics
2. Hypotheses
3. Inference
4. True
5. Descriptive
6. Classification
7. Bar
8. Line
9. Pie
10. Histogram
11. Relative
12. True

Unit - 4.2: Descriptive Measures

Notes

Objectives:

At the end of this unit, you will be able to understand

- Measure of the Central Tendency – I
- Measure of the Central Tendency – II
- Measure of Dispersion
- Kurtosis, Skewness

Introduction

Measures of central tendency are a single value which can be considered as representative of a set of observations. The value around which the observations can be considered as centered is known as an Average or average value or a location center. Since such representative values tend to lie centrally within a set of observations when arranged according to magnitudes, these averages are then called measures of central tendency.

4.2.1 Measure of the Central Tendency – I

Central Tendency Measures

Central tendency has three main measures: mode, median and mean. Each of those measurements represents a specific indication of the distribution's typical or central value.

- **Mean** - The mean is the average of the numbers. It is easy to calculate: add up all the numbers, then divide by how many numbers there are. In other words it is the sum divided by the count
- **Median** - Within a sorted, ascending or descending list of numbers, the median is the middle number and may be more representative of that set of data than the average. The median is often used as opposed to the mean when the series includes outliers that may distort the average of values.
- **Mode** - The mode is the number most frequently seen in a dataset. A collection of numbers may have one mode, one mode, or no mode at all. Other popular central tendency measurements include a set's mean, or mean, and a set 's median, middle value.

Average

An average is a single figure that sums up the characteristics of a whole group of figures.

In the words of Clark "average is an attempt to find one single figure to describe whole of figures. An average is described as a measure of central tendency as it is more or less a central value around which various values cluster.

In the world of CROXTON and COWDEN "an average is a single value within the

Notes

range of the data that is used to represent all of the values in the series. Since an average is somewhere within the range of the data, it is called a measure of central value.

Objectives served by Averages

Averages serve the following purposes:

1. To obtain a clear and concise picture of large number of numerical data.
2. To compare different groups by the means of averages.
3. To obtain a clear picture of a whole group studying sample data.
4. To provide definite rates to the relationship between different groups.

Characteristics of good average

1. It is rigidly defined and its value is always definite.
2. It is easy to understand and calculate, hence it is very popular.
3. It is based on all the observations; so that it can become a good representative.
4. It can be easily used for comparisons.
5. It is capable of further algebraic treatments, like finding the sum of the observation values. Finding the mean and total number of the observations, and finding the combined arithmetic mean when different groups are given etc.
6. It is not affected much by sampling fluctuations.

Essentials of a good Average

The essentials of a good average are as follows:

1. It must be defined rigidly.
2. It must be based on all the observation of the data.
3. It must be readily comprehensible or understandable.
4. It must be capable of being calculated with reasonable ease and rapidity.
5. It must be affected as little as possible by fluctuations of sampling.
6. It must be readily amenable to arithmetic or algebraic treatment.

Arithmetic Mean

Arithmetic mean is defined as the value obtained by dividing the total values of all items in the series by their number. In other word is defined as the sum of the given observations divided by the number of observations, i.e., add values of all items together and divide this sum by the number of observations.

Symbolically – $\bar{x} = \frac{x_1 + x_2 + x_3 + \dots + x_n}{n}$

Properties of Arithmetic Mean

1. The sum of the deviations, of all the values of x , from their arithmetic mean, is zero.
2. The product of the arithmetic mean and the number of items gives the total of all items.

- Finding the combined arithmetic mean when different groups are given.

Demerits of Arithmetic Mean

- Arithmetic mean is affected by the extreme values.
- Arithmetic mean cannot be determined by inspection and cannot be located graphically.
- Arithmetic mean cannot be obtained if a single observation is lost or missing.
- Arithmetic mean cannot be calculated when open-end class intervals are present in the data.

Arithmetic Mean for Ungrouped Data

A) Individual Series

1. Direct Method

The following steps are involved in calculating arithmetic mean under an individual series using direct method:

- Add up all the values of all the items in the series.
- Divide the sum of the values by the number of items. The result is the arithmetic mean.

The following formula is used: $X = \frac{\sum x}{N}$

Where, X = Arithmetic mean $\sum x$ = Sum of the values N = Number of items.

Illustration 1 – Value(x) – 125 128 132 135 140 148 155 157 159 191

Calculate the arithmetic mean

Solution –

Total number of terms = $N = 10$

Mean = $\frac{\sum x}{N} = \frac{125 + 128 + 132 + 135 + 140 + 148 + 155 + 157 + 159 + 191}{10} = 144$

$X = \frac{\sum x}{N} = \frac{1440}{10}$

$= 144$

2. Short-cut Method or Indirect method

The following steps are involved in calculating arithmetic mean under individual series using short-cut or indirect method:

- Assume one of the values in the series as an average. It is called as working mean or assumed average.
- Find out the deviation of each value from the assumed average.
- Add up the deviations
- Apply the following formula. $X = A + \frac{\sum d}{N}$

where, X = Arithmetic mean A = Assumed average $\sum d$ = Sum of the deviations

Notes

Notes

N = Number of items

Illustration - 1

Calculate the arithmetic average of the data given below using short-cut method

Roll No	1	2	3	4	5	6	7	8	9	10
Marks	43	48	65	57	31	60	37	48	78	59

Solution –

Roll No	Marks Obtained	D = 60
1	43	-17
2	48	-12
3	65	5
4	57	-3
5	31	-29
6	60	0
7	37	-23
8	48	-12
9	78	18
10	59	-1
		$\Sigma d = -74$

$$X = a + \Sigma d/N$$

$$+ 60 + (-74/10) = 52.6 \text{ marks}$$

2.1.5 Combined Arithmetic Mean

Arithmetic mean and number of items of two or more related groups are known as combined mean of the entire group. The combined average of two series can be calculated by the given formula –

$$n_1x_1 + n_2x_2 / n_1 + n_2$$

Where, n_1 = No. of items of the first group, n_2 = No. of items of the second group

x_1 = A.M of the first group, x_2 = A.M of the second group,

Example - From the following data ascertain the combined mean of a factory consisting of 2 branches namely branch A and Branch B. In branch A the number of workers is 500, and their average salary of 300. In branch B the number of workers is 1,000 and their average salary is 250

Solution:

Let the no. of workers in branch A be $n_1 = 500$

Let the no. of worker in branch B be $n_2 = 1000$

Average salary $x_1 = 300$

Average salary $x_2 = 250$

$n_1x_1 + n_2x_2 / n_1 + n_2$

$= 500(300) + 1000(250) / 500 + 1000$

$= 1,50,000 + 2,50,000 / 1500$

$= 266.66$

Weighted Arithmetic Mean

Some times, some observations get relatively more importance than other observations. The weight for such observation must be given on the basis of their relative importance. In weighted arithmetic mean, for finding an average the value of each item is multiplied by its weight and then the product are divided by the number of weights.

Symbolically $= \sum wx / \sum w$

Example – Calculate simple and weighted average from the following data –

Month	Jan	Feb	March	April	May	June
Price	42.5	51.25	50	52	44.25	54
No. of tonnes	25	30	40	50	10	45

Solution:

Month	Price Per Tonn	No. of tonnes (in 000)(x)	WX purchased (w)
Jan	42.5	25	1062.5
Feb	51.25	30	1537.5
March	50	40	2000
April	52	50	2600
May	44.25	10	442.5
June	54	45	2430
N = 6	X = 294	$\sum w = 200$	$\sum wx = 10027.5$

Simple AM

$X = \sum x / n = 294 / 6 = 49$

Weighted AM

$X_w = \sum wx / \sum w = 10027.5 / 200 = 50.137$

The correct average price paid is `50.30 and not `49 i.e., weight arithmetic mean is correct than simple arithmetic mean.

Notes

Median

Median is defined as the value of the item dividing the series into two equal halves, where one half contains all values less than (or equal to) it and the other half contains all values greater than (or equal to) it. It is also defined as the “central value of the variable. In median, the value of items must be arranged in order of their size or magnitude to find out the median.

Median is a positional average. The term position refers to the place of a value in the series, where the place of median is such that it is equal to the number of items lying on the either side; therefore it is also called as locative average.

Merits of Median

Following are the advantages of median:

1. It is rigidly defined.
2. It is easy to calculate and understand.
3. It can be located graphically.
4. It is not affected by extreme values like the arithmetic mean.
5. It can be found by mere inspection.
6. It can be used for qualitative studies.
7. Even if the extreme values are unknown, median can be calculated if one knows the number of items.

Demerits of Median

Following are the disadvantages of median:

1. In the case of individual observations, the values are to be arranged in order of their size to locate median. Such an arrangement of data is a tedious task if the number of items is large.
2. If the median is multiplied by the number of items, the total value of all the items cannot be obtained as in the case of the arithmetic average.
3. It is not suitable for complex algebraic or mathematical treatment.
4. It is more affected by sampling fluctuations.

Application of Median

Example – Determine the median from the following –

25, 15, 23, 40, 27 25 23 25 20

Solution - Arranging the figures in ascending order –

S. No	Value or Size
1	15
2	20
3	23

4	23
5	25
6	25
7	25
8	27
9	40

Median = $10/2$
 = 5th term
 = 25

Example:

The following steps are involved in calculating median in continuous series:

- 1) Find out the cumulative frequency
- 2) Find out the median item, i.e., $N/2$ th item.
- 3) Find out the group or class containing the median
- 4) Estimate the median applying the following formula.

$$Me = i + \frac{\frac{n}{2} - cf}{fm} \times i$$

where me = Median

i = Lower limit of the median class

cf = Cumulative frequency of the class preceding the median class

i = Size of class interval

fm = Middle one of frequency sequence.

Example 1:

Calculate the median mark from the following frequency distribution.

Mark	No. of students
0-10	5
0-20	13
0-30	20
0-40	32
0-50	60
0-60	80
0-70	90

Notes

Notes

Solution:

Mark	F	CF
0-10	5	5
10-20	6	13
20-30	7	20
30-40	12	32
40-50	28	60
50-60	20	80
60-70	10	90

$$M = \frac{N}{2} = \frac{90}{2} = 45$$

$$Me = 1 + \frac{\frac{n}{2} - cf}{fm} \times xi = 40 + \frac{50 - 40}{28}$$

$$= 40 + \frac{10}{28} \times 13$$

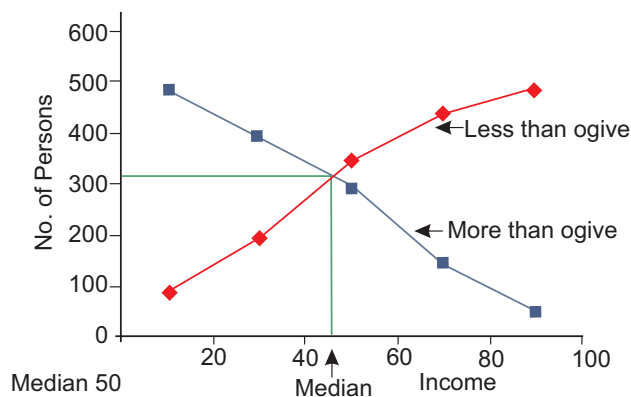
$$= 40 + 4.64 = 44.64$$

Find the median from the following series. Also draw less than ogive, more than ogive and locate median on a graph.

Income (₹)	No. of Persons
0-20	82
20-40	112
40-60	150
60-80	95
80-100	48

Solution:

C.I.	F	Class (Less then)	L.C.F.	Class (More then)	M.C.F.
0-20	82	20	82	0	487
20-40	112	40	194	20	405
40-60	150	60	344	40	293
60-80	95	80	439	60	143
80-100	48	100	487	80	48



Notes

Mode

The word “mode” is derived from the French word “la mode” which means fashion. So it can be regarded as the most fashionable item in the series or the group.

Croxtan and Cowden regard mode as “the most typical of a series of values”. As a result it can sum up the characteristics of a group more satisfactorily than the arithmetic mean or median. Mode is defined as the value of the variable occurring most frequently in a distribution. In other words it is the most frequent size of item in a series.

Merits of Mode

The following are the merits of mode:

1. The most important advantage of mode is that it is usually an actual value.
2. In the case of discrete series, mode can be easily located by inspection.
3. Mode is not affected by extreme values.
4. Mode can be determined even if extreme values are not given.
5. It is easy to understand and this average is used by people in their every day speech.

Demerits of Mode

The following are the demerits of mode:

1. It is not based on all the observation of the data
2. In a number of cases there will be more than one mode in the series.
3. If mode is multiplied by the number of items, the product will not be equal to the total value of the items.
4. It will not truly represent the group if there are a small number of items of the same size in a large group of items of different sizes
5. It is not suitable for further mathematical treatment

Applications of Mode

Mode in Ungrouped Data

a) Individual Series

The mode of this series can be obtained by mere inspection. The number which occurs most often is the mode.

Notes

Illustration - 1

Locate mode in the data 7, 12, 8, 5, 9, 6, 10, 9, 4, 9, 9

Solution:

On inspection, it is observed that the number 9 has maximum frequency i.e., repeated maximum of 4 times than any other number. Therefore mode (Z) = 9

b) Discrete Series

The mode is calculated by applying grouping and analysis table.

- Grouping Table: Consisting of six columns including frequency column, 1st column is the frequency 2nd and 3rd column is the grouping two way frequencies and 4th, 5th and 6th column is the grouping three way frequencies.
- Analysis table: consisting of two columns namely tally bar and frequency

Steps in Calculating Mode in Discrete Series

The following steps are involved in calculating mode in discrete series:

- Group the frequencies by two's.
- Leave the frequency and group the other frequencies into two's.
- Group the frequencies into three's.
- Leave the frequency of the first size and add the frequencies of other sizes in three's.
- Leave the frequencies of the first two sizes and add the frequencies of the other sizes in three's.
- Prepare an analysis table to know the size occurring the maximum number of times. Find out the size, which occurs the largest number of times. That particular size is the mode.

c) Continuous Series

The following steps are involved in calculating mode in continuous series.

- Find out the modal class. Modal class can be easily found out by inspection. The group containing maximum frequency is the modal group. Where two or more classes appear to be a modal class group, it can be decided by grouping process and preparing an analyzed table as was discussed in question number 2.102.
- The actual value of mode is calculated by applying the following formula.

$$Mo = l + \frac{fm - f_1}{2fm - f_1 - f_2}i$$

Example:

Marks	F	CF
0-10	5	5

Calculate the modal wages,

Daily wages in ₹ (x) : 20-25 25-30 30-35 35-40 40-45 45-50

No. of workers (f) : 1 2 8 12 7 5

Solution:

Here, the maximum frequency is 12, corresponding to the class interval (35-40) which is the modal class, Therefore, L1=35 L2=40 F1=12 FM=8 F2=7

X	F
20-25	1
25-30	3
30-35	8
35-40	12
40-45	7 f2
45-50	5

$$\text{Mode} = l + \frac{fm - f_1}{2fm - f_1 - f_2} \times i = 35 + \left(\frac{12 - 8}{2(12) - 8 - 7} \right) 5$$

$$= 35 + \left(\frac{4}{24 - 15} \right) 5 = 35 + \left(\frac{20}{9} \right) = 35 + 2.22 = 37.22$$

Example 2:

Less than	10	20	30	40	50	60	70	80
Frequency:	4	16	40	76	96	112	120	125

Solution:

Need to ascertain lower limit of the continuous class (LL = UL -) Class length (CL) = 20-10 = 10 i.e., (10-10 = 0.....)

Value (x)	Class		f
Less than 10	0-10	4	4
Less than 20	10-20	16-4=12	12
Less than 30	20-30	40-16=24	24
Less than 40	30-40	76-40=36	36
Less than 50	40-50	96-76=20	20
Less than 60	50-60	112-96=16	16
Less than 70	60-70	120-112=8	8
Less than 80	70-80	125-120=5	5

$$Z = l + \frac{fm - f_1}{2fm - f_1 - f_2} \times i = 30 + \left(\frac{36 - 24}{2(36) - 24 - 20} \right) 10$$

Notes

Notes

$$= 30 + \left(\frac{12}{72-44}\right) 10 = 30 + \left(\frac{120}{28}\right) = 30 + 4.285 \quad Z = 37.22$$

Empirical Relationship between Mean, Median and Mode

When mode is ill defined, it is difficult to find the value of mode, a sort of empirical relationship exist among the mean, median and mode in such a way that the median lies between the mode and the mean. The mode departs (to the left i.e., positive skewed) $\frac{2}{3}$ difference from the median and the mean departs (to the right i.e., negatively skewed) $\frac{1}{3}$ difference from the median. Karl Pearson's expressed this relationship as $Z = 3M - 2X$ (when it is positive skewness).

Example - M is 28, AM is 29 find Mode

Solution :

$$Z = 3M - 2X$$

$$= 3(28) - 2(29)$$

$$= 84 - 78$$

$$= 26$$

$$29 > 28 > 26$$

$$- M = ? \quad AM = 29 \quad Z = 36.5$$

Solution:

$$Z = 3M - 2X$$

$$= 36.5 = 3(M) - 2(29)$$

$$= 36.5 = 3M - 78$$

$$= 3M = -78 - 36.5$$

$$M = -114.5 / -3$$

$$= 38.16$$

Key Takeaways

- Measures of central tendency: It is a single value which can be considered as representative of a set of observations and around which the observations can be considered as Centered is called an 'Average' (or average value) or a Center of location.
- Average: It is described as a measure of central tendency as it is more or less a central value around which various values cluster. In the world of CROXTON and COWDEN "an average is a single value within the range of the data that is used to represent all of the values in the series."
- Median: It is defined as the value of that item which divides the series into two equal halves, one half contains all values less than (or equal to) it and the other half containing all values greater than (or equal to) it. It is also defined as the "central value of the variable."

4.2.2 Measure of the Central Tendency – II

There are four fundamental characteristics in which similar frequency distribution may differ. One of the characteristic is central tendency. A central tendency measure is a single value that attempts to describe a set of data by identifying the central position within that set of data. As a result, measures of central tendency are also known as measures of central location. They're also known as summary statistics. The mean (also known as the average) is probably the most familiar measure of central tendency, but there are others, such as the median and the mode. The mean, median, and mode are all valid measures of central tendency, but depending on the circumstances, some measures of central tendency are more appropriate to use than others.

Measures of central tendency are averages of the first order. Following are some general rules:

- The mean is the most commonly used and generally regarded as the best measure of central tendency. However, in some cases, either the median or the mode is preferable.
- When determining central tendency, the median is the preferred measure:
 - ◆ In the data distribution, there are a few outliers (Remember that a single outlier can have a significant impact on the mean).
 - ◆ Your data contains some missing or undetermined values.
 - ◆ There is an unrestricted distribution. (For example, if you have a data field that measures the number of children and your options are 0, 1, 2, 3, 4, 5 or "6 or more," the "6 or more" field is open ended and makes calculating the mean impossible because we do not know exact values for this field.
 - ◆ You have data that has been measured on an ordinal scale.
- When data is measured on a nominal (or, in some cases, ordinal) scale, mode is the preferred measure.

Arithmetic Mean

The mean (or average) is the most well-known and widely used measure of central tendency. It can be applied to both discrete and continuous data, but it is most commonly applied to continuous data. The sum of all the values in the data set divided by the number of values in the data set equals the mean. So, if we have 'n' values in a data set with values $x_1, x_2, \dots, x_n$, the sample mean, usually denoted by $\bar{x}$ (pronounced "x bar"), is:

$$\bar{x} = \frac{x_1 + x_2 + \dots + x_n}{n}$$

This formula is usually written in a slightly different style, with the Greek capital letter Σ , pronounced "sigma," which means "sum of...":

$$\bar{x} = \frac{\sum x}{n}$$

You may have noticed that the sample mean is mentioned in the preceding

Notes

formula. So, what is the significance of the term “sample mean”? This is because, in statistics, samples and populations have very different meanings, and these differences are very important, even though they are calculated in the same way in the case of the mean. To indicate that we are calculating the population mean rather than the sample mean, we use the Greek lower case letter “mu,” which is represented as μ :

$$\mu = \frac{\sum x}{n}$$

The mean is essentially a data set model. It is the most commonly used value. However, you will notice that the mean is not always one of the actual values in your data set. However, one of its most important characteristics is that it minimises error in predicting any single value in your data set. That is, it is the value in the data set that produces the least amount of error when compared to all other values in the data set. The mean has the important property of including every value in your data set as part of the calculation. Furthermore, the mean is the only measure of central tendency in which the sum of the deviations from the mean is always zero.

When mean should not be used?

The mean has one major drawback: it is especially vulnerable to the influence of outliers. These are values that are out of the ordinary in comparison to the rest of the data set because they are unusually small or large in numerical value. Consider the following wages for factory workers:

Staff	1	2	3	4	5	6	7	8	9	10
Salary	1500	1800	1600	1400	1500	1500	1200	1700	9000	9500

The average salary for these ten employees is Rs. 3070. However, an examination of the raw data suggests that this mean value may not be the best way to accurately reflect a worker's typical salary, as most workers earn between Rs.1200 and Rs. 1800. The two high salaries have skewed the mean. As a result, in this situation, we would like to have a more accurate measure of central tendency. As we will see later, taking the median is a better measure of central tendency in this case.

When our data is skewed, we usually prefer the median over the mean (or mode) (i.e., the frequency distribution for our data is skewed). When the data is perfectly normal, the mean, median, and mode are identical, according to the normal distribution, which is the most commonly used in statistics. Furthermore, they all represent the most common value in the data set. However, as the data becomes skewed, the mean loses its ability to provide the best central location for the data because the skewed data pulls it away from the average value. The median, on the other hand, best retains this position and is not as heavily influenced by the skewed values.

Median

The median is the middle score of a set of data arranged in order of magnitude. Outliers and skewed data have less of an impact on the median. Assume we have the following data to calculate the median:

65	55	89	56	35	14	56	55	87	45	92
----	----	----	----	----	----	----	----	----	----	----

Notes

We must first rearrange the data in descending order of magnitude (smallest first):

14	35	45	55	55	56	56	65	87	89	92
----	----	----	----	----	-----------	----	----	----	----	----

Our median score is the midpoint - in this case, 56. (highlighted in bold). It is the middle score because there are five scores before it and five scores after it. When you have an odd number of scores, this works fine, but what happens when you have an even number of scores? What if you only had ten points? Simply take the middle two scores and average the result. So, consider the following example:

65	55	89	56	35	14	56	55	87	45
----	----	----	----	----	----	----	----	----	----

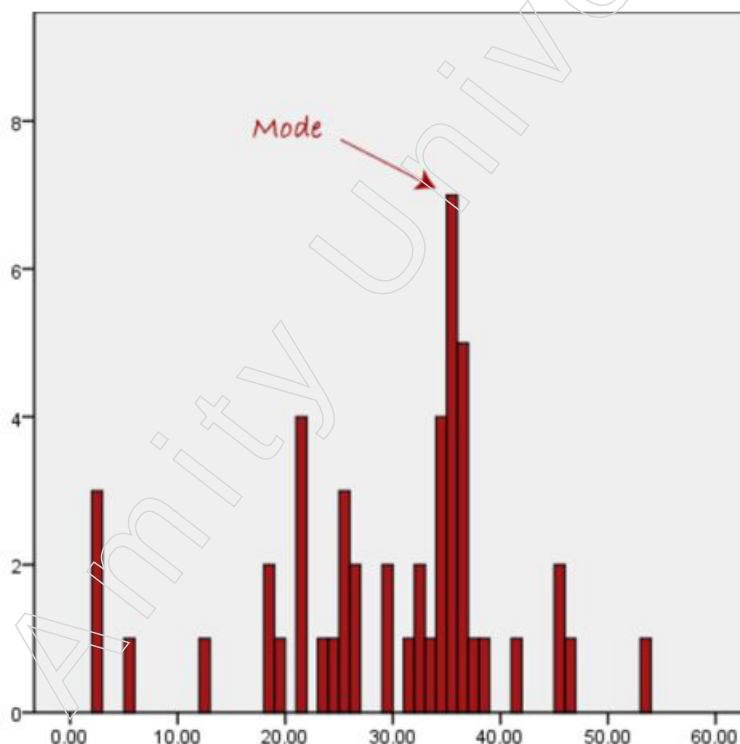
We reorder the data in the following order of magnitude (smallest first):

14	35	45	55	55	56	56	65	87	89
----	----	----	----	-----------	-----------	----	----	----	----

Only now do we need to average the fifth and sixth scores in our data set to get a median of 55.5.

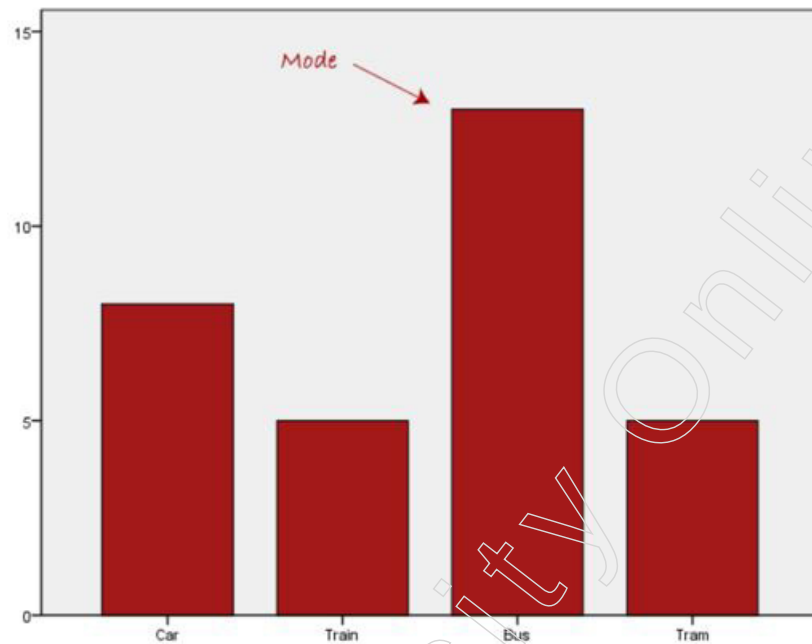
Mode

In our data set, the mode is the most frequent score. The highest bar in a bar chart or histogram is represented by this symbol on a histogram. As a result, the mode may be regarded as the most popular option at times. A mode is illustrated below as an example:

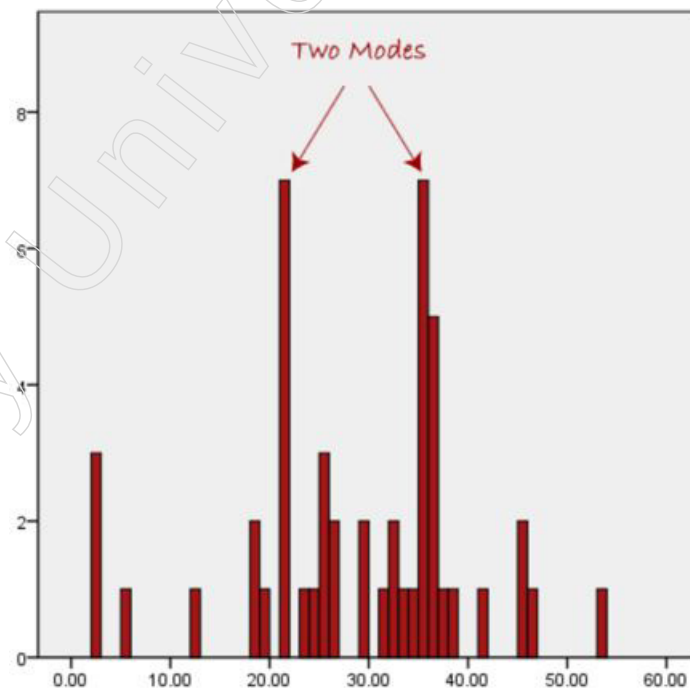


The mode is typically used for categorical data where we want to know which category is the most common, as shown below:

Notes



We can see from the data above that the bus is the most common mode of transportation in this data set. However, because the mode is not unique, we have problems when we have two or more values that share the highest frequency, as shown below:

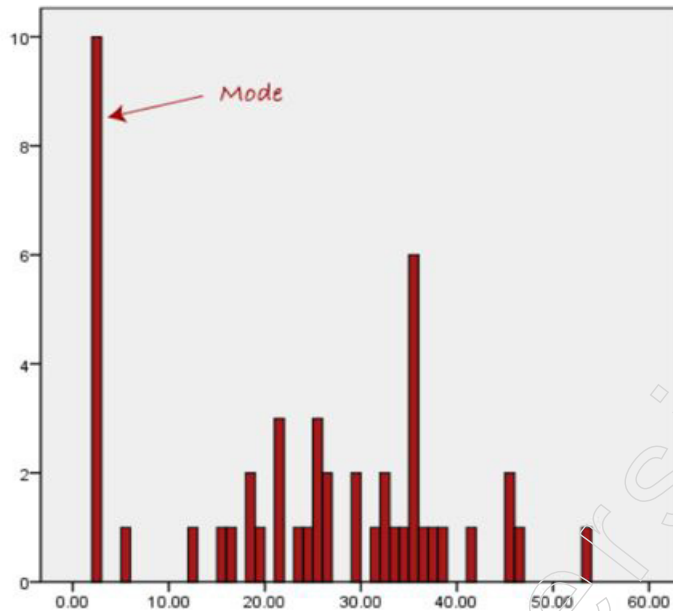


We are now at a loss as to which mode best describes the data's central tendency. This is especially problematic with continuous data because we are less likely to have any one value that is more frequent than the other. Consider measuring the weights of 30 people (to the nearest 0.1 kg). How likely is it that we will come across two or more people who are exactly the same weight (e.g., 67.4 kg)?

The answer is probably very unlikely - many people may be close, but with such a

Notes

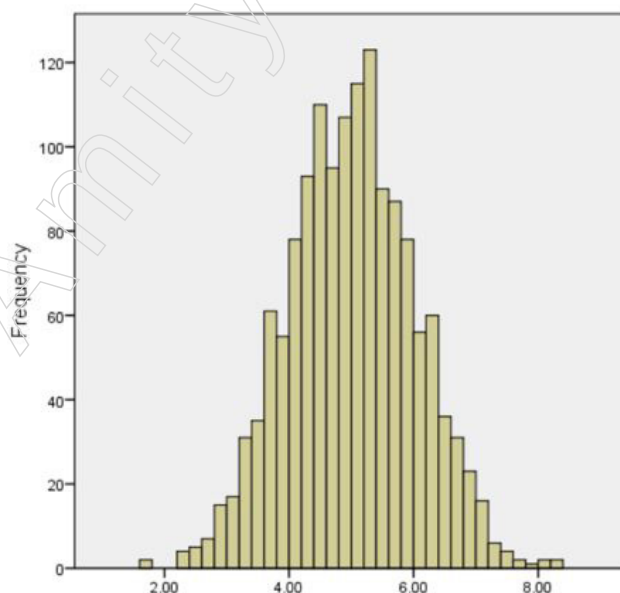
small sample (30 people) and such a wide range of possible weights, it is unlikely that you will find two people who are exactly the same weight; that is, to the nearest 0.1 kg. As a result, the mode is rarely used with continuous data. Another issue with the mode is that it does not provide a good measure of central tendency when the most common mark is far from the rest of the data in the data set, as shown in the diagram below:



The mode has a value of 2 in the diagram above. However, we can clearly see that the mode is not representative of the data, which is primarily concentrated in the 20 to 30 value range. It would be misleading to use the mode to describe the central tendency of this data set.

Skewed Distributions

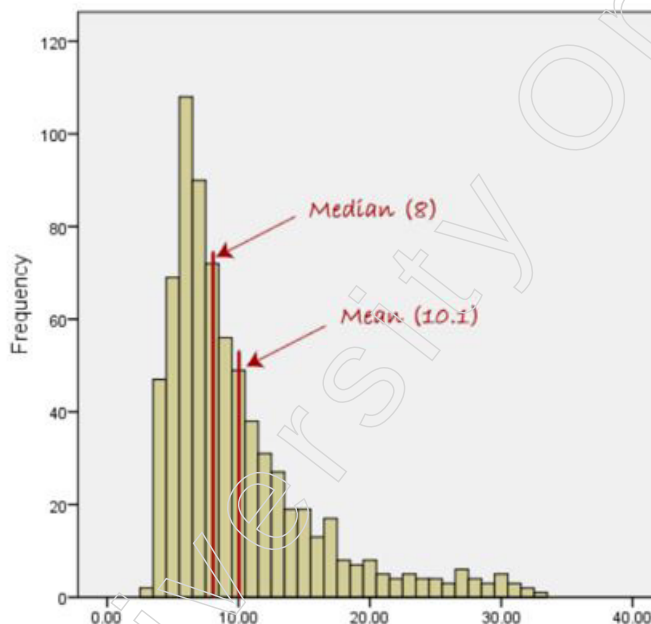
Because this is a common assumption underlying many statistical tests, we frequently test whether our data is normally distributed. The following is an example of a normally distributed set of data:



Notes

When you have a normally distributed sample, you can use either the mean or the median to calculate central tendency. In fact, the mean, median, and mode of any symmetrical distribution are all equal. However, in this case, the mean is widely regarded as the best measure of central tendency because it is the only measure that uses all of the values in the data set to calculate its value, and any change in any of the scores will affect the mean's value. This is not true for the median or mode.

However, if our data is skewed, as in the right-skewed data set below:



The mean is being dragged in the direction of the skew. In these cases, the median is generally regarded as the best representative of the data's central location. The greater the difference between the median and the mean, the greater the emphasis should be placed on using the median rather than the mean. Income (salary) is a classic example of the above right-skewed distribution, with higher-earners providing a false representation of typical income when expressed as a mean rather than a median.

When dealing with a normal distribution and normality tests reveal that the data is not normal, it is customary to use the median rather than the mean. However, this is more of a guideline than a rule of thumb. Researchers may wish to report the mean of a skewed distribution if the median and mean are not noticeably different (a subjective assessment), and if it allows for easier comparisons to previous research.

Please see the summary table below to find out what the best measure of central tendency is for each type of variable.

Sr. No.	Type of Variable	Best Measure of Central Tendency
1	Nominal	Mode
2	Ordinal	Median
3	Interval/Ratio (not skewed)	Mean
4	Interval/Ratio (skewed)	Median

4.2.3 Measure of Dispersion

Different series may possess different dispersions of items around the average. Measures of central tendency are averages of the first order. Measures of dispersion are averages of the second order. A measure of dispersion gives an idea about the extent of lack of uniformity in the sizes and qualities of the items in a series. It helps us to know the degree of uniformity and consistency in the series. If the difference between items is large the dispersion or variation is large and vice versa.

A measure of dispersion or variation in any data shows the extent to which the numerical values tend to spread about an average. If the difference between items is small, the average represents and describes the data adequately. For large differences it is proper to supplement information by calculating a measure of dispersion in addition to an average. It is useful to determine data for the knowledge it may serve:

- To compare the current results with the past results.
- To compare two or more sets of observations.
- To suggest methods to control variation in the data.

A study of variations helps us in knowing the extent of uniformity or consistency in any data. Uniformity in production is an essential requirement in industry. Quality control methods are based on the laws of dispersion.

Absolute and Relative Measures of Dispersion

The measures of dispersion can be either 'absolute' or 'relative'. Absolute measures of dispersion are expressed in the same units in which the original data are expressed. For example, if the series is expressed as Marks of the students in a particular subject; the absolute dispersion will provide the value in Marks. The only difficulty is that if two or more series are expressed in different units, the series cannot be compared on the basis of dispersion.

Definition: 'Relative' or 'Coefficient' of dispersion is the ratio or the percentage of a measure of absolute dispersion to an appropriate average. The basic advantage of this measure is that two or more series can be compared with each other despite the fact they are expressed in different units.

Definition: A precise measure of dispersion is one that gives the magnitude of the variation in a series, i.e. it measures in numerical terms, the extent of the scatter of the values around the average.

When dispersion is measured in terms of the original units of a series, it is absolute dispersion or variability. It is difficult to compare absolute values of dispersion in different series, especially when the series in different units or have different sets of values. A good measure of dispersion should have properties similar to those described for a good measure of central tendency.

Measures of Dispersion	Relative Variability
The range	Relative range
The Quartile Deviation	Deviation Relative Quartile Deviation

Notes

Notes

The Mean Deviation	Deviation Relative Mean deviation
The Median Deviation	Deviation Coefficient of Variation
The Standard Deviation	
Graphical Method	

Types of Measures of Dispersion

In statistics, dispersion measures help to understand data variability, i.e. how homogeneous or heterogeneous the data is. It shows how squeezed or dispersed the variable is, in simple terms.

There are two main types of dispersion methods in statistics which are the absolute Measure of Dispersion and the relative Measure of Dispersion

There practical implications involve -

- A measure of spread, sometimes also called a measure of dispersion, is used to describe the variability in a sample or population.
- A measure of spread gives us an idea of how well the mean, for example, represents the data.
- It is usually used in conjunction with a measure of central tendency, such as the mean or median, to provide an overall description of a set of data.
- The range is the difference between the highest and lowest scores in a data set and is the simplest measure of spread.
- Quartiles tell about the spread of a data set by breaking the data set into quarters, just like the median breaks it in half.

Range

The 'Range' of the data is the difference between the largest value of data and smallest value of data.

This is an absolute measure of variability. However, if we have to compare two sets of data, 'Range' may not give a true picture. In such case, relative measure of range, called coefficient of range is used. This is given by,

Formulae: $\text{Range} = L - S$

Where L – Largest value and S- Smallest Value

In individual observations and discrete series, L and S are easily identified. In continuous series, the following two methods are used as follows:

Method 1: L - Upper boundary of the highest class.

S - Lower boundary of the lowest class.

Method 2: L - Mid value of the highest class.

S - Mid Value of the lowest class.

Example 1:

Find the set of observations 10 5 8 11 12 9

Solution: $L = 12$ $S = 5$

Range = $L - S$

= $12 - 5$

= 7

Coefficient of range = $L - S / L + S$

= $12 - 5 / 12 + 5$

= $7/17$

= 0.4118

Example 2:

Compute the range and the co-efficient of range from the following distribution.

C.I	Frequency (f)
120 - 130	2
130 - 140	9
140 - 150	16
150 - 160	12
160-170	5

Solution:

In finding the range the frequencies are never taken into account. The upper limit of the highest class and the lower limit of the smallest class are only taken into account

Range = $L - S$

= $170 - 120 = 50$

Co-efficient of Range = $L-S/L+S = 170 - 120/170 + 120$

= $50/290$

= 0.1724

Interquartile Range and Deviations

Inter-quartile range and deviations are described in the following sub sections.

Inter-quartile Range

Inter-quartile range is a difference between upper quartile (third quartile) and lower quartile

(First quartile). Thus, Inter Quartile Range = $(Q_3 - Q_1)$

Quartile deviation

Definition: Quartile Deviation is the average of the difference between upper quartile and lower quartile.

Notes

Formulae: Thus, Quartile Deviation = $QD = (Q_3 - Q_1)/2$

Quartile Deviation (QD) also gives the average deviation of upper and lower quartiles from Median.

$$QD = (Q_3 - Q_1)/2 = (Q_3 - Q_1) / (Q_3 + Q_1)$$

Example 1:

Weekly wages of labourers is given below. Calculate Q.D. and coefficient of Q.D.

Weekly wages	100	200	400	500	600	Total
No. of Weeks:	5	8	21	12	6	52

Solution:

Weekly wages	No. of Weeks:	Cumulative Frequency
100	5	5
200	8	13
400	21	34
500	12	46
600	6	52
		N = 52

$$Q_1 = N+1 / 4$$

$$= 52+1/4$$

$$13.25$$

$$Q_1 = 13\text{th value} + 0.25 (14\text{th value} - 13\text{th value})$$

$$= 200 + 0.25 (400-200)$$

$$= 200 + 0.25 \times 200$$

$$= 200 + 50$$

$$= 250$$

$$Q_3 = 3(N+1 / 4)$$

$$= 3 \times 13.25$$

$$= 39.75$$

$$Q_3 = 39\text{th value} + 0.75 (40\text{th value} - 39\text{th value})$$

$$= 500 + 0.75 (500-500)$$

$$= 500 + 0.75 \times 0$$

$$= 500.$$

$$Q.D. = Q_3 - Q_1 / 2$$

$$= 500 - 250/2$$

$$= 250/2$$

$$= 125$$

$$\text{Coefficient of Q.D.} = \frac{Q_3 - Q_1}{Q_3 + Q_1}$$

$$\therefore = \frac{500 - 250}{500 + 250}$$

$$= 250/750$$

$$= 0.333$$

Example 2:

Determine the interquartile range and percentile range of the following distribution:

C.I	Frequency (f)
11 - 13	8
13 - 15	10
15 - 17	15
17 - 19	20
19 - 21	12
21 - 23	11
23 - 25	4

Solution:

Class Intervals	Frequency	Less than C.F.
11 - 13	8	8
13 - 15	10	18
15 - 17	15	33
17 - 19	20	53
19 - 21	12	65
21-23	11	76
23-25	4	80

1. Calculation of Interquartile Range

Calculation of Q_1

Since $\frac{N}{4} = \frac{80}{4} = 20$, the first quartile class is 15–17

$\therefore l_{Q_1} = 15, f_{Q_1} = 15, h = 2$ and $C = 18$

$$\text{Hence, } Q_1 = 15 + \frac{20 - 18}{15} \times 2 = 15.27$$

Calculation of Q_3

Since $\frac{3N}{4} = \frac{3 \times 80}{4} = 60$, the third quartile class is 19-21

Notes

$$\therefore I_{Q_3} = 19, f_{Q_3} = 12, h = 2 \text{ and } C = 53$$

$$\text{Hence, } Q_3 = 19 + \frac{60 - 53}{12} \times 2 = 20.17$$

$$\text{Thus, the interquartile range} = 20.17 - 15.27 = 4.90$$

2. Calculation of Percentile Range

Calculation of P_{10}

$$\text{Since, } \frac{10N}{100} = \frac{10 \times 80}{100} = 8, P_{10} \text{ lies in the class interval } 11-13$$

$$\therefore I_{P_{10}} = 11, f_{P_{10}} = 8, h = 2 \text{ and } C = 0$$

$$\text{Hence, } P_{10} = 11 + \frac{8 - 0}{8} \times 2 = 13$$

Calculation of P_{90}

$$\text{Since } \frac{90N}{100} = \frac{90 \times 80}{100} = 72, P_{90} \text{ lies in the class interval } 21-23$$

$$\therefore I_{P_{90}} = 21, f_{P_{90}} = 11, h = 2 \text{ and } C = 65$$

$$\text{Hence, } P_{90} = 21 + \frac{72 - 65}{11} \times 2 = 22.27$$

$$\text{Thus, the percentile range} = P_{90} - P_{10} = 22.27 - 13.0 = 9.27$$

Mean Deviation

Mean deviation is the arithmetic mean of the absolute deviations of the values about their arithmetic mean or median or mode. Mean Deviation (MD) is an average value of absolute deviation of observations from the data mean (or the median or the mode). It gives how spread/dispersed the data is.

If $x_1, x_2, \dots, x_n$ are N observations, then,

$$\text{Mean Deviation MD} = \frac{\sum d_i}{N} = \frac{\sum |x_i - \text{Average}|}{N}$$

Where,

$$d_i = \text{Deviation of each observation} = x_i - \text{Average}$$

Average used for calculating deviation can be the mean, the median or the mode. However, usually the mean is used. There is also an advantage of taking deviations from the median, because 'Mean Deviation' from median is lowest as compared to any other 'Mean Deviations'. Since absolute values of deviations ignoring sign are taken for calculating Mean Deviation, the mean deviation is not amenable to further algebraic treatment.

Mean Deviation Application

Definition: The relative measure corresponding to the 'Mean Deviation' is coefficient of Mean Deviation'. It is defined as:

Coefficient of mean deviation = Mean Deviation/ Mean or Median or Mode It can also be expressed in percentage by multiplying it with 100.

Formulae:

Coefficient of Mean deviation (about mean) = Mean deviation about Mean / Mean

$$= \Sigma |x - \bar{x}| / N$$

Coefficient of Mean deviation (about Median) = Mean deviation about Median / Median

$$= \Sigma f |x - M| / N$$

Coefficient of Mean deviation (about Mode) = Mean deviation about Mode / Mode

$$= \Sigma f |x - z| / N$$

Example:

Calculate mean deviation about the mean for the following:

12 7 9 7 7 4 10 9 15 20

Solution:

$$X = 12 + 7 + 9 + 7 + 7 + 4 + 10 + 9 + 15 + 20 / 10$$

$$= 100 / 10$$

$$= 10$$

$$\text{Mean deviation about mean} = \Sigma |x - \bar{x}| / N$$

$$= 2 + 3 + 1 + 3 + 3 + 6 + 0 + 1 + 5 + 10 / 10$$

$$= 34 / 10$$

$$= 3.4$$

Example 1:

MD in Individual series

Calculate mean deviation and its coefficient for the following data

Value (x) 125 128 132 135 140 145 155 157 159 161

Notes

Solution:

Steps 1: First compute AM			Step 2: Deviation From X		Mean deviation
Sl. No.	Value (x)	Formula	(X-X) = Dx		
A	125	$X = \frac{\sum x}{n}$	125-144= -19	ignoring negative sign	$MD = \frac{\sum Dx}{n}$
B	128		128-144= -16		
C	132	$MD = \frac{1440}{10}$	132-144= -12		$MD = \frac{120}{10}$
D	135		135-144= -9		
E	140	X=144	140-144= -4		MD= 12
F	148		148-144= +4		
G	155		155-144= +11		Coefficient of $MD = \frac{MD}{x}$
H	157		157-144= 13		
I	159		159-144= +15		$\frac{12}{144} = 0.083$
J	161		161-144= +17		
n=10	$\sum x = 1.440$		$\sum Dx = 120$		

$$= 30 + \left(\frac{12}{72.44}\right) 10$$

$$= 30 + \left(\frac{120}{28}\right) = 30 + 4.285$$

$$z = 34.285$$

Example 2:

MD in Discrete Series

Calculate mean deviation and its co-efficient for the following data.

X	35	40	45	50	55	60	65	70	75	80	85	90	95
f	3	8	12	9	4	7	15	5	10	7	5	3	2

Solution:

X	f	fx	AM	(X-X) = Dx	fdx	Mean deviation
35	3	105	$\bar{x} = \frac{\sum fx}{n}$	35-61.95 = -26.95	80.85	$MD = \frac{\sum Dx}{n}$
40	8	320		40-61.95 = -21.95	175.60	
45	12	540	$= \frac{5.575}{90}$	45-61.95 = -16.95	203.40	$MD = \frac{12.17.28}{90}$
50	9	450		50-61.95 = -11.95	107.55	
55	4	220	= 61.95	55-61.95 = -6.95	27.88	MD = 13.525
60	7	420		60-61.95 = -1.95	13.65	
65	15	975		65-61.95 = 3.05	45.75	Coefficient of
70	5	350		70-61.95 = 8.05	40.25	
75	10	750		75-61.95 = 13.05	130.50	$MD = \frac{MD}{\bar{x}}$
80	7	560		80-61.95 = 18.05	126.35	
85	5	425		85-61.95 = 23.05	115.25	$= \frac{13.525}{61.95}$
90	3	270		90-61.95 = 28.05	84.15	
95	2	190		95-61.95 = 33.05	66.10	= 0.218
	N=90	$\sum fx = 5575$			$\sum fdx = 1217.28$	

Example 3:

MD in Continuous Series

Calculate mean deviation and its co-efficient for the following data:

X	f
10-20	5
20-30	4
30-40	7
40-50	12
50-60	10
60-70	8
70-80	4

Solution

X	f	Mid Point X	fx	AM	(X-$\bar{x}$) = dx	fdx	
10-20	5	15	75	$\bar{x} = \frac{\sum fx}{n}$	31.6	80.85	$MD = \frac{\sum Dx}{n}$
20-30	4	25	100		21.6	175.60	
30-40	7	35	245	$\bar{x} = \frac{2,330}{50}$	11.6	203.40	$MD = \frac{689.6}{50}$
40-50	12	45	540		1.6	107.55	
50-60	10	55	550	= 46.6	8.4	84.0	Co-efficient of $\frac{md}{\bar{x}}$
60-70	8	65	520		18.4	147.2	
70-80	4	75	300		28.4	113.6	$= \frac{13.792}{46.6}$
N= 50			$\sum fx = 2,330$		$\sum fdx = 689.6$		

4.2.4 Kurtosis, Skewness**Measures of Skewness**

In addition to measures of central tendency and measures of variation, there are two attributes of frequency distribution of a data set that may be of interest to managers for effective decision-making. These are the Skewness and Kurtosis.

When the distribution stretches more to the right than it does to the left, the distribution is said to be 'right skewed' or 'positively skewed'. Similarly, a left-skewed distribution is the one that stretches asymmetrically to the left. Thus, the skewness is a measure of the extent of symmetry or asymmetry of the distribution. In symmetrical distribution, with single mode, we have (mode = mean = median). In such case skewness is zero. In case of positive skewness (i.e. right skewness) the mean is to the right of median, which in turn lies to the right of the mode. The opposite is for negative skewness. Skewness can be measured either in absolute term as 'mean minus mode' or in relative terms. Some of the relative measures are as follows:

1. Kari Pearson's coefficient of skewness (SKp). It is defined as:

Notes

Notes

$$S_{kk} = \frac{(P_{90} + P_{10}) - 2 \cdot M_d}{(P_{90} - P_{10})}$$

2. Bowley's Coefficient of Skewness (SKB) (quartile coefficient of skewness). It is defined as:

$$\begin{aligned} (SK_B) &= \frac{(Q_3 - Q_2) - (Q_2 - Q_1)}{(Q_3 + Q_1) + (Q_2 - Q_1)} \\ &= \frac{(Q_3 + Q_1) - 2 \cdot M_d}{(P_{90} - P_{10})} \end{aligned}$$

Where, Q is quartile.

3. Kelly's coefficient of skewness (Skk). It is defined as:

$$\text{Skewness} = \left(\frac{\sum \frac{(x_i - \mu)^3}{\sigma}}{N} \right)$$

Where, P is percentile.

Skewness is also defined in term of the moment about mean. One such measure is defined as:

$$\text{Absolute Kurtosis} = \left(\frac{\sum \frac{(x_i - \mu)^4}{\sigma}}{N} \right)$$

Relative kurtosis = Absolute kurtosis - 3

4. Lorenz Curve: It is a special type of graph, designed to describe as to how much a certain distribution varies from a completely uniform distribution. It is a cumulative percentage curve comparing the population and factor under study. For example, we could plot a graph of percentage of population and percentage of their wealth. Lorenz curve is very useful for comparing two populations particularly when their means and SD are same

Karl-Pearson's Coefficient of Skewness

Pearson's coefficient of skewness is a method developed by Karl Pearson to find skewness in a sample using descriptive statistics like the mean and mode. Skewness is one measure of the shape of a set of data. Pearson's coefficient of skewness is calculated by multiplying the difference between the mean and median, multiplied by three. The result is divided by the standard deviation.

The formula when using the mode is -

$$SK_1 = \frac{\bar{X} - Mo}{S}$$

Where x = the mean, Mo = the mode and s = the standard deviation for the sample.

The formula when using the median is -

$$SK_2 = \frac{3(\bar{X} - Md)}{S}$$

Where $\bar{x}$ = the mean, Mo = the mode and s = the standard deviation for the sample.

Notes

3.1.17 Bowley's Coefficient of Skewness

Bowley skewness is a method to figure out whether there is a positively-skewed or negatively skewed distribution. Bowley Skewness is used as an alternative to find out more about the asymmetry of an distribution. It is very useful if there are extreme data values ie. the outliers or if there is an open-ended distribution.

$$\text{Bowley Skewness} = \frac{Q_3 + Q_1 - 2Q_2}{(Q_3 - Q_1)}$$

Skewness = 0 means that the curve is symmetrical.

Skewness > 0 means the curve is positively skewed.

Skewness < 0 means the curve is negatively skewed.

In a symmetric distribution, like the normal distribution, the first (Q_1) and third (Q_3) quartiles are at equal distances from the mean (Q_2). In other words, ($Q_3 - Q_2$) and ($Q_2 - Q_1$) will be equal. If you have a skewed distribution then there will be a difference between those two values.

Limitations of Bowley Skewness

Bowley Skewness is an absolute measure of skewness. It gives a result in the units that the distribution is in. That's compared to the Pearson Mode Skewness, which gives the results in a dimensionless unit — the standard deviation. This means that one cannot compare the skewness of different distributions with different units using Bowley Skewness.

Example:

Find the Bowley's coefficient of the data

Pets	Families	Cumulative Frequency
0	60	60
1	60	120
2	50	170
3	20	190
4	25	215
5	10	225
6 or more	5	230

Solution –

Step 1: Finding the quartiles for the data set. Looking at for the “nth” observation using the following formulas:

$$Q_1 = (\text{total cum freq} + 1 / 4)^{\text{th}} \text{ observation} = (230 + 1 / 4) = 57.75$$

$$Q_2 = (\text{total cum freq} + 1 / 2)^{\text{th}} \text{ observation} = (230 + 1 / 2) = 115.5$$

$$Q_3 = 3 (\text{total cum freq} + 1 / 4)^{\text{th}} \text{ observation} = 3(230 + 1 / 4) = 173.25$$

Notes

Step 2: Looking in the table to find the nth observations as calculated in Step 1:

$$Q^1 = 57.75^{\text{th}} \text{ observation} = 0$$

$$Q^2 = 115.5^{\text{th}} \text{ observation} = 1$$

$$Q^3 = 173.25^{\text{th}} \text{ observation} = 3$$

Step 3: Plugging the above values into the formula:

$$S_{kq} = Q_3 + Q_1 - 2Q_2 / Q_3 - Q_1$$

$$S_{kq} = 3 + 0 - 2 / 3 - 0 = 1/3$$

$$S_{kq} = + 1/3, \text{ so the distribution is positively skewed.}$$

Measure of Kurtosis

Kurtosis is a measure of peaked-ness of distribution. Larger the kurtosis, more and more peaked will be the distribution. The kurtosis is calculated either as an absolute or a relative value. Absolute kurtosis is always a positive number. Absolute kurtosis of a normal distribution (symmetric bell shaped distribution) is taken as 3. Relative kurtosis can be calculated as follows:

$$\text{Absolute Kurtosis} = \left(\frac{\sum \frac{(x_i - \mu)^4}{\sigma}}{N} \right)$$

$$\text{Relative kurtosis} = \text{Absolute kurtosis} - 3$$

- Relative kurtosis can be negative. Managers usually work with relative kurtosis.
- Negative kurtosis indicates a flatter distribution than the normal distribution, and called as platykurtic.
- A positive kurtosis means more peaked curve, called Leptokurtic.
- A peak of normal distribution is called Mesokurtic.

Example:

The first four central moments of a distribution are 0, 2.5, 0.7 and 18.75. Test the skewness and kurtosis of the distribution.

Testing Skewness

We are given $\mu_1 = 0$, $\mu_2 = 2.5$, $\mu_3 = 0.7$ and $\mu_4 = 18.75$

Skewness is measured by the coefficient β_1

$$\beta_1 = \frac{\mu_3^2}{\mu_2^3}$$

Here $\mu_2 = 2.5$, $\mu_3 = 0.7$

$$\text{Substituting the values, } \beta_1 = \frac{(0.7)^2}{(2.5)^3} = + 0.031$$

Since $\beta_1 = + 0.031$, the distribution is slightly skewed.

Testing Kurtosis:

Notes

For testing kurtosis we compute the value of β_2 .

When a distribution is normal or symmetrical, $\beta_2 = 3$.

When a distribution is more peaked than the normal, β_2 is more than 3 and when it is less peaked than the normal, β_2 is less than 3.

$$\beta_1 = \frac{\mu_4}{\mu_2^2}$$

$$\mu_4 = 18.75, \mu_2 = 2.5$$

$$\beta_1 = \frac{18.75}{(2.5)^3} = \frac{18.75}{6.25} = 3$$

Since β_2 is exactly three, the distribution is mesokurtic.

Key Takeaways:

Measure of dispersion: IT gives an idea about the extent of lack of uniformity in the sizes and qualities of the items in a series. It helps us to know the degree of uniformity and consistency in the series. If the difference between items is large the dispersion or variation is large and vice versa.

Range: The 'Range' of the data is the difference between the largest value of data and smallest value of data.

Inter-quartile range: It is a difference between upper quartile (third quartile) and lower quartile (first quartile). Quartile Deviation is the average of the difference between upper quartile and lower quartile.

Variance: It is the average squared deviation of the data from their mean. For sample data, we take the average by dividing with $(n-1)$ where n is a sample size. This is to cater for degree of freedom. For population data, we average by dividing with the population size N .

Variance analysis: It helps in managing budgets by controlling budgeted versus actual costs. Without the standard deviation, you can't compare two data sets effectively.

Coefficient of Variation: It is defined as the ratio of SD and mean, multiplied by 100.

The Standard Deviation (SD) of a set: It is the positive square root of the variance of the set. This is also referred as Root Mean Square (RMS.) value of the deviations of the data points. SD of sample is the square root of the sample variance.

Skewness: It is a measure of the extent of symmetry or asymmetry of the distribution. Kurtosis is a measure of peaked-ness of distribution.

Check your Understanding

1. The _____ is the average of the numbers.
2. Withing a sorted, ascending or descending list of numbers, the _____ is that middle number and may be more representative of that set of data than the average.

Notes

3. The _____ is the number most frequently seen in a dataset.
4. In _____ arithmetic mean, for finding an average the value of each item is multiplied by its weight and then the product are divided by the number of weights.
5. The word “mode” is derived from the French word “la mode” which means fashion. True / False
6. A central _____ measure is a single value that attempts to describe a set of data by identifying the central position within that set of data.
7. When you have a _____ distributed sample, you can use either the mean or the median to calculate central tendency.
8. Different series may possess different dispersions of items around the average. True / False
9. Absolute measures of _____ are expressed in the same units in which the original data are expressed.
10. _____ or Coefficient of dispersion is the ratio or the percentage of a measure of absolute dispersion to an appropriate average.
11. Negative kurtosis indicates a flatter distribution than the normal distribution, and called as _____.
12. A positive kurtosis means more peaked curve, called _____.
13. A peak of normal distribution is called Leptokurtic. True / False

Multiple Choice

1. The mean height of 25 male workers in a factory is 61 cm and the mean height of 35 female workers in the same factory is 58cm. The combined mean height of 60 workers in the factory is?
 - a. 59.25
 - b. 59.5
 - c. 59.75
 - d. 58.75
2. Mean of 100 items is 49. It was discovered that three items which should have been 60, 70, and 80 were wrongly read as 40, 20, and 50, respectively. The correct mean is
 - a. 48
 - b. 89
 - c. 50
 - d. 80
3. If the mode of the data is 18 and the mean is 24, what is the median?
 - a. 18
 - b. 24
 - c. 22

- d. 21
4. The mean of 30 given numbers, when it is given that the mean of 10 of them is 12 and the mean of the remaining 20 is 9, is equal to
- a. 11
- b. 10
- c. 9
- d. 5

Summary

- Central tendency has three main measures: mode, median and mean. Each of those measurements represents a specific indication of the distribution's typical or central value.
- Arithmetic Mean is defined as the value obtained by dividing the total values of all items in the series by their number. In other words, is defined as the sum of the given observations divided by the number of observations, i.e., add values of all items together and divide this sum by the number of observations.
- Arithmetic mean for ungrouped data of individual series can be calculated using either the direct method or Short-cut method(indirect method).
- Sometimes, some observations get relatively more importance than other observations, the weight for such observation must be given on the basis of their relative importance. In weighted arithmetic mean, for finding an average the value of each item is multiplied by its weight and then the product are divided by the number of weights.
- Median is defined as the value of the item dividing the series into two equal halves, where one half contains all values less than (or equal to) it and the other half contains all values greater than (or equal to) it. It is also defined as the 'central value' of the variable.
- Mode is defined as the value of the variable occurring most frequently in a distribution.
- A central tendency measure is a single value that attempts to describe a set of data by identifying the central position within that set of data. As a result, measures of central tendency are also known as measures of central location. They're also known as summary statistics. The mean (also known as the average) is probably the most familiar measure of central tendency, but there are others, such as the median and the mode. The mean, median, and mode are all valid measures of central tendency, but depending on the circumstances, some measures of central tendency are more appropriate to use than others.
- The skewness is a measure of the extent of symmetry or asymmetry of the distribution.
- A measure of dispersion gives an idea about the extent of lack of uniformity in the sizes and qualities of the items in a series. It helps us to know the degree of uniformity and consistency in the series.

Notes

- Absolute measures of dispersion are expressed in the same units in which the original data are expressed. The only difficulty is that if two or more series are expressed in different units, the series cannot be compared on the basis of dispersion.
- Relative or Coefficient of dispersion is the ratio or the percentage of a measure of absolute dispersion to an appropriate average.
- The range of the data is the difference between the largest value of data and the smallest value of data.
- Mean deviation is the arithmetic mean of the absolute deviations of the values about their arithmetic mean or median or mode. There is also an advantage of taking deviations from the median, because 'Mean Deviation' from median is lowest as compared to any other 'Mean Deviations'.
- In addition to measures of central tendency and measures of variation, there are two attributes of frequency distribution of a data set that may be of interest to managers for effective decision-making. These are the Skewness and Kurtosis.
- Pearson's coefficient of skewness is calculated by multiplying the difference between the mean and median, multiplied by 3. The result is then divided by the standard deviation.
- Bowley skewness is a method to figure out whether there is a positively skewed or negatively skewed distribution.
- Kurtosis is a measure of peaked-ness of distribution. Larger the kurtosis, more and more peaked will be the distribution. The kurtosis is calculated either as an absolute or a relative value. Absolute kurtosis is always a positive number.

Activity

1. Using Karl-Person's Coefficient of Skewness formula; solve for the median and mode.
 - a. Skewness of Distribution = 0.32
 - b. Standard Deviation = 6.5
 - c. Mean = 29.6
2. What is the coefficient of skewness if beta one is 9 and beta two is 11?
3. The median of a skewed distribution is 8, 3rd quartile is 12, 1st quartile is 8, and inter-quartile range is 4. What is the relative coefficient of skewness?

Question and Exercises

1. What are the objectives served by averages?
2. List the disadvantages of arithmetic mean.
3. What are the steps to finding the mean using indirect method?
4. What is the formula for calculating weighted average?
5. List the advantages of median.
6. What are the two main types of dispersion methods in statistics?

7. How does one calculate mean deviation?

Notes

Glossary

- **Arithmetic Mean:** is defined as the value obtained by dividing the total values of all items in the series by their number.
- **Mean:** an average, it is a single figure that sums up the characteristics of a whole group of figures.
- **Median:** is defined as the value of the item dividing the series into two equal halves, where one half contains all values less than (or equal to) it and the other half contains all values greater than (or equal to) it.
- **Mode:** is defined as the value of the variable occurring most frequently in a distribution.
- **Range:** is the difference between the largest value of data and the smallest value of data.
- **Skewness:** is a measure of the extent of symmetry or asymmetry of the distribution.

Further Readings

1. Bekes, Gabor. Data Analysis for Business, Economics, and Policy. Cambridge University Press (May 6, 2021)
2. Huberman, A. Michael; Miles, Matthew B.; Saldana, Johnny. Qualitative Data Analysis: A Methods Sourcebook
3. Knaflic, Cole Nussbaumer. Storytelling with Data: A Data Visualization Guide for Business Professionals. Wiley; 1st edition (November 2, 2015)

Check your Understanding: Answers

- | | | |
|--------------|-----------------|-----------------|
| 1. Mean | 2. Median | 3. Mode |
| 4. Weighted | 5. True | 6. Tendency |
| 7. Normally | 8. True | 9. Dispersion |
| 10. Relative | 11. Platykurtic | 12. Leptokurtic |
| 13. False | | |

Multiple Choice

1. A
2. C
3. B
4. C

Module - 5: Forecasting Techniques

Structure:

5.1 Correlation

1.1.1 Correlation-Coefficient_Introduction

1.1.2 Correlation_Coefficient_Application

1.1.3 Introduction_Rank_Correlation

1.1.4 Comparison_Pearson_Spearman_Correlation

1.1.5 Application_Rank_Correlation

5.2 Regression

2.1.1 Introduction_Linear_Regression_model

2.1.2 Population_sample_Regression

2.1.3 Method_Least_Square_Understanding

2.1.4 Maths_Behind_Least_Square

Unit - 5.1: Correlation

Notes

Objectives:

- Correlation-Coefficient Introduction
- Correlation-Coefficient Application
- Introduction Rank Correlation
- Application Rank Correlation

Introduction

Correlation is a degree of linear association between two random variables. In these two variables, we do not differentiate them as dependent and independent variables. It may be the case that one is the cause and other is an effect i.e. independent and dependent variables respectively. On the other hand, both may be dependent variables on a third variable. In some cases there may not be any cause effect relationship at all. Therefore, if we do not consider and study the underlying economic or physical relationship, correlation may sometimes give absurd results

Frequently, data is available in the form of some kind of ranking for different variables. Other times, data can have instances where it is difficult to measure the cause-effect variables. An example of this instance is when selecting a candidate for scenario. There are a number of factors on which the experts need to base their assessment on, therefore, it is not possible to measure many of these parameters in physical units e.g., sincerity, loyalty, integrity, tactfulness, initiative, etc. The purpose of computing a correlation coefficient in such situations is to determine the extent to which the two sets of ranking are in agreement. We will discuss the applications of correlation coefficient in depth in this unit.

5.1.1 Correlation-Coefficient Introduction

Correlation-Coefficient

Correlation of Grouped Data

Many times the observations are grouped into a 'two way' frequency distribution table. These are called bivariate frequency distribution. It is a matrix where rows are grouped for X variable and columns are grouped for Y variable. Each cell say (i, j) represents the frequency or count that falls in both groups of a particular range of values of X_i and Y_j . In this case correlation coefficient is given by

$$r = \frac{\sum f \times m_x \times m_y - \frac{1}{n} \sum (f \times m_x) \sum (f \times m_y)}{\sqrt{\sum (f \times m_x^2) - \frac{\sum f \times m_x^2}{2}} \sqrt{\sum (f \times m_y^2) - \frac{\sum f \times m_y^2}{2}}}$$

Where m_x and m_y are class marks of frequency distributions of X and Y variables, f_x and f_y are marginal frequencies of X and Y and f_{xy} are joint frequencies of X and Y respectively.

Notes

Example: Calculate coefficient of correlation for the following data.

x	0-500	500-1000	1000-1500	1500-2000	2000-2500	Total
y						
0-200	12	6	-	-	-	18
200-400	2	18	4	2	1	27
400-600	-	4	7	3	-	14
600-800	-	1	-	2	1	4
800-1000	-	-	1	2	3	6
Total	14	29	12	9	5	69

Solution:

Let the assumed mean for X be $a = 1250$ and the scaling factor $g = 500$. Therefore, we can calculate $f \times d_y$ and $f \times d_x^2$ from the marginal distribution of X as,

X	Class Mark m_x	$d_x = \frac{mx - a}{g}$	Frequency f	$f \times d_x$	$f \times d_x^2$
0-500	250	-2	14	-28	66
600-1000	750	-1	29	-29	29
1000-1500	1250	0	12	0	0
1500-2000	1750	1	9	9	9
2000-2500	2250	2	5	10	20
Total				-38	114

Definition: The correlation coefficient measures the degree of association between two variables X and Y.

The coefficient is given as:

$$r = \frac{\text{Cov } x, \text{cov } y}{\sigma_x \sigma_y}$$

$$r = \frac{\frac{1}{n} \sum (x - \bar{x})(y - \bar{y})}{\sigma_x \sigma_y}$$

Where r is the 'Correlation Coefficient' or 'Product Moment Correlation Coefficient' between X and Y. σ_X and σ_Y are the standard deviations of X and Y respectively. 'n' is the number of the pairs of variables X and Y in the given data.

The expression $-1/n \sum (X - \bar{X})(Y - \bar{Y})$ is known as a covariance between the variables X and Y. It is denoted as $\text{Cov}(x, y)$. The Correlation Coefficient r is a dimensionless number whose value lies between +1 and -1. Positive values of r indicate positive (or direct) correlation between the two variables X and Y i.e. both X and Y increase or decrease together.

Negative values of r indicate negative (or inverse) correlation, thereby meaning that an increase in one variable X or Y results in a decrease in the value of the other variable. A zero correlation means that there is no association between the two

variables.

formula can be modified as:

$$r = \frac{\frac{1}{n} \sum (x - \bar{x})(y - \bar{y})}{\sigma_x \sigma_y} = \frac{\frac{1}{n} \sum (xy - x\bar{y} - \bar{x}y + \bar{x}\bar{y})}{\sigma_x \sigma_y}$$

$$= \frac{\frac{N}{\sum x^2} - \left(\frac{\sum x}{n}\right)^2}{\sqrt{\frac{\sum x^2}{n} - \left(\frac{\sum x}{n}\right)^2} \sqrt{\frac{\sum y^2}{n} - \left(\frac{\sum y}{n}\right)^2}}$$

$$= \frac{E[XY] - E[X]E[Y]}{\sqrt{E[X^2] - (E[X])^2} \sqrt{E[Y^2] - (E[Y])^2}}$$

Equations (2) and (3) are alternate forms of equation (1). These have advantage that we don't have to subtract each value from the mean.

5.1.2 Correlation Coefficient Application

Example:

The data of advertisement expenditure (X) and sales (Y) of a company for past 10 year period is given below. Determine the correlation coefficient between these variables and comment the correlation.

X	50	50	50	40	30	20	20	15	10	5
Y	700	650	600	500	450	400	300	250	210	200

Solution:

We shall take U to be the deviation of X values from the assumed mean of 30 divided by 5. Similarly, V represents the deviation of Y values from the assumed mean of 400 divided by 10.

Sl. No.	(X=X _i)	(Y=y _i)	(U=u _i)	(V=v _i)	(u _i =v _i)	u _i ²	v _i ²
1	50	700	4	30	120	16	900
2	50	650	4	25	100	16	625
3	50	600	4	20	80	16	400
4	40	500	2	10	20	4	100
5	30	450	0	5	0	0	25
6	20	400	-2	0	0	4	0
7	20	300	-2	-10	20	4	100
8	15	250	-3	-15	45	9	225
9	10	210	-4	-19	76	16	361
10	5	200	-5	-20	100	25	400
Total			-2	26	561	110	3136

Short cut procedure for calculation of correlation coefficient

Notes

Notes

$$r = \frac{\sum_{i=1}^n u_i v_i - \frac{1}{n} \sum_{i=1}^n u_i \sum_{i=1}^n v_i}{\sqrt{\sum_{i=1}^n u_i^2 - \frac{1}{n} \left(\sum_{i=1}^n u_i \right)^2} \sqrt{\sum_{i=1}^n v_i^2 - \frac{1}{n} \left(\sum_{i=1}^n v_i \right)^2}}$$

$$= \frac{561 - \frac{(-2)(26)}{10}}{\sqrt{110 - \frac{4}{10}} \sqrt{3136 - \frac{676}{10}}} = \frac{561 + 5.2}{\sqrt{109.6} \sqrt{3068.4}} = 0.976$$

Interpretation of r

- The correlation coefficient, r ranges from -1 to 1. A value of 1 implies that a linear equation describes the relationship between X and Y perfectly, with all data points lying on a line for which Y increases as X increases. A value of -1 implies that all data points lie on a line for which Y decreases as X increases. A value of 0 implies that there is no linear correlation between the variables.
- More generally, note that $(X_i - \bar{X})(Y_i - \bar{Y})$ is positive if and only if X_i and Y_i lie on the same side of their respective means. Thus the correlation coefficient is positive if X_i and Y_i tend to be simultaneously greater than, or simultaneously less than, their respective means.
- The correlation coefficient is negative if X_i and Y_i tend to lie on opposite sides of their respective means.
- The coefficient of correlation r lies between -1 and +1 inclusive of those values.
- When r is positive, the variables x and y increases or decrease together.
- $r = +1$ implies that there is a perfect positive correlation between variables x and y.
- When r is negative, the variables x and y move in the opposite direction.
- When $r = -1$, there is a perfect negative correlation.
- When $r = 0$, the two variables are uncorrelated.

5.1.3 Introduction Rank Correlation

Quite often the data is available in the form of some ranking for different variables. Also there are occasions where it is difficult to measure the cause-effect variables. For example, while selecting a candidate, there are number of factors on which the experts base their assessment. It is not possible to measure many of these parameters in physical units e.g. sincerity, loyalty, integrity, tactfulness, initiative, etc. Similar is the case during dance contests. However, in these cases the experts may rank the candidates. It is then necessary to find out whether the two sets of ranks are in agreement with each other. This is measured by Rank Correlation Coefficient. The purpose of computing a correlation coefficient in such situations is to determine the extent to which the two sets of ranking are in agreement. The coefficient that is determined from these ranks is known as Spearman's rank coefficient, r_S.

This is defined by the following formula:

$$rs = 1 - \frac{6 \times \sum_{i=1}^n d_i^2}{n(n^2 - 1)}$$

Where, n = Number of observation pairs

$$D = X_i - Y_i$$

= X_i = Values of variable X and = Y_i values of variable Y

Rank correlation for tied ranks

In case of a tie, i.e., when two or more individuals have the same rank, each individual is assigned a rank equal to the mean of the ranks that would have been assigned to them in the event of there being slight differences in their values. To understand this, let us consider the series 20, 21, 21, 24, 25, 25, 25, 26, 27, 28. Here the value 21 is repeated two times and the value 25 is repeated three times. When we rank these values, rank 1 is given to 20. The values 21 and 21 could have been assigned ranks 2 and 3 if these were slightly different from each other. Thus, each value will be assigned a rank equal to mean of 2 and 3, i.e., 2.5. Further, the value 24 will be assigned a rank equal to 4 and each of the values 25 will be assigned a rank equal to 6, the mean of 5, 6 and 7 and so on.

Since the Spearman's formula is based upon the assumption of different ranks to different individuals, therefore, its correction becomes necessary in case of tied ranks. It should be noted that the means of the ranks will remain unaffected.

Application of Rank Correlation

When two or more items have the same rank, a correction has to be applied to $\sum d_i^2$.

For example, if the ranks of X are 1, 2, 3, 3, 5,..... showing that there are two items with the same 3rd rank and fourth rank is skipped, then instead of writing 3, we write $3\frac{1}{2}$ for both. Thus the sum of these ranks which is 7 ($3+4=3\frac{1}{2} + 3\frac{1}{2}=7$) remains same keeping the mean of ranks unaffected. But in such cases the standard deviation is affected. Therefore, correction is required for the Rank Correlation Coefficient. For this, $\sum d_i^2$ is increased by $\frac{(m^3-m)}{12}$ for each tie, where m is number of items in each tie.

We must remember that if there are more than one group of items with common rank, this correction factor is to be added that many times once for each group.

Example: Twelve salesmen are ranked for efficiency and length of service as below:

Salesman	A	B	C	D	E	F	G	H	I	J	K	L
Efficiency (X)	1	2	3	4	4	4	7	8	9	10	11	12
Length of Service (Y)	2	1	5	3	9	7	7	6	4	11	10	11

Find the value to spearman's rank coefficient

Solution:

Notes

Computations of Spearman's Rank Correlation as shown below:

Individual	Efficiency ($X=X_1$)	Length of Service ($Y=y_1$)	$d_1 = x_1 - y_1$	d_1^2
A	1	2	-1	1
B	2	1	1	1
C	3	5	-2	4
D	$(4+5+6)/3=5$	3	2	4
E	$(4+5+6)/3=5$	9	-4	16
F	$(4+5+6)/3=5$	$(7+8)/3=7.5$	-2.5	6.25
G	7	$(7+8)/3=7.5$	-0.5	0.25
H	8	6	2	4
I	9	4	5	25
J	10	$(11+12)/2=11.5$	-1.5	2.25
K	11	10	1	1
L	12	$(11+12)/2=11.5$	0.5	0.25
Total				65

Now, $n = 12$, $\sum_{i=1}^N d_i^2 = 65$

Using the formula

$$r_s = 1 - \frac{\left\{ \sum_{i=1}^N d_i^2 + \frac{1}{12} \times (3^3 - 3) \frac{1}{12} \times (2^3 - 2) + \frac{1}{12} \times (2^3 - 2) \right\}}{n(n^2 - 1)}$$

$$= 1 - \frac{6 \times \{65 + 2 + 0.5 + 0.5\}}{12(144 - 1)} = 0.762$$

We can conclude that there is a high degree of correlation between efficiency and length of service.

5.1.4 Comparison Pearson Spearman Correlation

a. Rank Correlation when Ranks are given

Example: Ranks obtained by a set of ten students in a mathematics test (variable X) and a physics test (variable Y) are shown below:

Rank for Variable X	1	2	3	4	5	6	7	8	9	10
Rank for Variable Y	3	1	4	2	6	9	8	10	5	7

To determine the coefficient of rank correlation, S r.

Solution:

Computations of Spearman's Rank Correlation as shown below:

Notes

Individual	Rank in Maths ($X=X_2$)	Rank In Physics ($Y=y_1$)	$d_1=x_1-y_1$	d_1^2
1	1	3	+2	4
2	2	1	-1	1
3	3	4	+1	1
4	4	2	-2	4
5	5	6	+1	1
6	6	9	+3	9
7	7	8	+1	1
8	8	10	+2	4
9	9	5	-4	16
10	10	7	-3	9
Total				50

Now, $n = 10$, $\sum_{i=1}^N d_i^2 = 50$

Using the Formula

$$r_s = 1 - \frac{6 \times \sum_{i=1}^N d_i^2}{N(n^2-1)} = 1 - \frac{6 \times 50}{10(100-1)} = 0.697$$

We can say that there is a high degree of correlation between the performance in mathematics and physics.

a. Rank Correlation when Ranks are not given

Example: Find the rank correlation coefficient for the following data.

X	75	88	95	70	60	80	81	50
Y	120	134	115	110	140	142	100	150

Solution: Let R1 and R2 denotes the ranks in X and Y respectively.

X	Y	R1	R2	d=R1-R2	d ²
75	120	5	5	0	0
88	134	2	4	-2	4
95	150	1	1	0	0
70	115	6	6	0	0
60	110	7	7	0	0
80	140	4	3	1	1
81	142	3	2	1	1
50	100	8	8	0	0
					6

Notes

$$\text{Coefficient of Correlation } P = 1 - \frac{6 \sum d^2}{N(n^2 - 1)} = 1 - \frac{6 \times 6}{8(64 - 1)} = +.93$$

In this method the biggest item gets the first rank, the next biggest second rank and so on.

Rank correlation for tied ranks

In case of a tie, i.e., when two or more individuals have the same rank, each individual is assigned a rank equal to the mean of the ranks that would have been assigned to them in the event of there being slight differences in their values. To understand this, let us consider the series 20, 21, 21, 24, 25, 25, 25, 26, 27, 28. Here the value 21 is repeated two times and the value 25 is repeated three times. When we rank these values, rank 1 is given to 20. The values 21 and 21 could have been assigned ranks 2 and 3 if these were slightly different from each other. Thus, each value will be assigned a rank equal to mean of 2 and 3, i.e., 2.5. Further, the value 24 will be assigned a rank equal to 4 and each of the values 25 will be assigned a rank equal to 6, the mean of 5, 6 and 7 and so on.

Since the Spearman's formula is based upon the assumption of different ranks to different individuals, therefore, its correction becomes necessary in case of tied ranks. It should be noted that the means of the ranks will remain unaffected.

5.1.5 Application Rank Correlation

When two or more items have the same rank, a correction has to be applied to $\sum d_i^2$.

For example, if the ranks of X are 1, 2, 3, 3, 5,..... showing that there are two items with the same 3rd rank and fourth rank is skipped, then instead of writing 3, we write $3\frac{1}{2}$ for both. Thus the sum of these ranks which is 7 ($3+4=3\frac{1}{2} + 3\frac{1}{2}=7$) remains same keeping the mean of ranks unaffected. But in such cases the standard deviation is affected. Therefore, correction is required for the Rank Correlation Coefficient. For this, $\sum d_i^2$ is increased by $\frac{(m^3 - m)}{12}$ for each tie, where m is number of items in each tie.

We must remember that if there are more than one group of items with common rank, this correction factor is to be added that many times once for each group.

Example: Twelve salesmen are ranked for efficiency and length of service as below:

Salesman	A	B	C	D	E	F	G	H	I	J	K	L
Efficiency (X)	1	2	3	4	4	4	7	8	9	10	11	12
Length of Service (Y)	2	1	5	3	9	7	7	6	4	11	10	11

Find the value to Spearman's rank coefficient

Solution:

Computations of Spearman's Rank Correlation as shown below:

Notes

Individual	Efficiency ($X=X_1$)	Length of Service ($Y=y_1$)	$d_1=x_1-y_1$	d_1^2
A	1	2	-1	1
B	2	1	1	1
C	3	5	-2	4
D	$(4+5+6)/3=5$	3	2	4
E	$(4+5+6)/3=5$	9	-4	16
F	$(4+5+6)/3=5$	$(7+8)/3=7.5$	-2.5	6.25
G	7	$(7+8)/3=7.5$	-0.5	0.25
H	8	6	2	4
I	9	4	5	25
J	10	$(11+12)/2=11.5$	-1.5	2.25
K	11	10	1	1
L	12	$(11+12)/2=11.5$	0.5	0.25
Total				65

Now, $n = 12$, $\sum_{i=1}^N d_i^2 = 65$

Using the formula

$$r_s = 1 - \frac{\left\{ \sum_{i=1}^N d_i^2 + \frac{1}{12} \times (3^3 - 3) \frac{1}{12} \times (2^3 - 2) + \frac{1}{12} \times (2^3 - 2) \right\}}{n(n^2 - 1)}$$

$$= 1 - \frac{6 \times \{65 + 2 + 0.5 + 0.5\}}{12(144 - 1)} = 0.762$$

We can conclude that there is a high degree of correlation between efficiency and length of service.

Check your Understanding

1. A table displaying correlation coefficients between variables is a correlation _____.
2. The purpose of computing a correlation coefficient in such situations is to determine the extent to which the two sets of ranking are in agreement. True / False
3. The coefficient that is determined from these ranks is known as _____ rank coefficient.
4. Many times the observations are grouped into a "two way" frequency distribution table. True / False

Multiple Choice

1. The sign of the correlation is determined by the
 - a. Standard Deviation

Notes

- b. Covariance between X and Y
 - c. X variable
 - d. Y variable
2. Correlation r contains no unit because
 - a. Absolute number
 - b. It is a relative number
 - c. It denotes variables
 - d. It show the value of attributes
3. Value of r lying outside the range of -1 and $+1$, indicates
 - a. Zero correlation
 - b. Strong correlation
 - c. Error in calculation
 - d. Weak correlation
4. A scatter diagram displaying plotted points on an upward sloping straight line denotes that X and Y has
 - a. Positive correlation
 - b. Zero correlation
 - c. Perfect positive correlation
 - d. Perfect negative correlation

Summary

- A table displaying correlation coefficients between variables is a correlation matrix. The association between two variables is presented by each cell in the table. To summarise results, as an input into a more advanced analysis, and as a diagnosis for advanced analysis, a correlation matrix is used.
- Quite often the data is available in the form of some ranking for different variables. Also, there are occasions where it is difficult to measure the cause-effect variables. It is necessary to find out whether the two sets of ranks are in agreement with each other. This is measured by Rank Correlation Coefficient. The purpose of computing a correlation coefficient in such situations is to determine the extent to which the two sets of ranking are in agreement. The coefficient that is determined from these ranks is known as Spearman's rank coefficient.

Activity

1. Sara and Lily are studying for a text. Sara has spent eight hours studying for this test where as Lily has only put in two hours. The tests were graded and Sara received an A on the test and Lily received a C. Seeing this Lily wondered if there was a correlation between the hours studied and the test score. Below is the data that Lily collected. Using that find the correlation coefficient.

X (time)	8	2	6	4	2
----------	---	---	---	---	---

Y (grade)	98	74	87	82	72
-----------	----	----	----	----	----

Notes

2. The scores of ten students in physics and biology are as follows. Calculate the Spearman Rank Correlation
 - a. Physics: 35, 23, 47, 17, 10, 43, 9, 6, 28, 32
 - b. Biology: 30, 33, 45, 23, 8, 49, 12, 4, 31, 42

Questions and Exercises

1. What are the key decisions that need to be made when creating a correlation matrix?
2. What is Spearman's rank coefficient formula?
3. How does one determine the rank for tied ranks?

Glossary

- **Bivariate Frequency Distribution:** observations that are grouped into a "two way" frequency distribution table.
- **Correlation Matrix:** A table displaying correlation coefficients between variables.

Further Reading

1. Athanasopoulos, George; Hyndman, Rob J. Forecasting: Principles and Practice. [Otexts; 3rd ed. edition (May 31, 2021)]
2. Gilliland, Michael. Business Forecasting (Wiley and SAS Business Series). Wiley; 1st edition (December 29, 2015)

Check your Understanding: Answers

1. Matrix
2. True
3. Spearman's
4. True

Multiple Choice

1. A
2. B
3. C
4. C

Notes

Unit - 5.2: Regression

Objectives:

At the end of this unit, you will be able to understand

- Introduction to Linear Regression Model
- Population Sample Regression
- Method Least Square Understanding
- Math Behind Least Square

Introduction

There is a need for a statistical model that will extract information from the given data to establish the regression relationship between independent and dependent relationship. The model should capture systematic behaviour of data. The non-systematic behaviour cannot be captured and called as errors. The error is due to random component that cannot be predicted as well as the component not adequately considered in statistical model. Good statistical model captures the entire systematic component leaving only random errors.

In any model we attempt to capture everything which is systematic in data. Random errors cannot be captured in any case. Assuming the random errors are 'Normally distributed' we can specify the confidence level and interval of random errors. Thus, our estimates are more reliable. If the variables in a bivariate distribution are correlated, the points in scatter diagram approximately cluster around some curve. If the curve is straight line we call it as linear regression. Otherwise, it is curvilinear regression. The equation of the curve which is closest to the observations is called the 'best fit'.

The best fit is calculated as per Legendre's principle of least sum squares of deviations of the observed data points from the corresponding values on the 'best fit' curve. This is called as minimum squared error criteria. It may be noted that the deviation (error) can be measured in X direction or Y direction. Accordingly we will get two 'best fit' curves. If we measure deviation in Y direction, i.e. for a given x value of data point (x, y) and then we measure corresponding y value on 'best fit' curve and then take the value of deviation in y , we call it as regression of Y on X. In the other case, if we measure deviations in X direction we call it as regression of X and Y.

Definition: According to Morris Myers Blair, regression is the measure of the average relationship between two or more variables in terms of the original units of the data.

5.2.1 Introduction to Linear Regression Model

Linear regression is a linear model, e.g. a model that assumes a linear relationship between the input variables (x) and the single output variable (y). More specifically, that y can be calculated from a linear combination of the input variables (x).

Notes

A simple and widely used kind of predictive analysis is linear regression. Two questions are discussed in the general concept of regression:

1. Does a collection of predictor variables do a good job of predicting an outcome (dependent) variable?
2. In particular, which variables are important predictors of the outcome variable and how they influence the outcome variable, as shown by the magnitude and sign of the beta estimates.

The relationship between one dependent variable and one or more independent variables is explained by these regression projections. The formula $y = c + b \cdot x$ describes the simplest form of the regression equation with one dependent and one independent variable, where y = expected dependent variable score, c = constant, b = coefficient of regression and x = score on the independent variable.

When there is a single input variable (x), the method is referred to as simple linear regression. When there are multiple input variables, literature from statistics often refers to the method as multiple linear regression.

Regression Lines

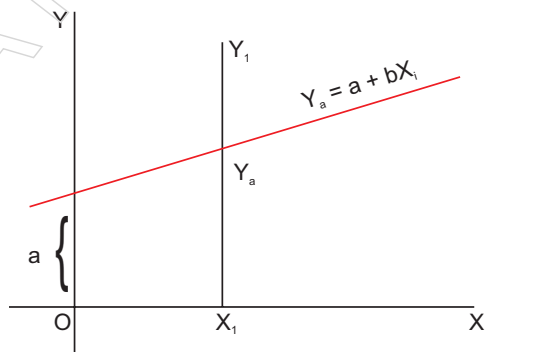
For a bivariate data (X_i, Y_i), $i = 1, 2, \dots, n$, we can have either X or Y as independent variable. If X is independent variable then we can estimate the average values of Y for a given value of X . The relation used for such estimation is called regression of Y on X . If on the other hand Y is used for estimating the average values of X , the relation will be called regression of X on Y .

For a bivariate data, there will always be two lines of regression. It will be shown later that these two lines are different, i.e., one cannot be derived from the other by mere transfer of terms, because the derivation of each line is dependent on a different set of assumptions.

Line of Regression of Y on X

The general form of the line of regression of Y on X is $Y_{Ci} = a + bX_i$, where Y_{Ci} denotes the average or predicted or calculated value of Y for a given value of $X = X_i$. This line has two constants, a and b .

The constant a is defined as the average value of Y when $X = 0$. Geometrically, it is the intercept of the line on Y -axis. Further, the constant b , gives the average rate of change of Y per unit change in X , is known as the regression coefficient.

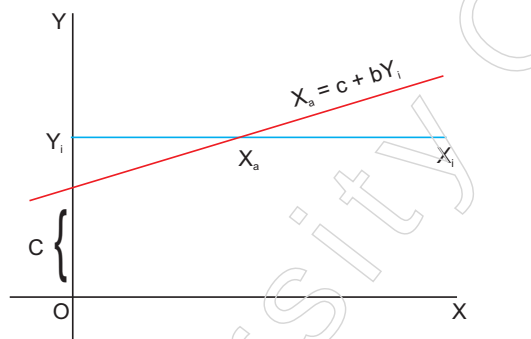


Notes

The above line is known if the values of a and b are known. These values are estimated from the observed data (X_i, Y_i) , $i = 1, 2, \dots, n$.

Line of Regression of X on Y

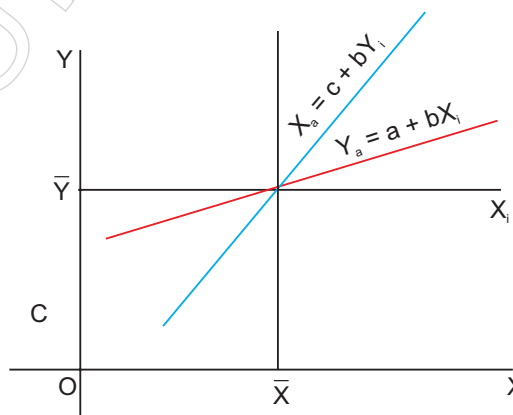
The general form of the line of regression of X on Y is $X_{Ci} = c + dY_i$, where X_{Ci} denotes the predicted or calculated or estimated value of X for a given value of $Y = Y_i$ and c and d are constants. d is known as the regression coefficient of regression of X on Y .



In this case, we have to calculate the value of c and d so that $S_1 = \sum (X_i - X_a)^2$ is minimised.

Graphing Regression Lines

This shows that the line of regression also passes through the point $\bar{X}, \bar{Y}$. Since both the lines of regression pass through the point $\bar{X}, \bar{Y}$, therefore $\bar{X}, \bar{Y}$ is their point of intersection as shown



We can write $c = \bar{X} - d\bar{Y}$

5.2.2 Population Sample Regression

It is the regression model is based on a sample of n bivariate observations drawn from a larger population of measurements.

$$\hat{y} = b_0 + b_1x$$

To construct an ordinary least-squares regression line, we use the means and standard deviations of our sample data to calculate the slope (b_1) and y-intercept (b_0).

$\mu_y = \beta_0 + \beta_1 x$, where μ_y is the population mean response, β_0 is the y-intercept, and β_1 is the slope for the population model.

In a population, there can be many different responses for a value of x . In simple linear regression, the model assumes that for each value of x the observed values of the response variable y are normally distributed with a mean that depends on x . To represent such means μ_y is used. It is also assumed that these means all lie on a straight line when plotted against x (a line of means). The sample data then fit the statistical model:

$$\text{Data} = \text{fit} + \text{residual}$$

$$y_i = (\beta_0 + \beta_1 x_i) + \epsilon_i$$

where the errors (ϵ_i) are independent and normally distributed $N(0, \sigma)$. Linear regression also assumes equal variance of y (σ is the same for all values of x). We use ϵ (Greek epsilon) to stand for the residual part of the statistical model. A response y is the sum of its mean and chance deviation ϵ from the mean. The deviations ϵ represents the “noise” in the data. In other words, the noise is the variation in y due to other causes that prevent the observed (x, y) from forming a perfectly straight line.

The sample data used for regression are the observed values of y and x . The response y to a given x is a random variable, and the regression model describes the mean and standard deviation of this random variable y . The intercept β_0 , slope β_1 , and standard deviation σ of y are the unknown parameters of the regression model and must be estimated from the sample data.

- The value of $\hat{y}$ from the least squares regression line is really a prediction of the mean value of y (μ_y) for a given value of x .
- The least squares regression line ($\hat{y} = b_0 + b_1 x$) obtained from sample data is the best estimate of the true population regression line
- ($\mu_y = \beta_0 + \beta_1 x$)

5.2.3 Method Least Square Understanding

Then generally used method to find the ‘best’ fit that a straight line of this kind can give is the least-square method. To use it efficiently, we first determine.

$$\sum x_i^2 = \sum X_i^2 - n\bar{X}^2$$

$$\sum y_i^2 = \sum Y_i^2 - n\bar{Y}^2$$

$$\sum x_i y_i = \sum X_i Y_i - n\bar{X} \cdot \bar{Y}$$

$$b = \frac{\sum x_i y_i}{\sum x_i^2}, a = \bar{Y} - b\bar{X}$$

These measures define a and b which will give the best possible fit through the original X and Y points and the value of r can then be worked out as under:

Notes

$$r = \frac{b\sqrt{\sum x_i^2}}{\sqrt{\sum y_i^2}}$$

Thus, the regression analysis is a statistical method to deal with the formulation of mathematical model depicting relationship amongst variables which can be used for the purpose of prediction of the values of dependent variable, given the values of the independent variable. Alternatively, for fitting a regression equation of the type $Y = a + bX$ to the given values of X and Y variables, we can find the values of the two constants viz., a and b by using the following two normal equations:

$$\begin{aligned}\sum y_i &= na + b \sum x_i \\ \sum X_i Y_i &= a \sum X_i + b \sum X_i^2\end{aligned}$$

and then solving these equations for finding a and b values. Once these values are obtained and have been put in the equation $Y = a + bX$, we say that we have fitted the regression equation of Y on X to the given data. In a similar fashion, we can develop the regression equation of X and Y viz., $X = a + bY$, presuming Y as an independent variable and X as dependent variable.

Method of Least Square Parabolic Trend

The mathematical form of a parabolic trend is given by $Y_t = a + bt + ct^2$ or $Y = a + bt + ct^2$ (dropping the subscript for convenience). Here a , b and c are constants to be determined from the given data. Using the method of least squares, the normal equations for the simultaneous solution of a , b , and c are:

$$\begin{aligned}\sum Y &= na + b \sum t + c \sum t^2 \\ \sum tY &= a \sum t + b \sum t^2 + c \sum t^3 \\ \sum t^2 Y &= a \sum t^2 + b \sum t^3 + c \sum t^4\end{aligned}$$

By selecting a suitable year of origin, i.e., define $X = t - \text{origin}$ such that $\sum X = 0$, the computation work can be considerably simplified. Also note that if $\sum X = 0$, then $\sum X^3$ will also be equal to zero. Thus, the above equations can be rewritten as:

$$\sum Y = na + c \sum X^2 \quad \dots(i)$$

$$\sum XY = b \sum X^2 \quad \dots(ii)$$

$$\sum X^2 Y = a \sum X^2 + c \sum X^4 \quad \dots(iii)$$

$$\text{From equation (ii), we get } b = \frac{\sum XY}{\sum X^2} \quad \dots(iv)$$

$$\text{Further, from equation (i) we get } a = \frac{\sum Y - c \sum X^2}{n} \quad \dots(v)$$

$$\text{And from equation (iii), we get } c = \frac{n \sum X^2 Y - (\sum X^2)(\sum Y)}{n \sum X^4 - (\sum X^2)^2} \quad \dots(vi)$$

Thus, equations (iv), (v) and (vi) can be used to determine the values of the constants a , b and c .

Method of Moving Average - Introduction

This method is based on the principle that the total effect of periodic variations at different points of time in its cycle gets completely neutralised, i.e., $S = 0$ in one year and $C = 0$ in the period of cyclical variations.

Method of Least Square - Exponential Trend

The general form of an exponential trend is $Y = a.b^t$, where a and b are constants to be determined from the observed data. Taking logarithms of both sides, we have $\log Y = \log a + t \log b$. This is a linear equation in $\log Y$ and t and can be fitted in a similar way as done in case of linear trend. Let $A = \log a$ and $B = \log b$, then the above equation can be written as $\log Y = A + Bt$.

The normal equations, based on the principle of least squares are:

$$\sum \log Y = nA + B \sum t$$

$$\text{and } \sum t \log Y = A \sum t + B \sum t^2$$

By selecting a suitable origin, i.e., defining $X = t - \text{origin}$, such that $\sum X = 0$, the computation work can be simplified. The values of A and B are given by

$$A = \frac{\sum \log Y}{n} \text{ and } B = \frac{\sum X \log Y}{\sum X^2}$$

respectively. Thus, the fitted trend equation can be written as $\log Y = A + BX$

or $Y = \text{Antilog } [A + BX] = \text{Antilog } [\log a + X \log b]$

$= \text{Antilog } [\log a.b^x] = a.b^x$.

A probability is the quantitative measure of risk. Statistician I.J. Good suggests, "The theory of probability is much older than the human species, since the assessment of uncertainty incorporates the idea of learning from experience, which most creatures do."

Development of probability theory in Europe is associated with gamblers in the famous European casinos, such as the one at Monte Carlo. It is also associated with astrology. This unit provides exposure to fundamental concepts, since probability is inseparable from statistical methods. Those, not familiar with the subject, are suggested to study details from any book or a book by this author on probability and solve few numerical problems to understand the logic.

In probability, the understanding and method to solve a given problem. The theory of probability is an indispensable tool in the analysis of situations involving risk. It is used in various fields such as quality control, management, engineering, physics, biology, and economics and so on.

Important Terms in Probability

Probability and sampling are inseparable parts of statistics. Before we discuss probability and sampling distributions, we must be familiar with some common terms used in theory of probability. Although these terms are commonly used in business, they have precise technical meaning.

Notes

5.2.4 Maths Behind Least Square

The Method of Least Squares is a method for determining the best fit line to data; the proof employs simple calculus and linear algebra. The fundamental issue is determining the best-fitting straight line

$$y = ax + b$$

given that, for $n \in \{1, \dots, N\}$, the pairs (x_n, y_n) are observed. The method easily generalises to determining the best form fit:

$$y = a_1 f_1(x) + \dots + c_k f_k(x);$$

It is not necessary for the functions f_k to be linear in x ; instead, y must be a linear combination of these functions.

To find the line of best fit for **N** points:

Step 1: For each (x,y) point calculate x^2 and xy

Step 2: Sum all x , y , x^2 and xy , which gives us Σx , Σy , Σx^2 and Σxy (Σ means "sum up")

Step 3: Calculate Slope **m**:

$$m = \frac{N \Sigma(xy) - \Sigma x \Sigma y}{N \Sigma(x^2) - (\Sigma x)^2}$$

(N is the number of points.)

Step 4: Calculate Intercept **b**:

$$b = \Sigma y - m \Sigma x / N$$

Step 5: Assemble the equation of a line

$$y = mx + b$$

Example: Sam found how many hours of sunshine vs how many ice creams were sold at the shop from Monday to Friday:

"x" Hours of Sunshine	"y" Ice Creams Sold
2	4
3	5
5	7
7	10
9	15

Let us find the best **m** (slope) and **b** (y-intercept) that suits that data

$$y = mx + b$$

Step 1: For each (x,y) calculate x^2 and xy :

x	y	x ²	xy
2	4	4	8
3	5	9	15
5	7	25	35
7	10	49	70
9	15	81	135

Step 2: Sum x, y, x² and xy (gives us Σx , Σy , Σx^2 and Σxy):

x	y	x ²	xy
2	4	4	8
3	5	9	15
5	7	25	35
7	10	49	70
9	15	81	135
Σx: 26	Σy: 41	Σx^2: 168	Σxy: 263

Also **N** (number of data values) = 5

Step 3: Calculate Slope **m**:

$$m = \frac{N \Sigma(xy) - \Sigma x \Sigma y}{N \Sigma(x^2) - (\Sigma x)^2}$$

$$= \frac{5 \times 263 - 26 \times 41}{5 \times 168 - 26^2}$$

$$= \frac{1315 - 1066}{840 - 676}$$

$$= \frac{249}{164} = 1.5183...$$

Step 4: Calculate Intercept **b**:

$$b = \Sigma y - m \Sigma x / N$$

$$= 41 - 1.5183 \times 26$$

$$= 0.3049...$$

Step 5: Assemble the equation of a line:

$$y = mx + b$$

$$y = 1.518x + 0.305$$

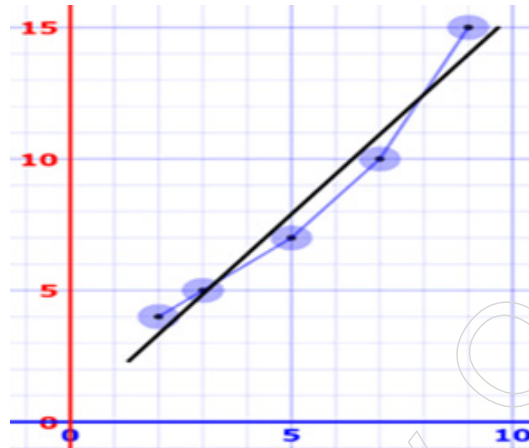
Let's see how it works out:

x	y	y = 1.518x + 0.305	error
2	4	3.34	-0.66
3	5	4.86	-0.14
5	7	7.89	0.89
7	10	10.93	0.93
9	15	13.97	-1.03

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Here are the (x,y) points and the line $y = 1.518x + 0.305$ on a graph:



Nice fit!

Sam hears the weather forecast which says “we expect 8 hours of sun tomorrow”, so he uses the above equation to estimate that he will sell

$$y = 1.518 \times 8 + 0.305 = 12.45 \text{ Ice Creams}$$

Sam makes fresh waffle cone mixture for 14 ice creams just in case.

Check your Understanding

Fill Up / True or False

1. There is a need for a statistical model that will extract information from the given data to establish the _____ relationship between independent and dependent relationship.
2. The non-systematic behaviour cannot be captured and called as _____.
3. The error is due to _____ component that cannot be predicted as well as the component not adequately considered in statistical model.
4. If the variables in a bivariate distribution are correlated, the points in scatter diagram approximately cluster around some curve. True / False
5. The equation of the curves which is closest to the observations is called the “_____”.
6. When there are multiple input variables, literature from statistics often refers to the method as _____ linear regression.
7. To construct an ordinary least-squares regression line, we use the _____ and standard deviations of our sample data to calculate the slope and y-intercept.
8. In the method of moving average, successive arithmetic averages are computed from overlapping groups of successive values of a time series. True / False

Multiple Choice Questions

1. Fit the straight line to the following data.

x	1	2	3	4	5
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y	1	2	3	4	5
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- a) $y=x'$
- b) $y=x+1$
- c) $y=2x$
- d) $y=2x+1$

2. Fit a second degree parabola to the following data.

x	1	2	3	4	5	6	7	8	9
y	2	6	7	8	10	11	11	10	9

- a) $y = -0.2673x^2 + 3.5232x - 0.9286$
- b) $y = 0.2673x^2 + 3.5232x - 0.9286$
- c) $y = 0.2673x^2 + 3.5232x + 0.9286$
- d) $y = -0.2673x^2 + 3.5232x + 0.9286$

3. Fit the straight line to the following data.

x	0	5	10	15	20
y	7	11	16	20	26

- a) $y = 0.94x + 6.6$
- b) $y = 6.6x + 0.94$
- c) $y = 0.04x + 5.6$
- d) $y = 5.6x + 0.04$

4. Fit the straight line curve to the following data.

x	75	80	93	65	87	71	98	68	84	77
y	82	78	86	72	91	80	95	72	89	74

- a) $y = 0.9288x + 7.78155$
- b) $y = 7.78155x + 0.9288$
- c) $y = 0.8288x + 6.78155$
- d) $y = 6.78155x + 0.8288$

Summary

- There is a need for a statistical model that will extract information from the given data to establish the regression relationship between independent and dependent relationship. The model should capture systematic behaviour of data. The non-systematic behaviour cannot be captured and called as errors. The error is due to random component that cannot be predicted as well as the component not adequately considered in statistical model.
- Linear regression is a linear model, e.g a model that assumes a linear relationship between the input variables(x) and the single output variable (y).

Notes

- When there is a single input variable (x), the method is referred to as simple linear regression. When there are multiple input variables, literature from statistics often refers to the method as multiple linear regression.
- To construct an ordinary least-squares regression line, we use the means and standard deviations of our sample data to calculate the slope and y-intercept.
- In the method of moving average, successive arithmetic averages are computed from overlapping groups of successive values of a time series.

Activity

1. Determine the forecast for the four year simple moving average

Year	1	2	3	4	5	6	7	8	9	10
Sales	20	21	23	22	25	24	27	26	28	30

Questions and Exercises

1. What are the two questions discussed in the general concept of regression?
2. What is the mathematical form of a parabolic trend?
3. What is the applicability of using the moving average method?
4. What is the mathematical equation for the method of least squares?

Glossary

- **Linear regression** is a linear model, e.g a model that assumes a linear relationship between the input variables(x) and the single output variable (y).
- **Moving average method**: successive arithmetic averages are computed from overlapping groups of successive values of a time series.
- **Population Regression**: is the regression model based on a sample of n bivariate observations drawn from a larger population of measurements.

Further Reading

1. Athanasopoulos, George; Hyndman, Rob J. Forecasting: Principles and Practice. Otexts; 3rd ed. edition (May 31, 2021)
2. Gilliland, Michael. Business Forecasting (Wiley and SAS Business Series). Wiley; 1st edition (December 29, 2015)

Check your Understanding: Answers

Fill Ups/ True or False

1. Regression
2. Errors
3. Random
4. True
5. Best fit
6. Multiple

7. Means

8. True

Multiple Choice

a. a

b. a

c. a

d. a

Notes